Public Housing Preferences: New York City, 1930–2010*

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Abstract

This paper studies how public housing reshaped neighborhoods and private housing markets. I link archival NYCHA records to decennial Census data from 1930 to 2010 and 19,463 geocoded New York Times rental listings. To estimate these effects, I use a stacked event study design that compares 166 New York City census tracts first treated before 1970 with slightly more distant tracts that were never treated. Public housing raised total population and housing, but these gains masked a persistent contraction of the private market. Six decades after completion, treated tracts had about 960 fewer privately housed residents and roughly 720 fewer private units. Growth in total Black and Hispanic populations largely reflected public-housing occupancy rather than private-market inflows. Median asking rents per room were roughly 50 percent lower in the long run, and Census contract rents fell by about 40 percent. The joint decline in private quantities and rents suggests that public housing reduced neighborhood demand rather than simply removing private housing supply. I then use a location-choice framework to recover rent-equivalent valuations of public-housing attributes. Greater construction investment is associated with smaller private-market discounts, while taller buildings are associated with larger discounts. The results highlight a central trade-off in affordable-housing design between project scale and construction quality.

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1 Introduction

Throughout the twentieth century, public housing was a cornerstone of housing policy in the United States. It provided affordable housing to low-income households, but it also represented a large and permanent intervention in urban neighborhoods. When federal public housing began in the mid-1930s, projects were often celebrated as an improvement over slum conditions. By the late 1960s, however, this perception had changed. Critics increasingly associated large high-rise developments with physical isolation, the concentration of disadvantaged households, and harm to surrounding neighborhoods (Jacobs, 1961; Newman, 1997; Plunz, 2016). These concerns remained salient decades later and extended to low-income housing more broadly. In a 2002 survey of 1,000 Chicago-area residents, 52 percent associated low-income housing with increased neighborhood crime, while 47 percent regarded its design as unattractive (Belden Russonello & Stewart, 2003, p. 9).

These perceptions, however, do not establish how public housing actually affects a neighborhood. Projects can affect neighborhoods through two related margins: a planning margin, which determines what is built, and a resident margin, which determines who moves in and who moves out. On the planning margin, construction adds subsidized housing but may replace existing housing, while the scale and design of a project can change how private households value the neighborhood. On the resident margin, public housing brings new tenants into the tract and may alter the location choices of households in the surrounding private market. Public housing may therefore expand total population and housing even as private population and housing contract. The magnitude of this response may depend on construction investment, height, site coverage, and the composition of a project's initial tenants (Radford, 2008; Goetz, 2012; Bloom et al., 2016; Allen and Van Riper, 2020).

New York's postwar construction boom provides a particularly useful setting in which to study these questions. NYCHA built 72,499 public-housing units in the 1950s and another 42,721 in the 1960s (Plunz, 2016).[1] This program did more than increase the supply of subsidized apartments. Through slum clearance and large-site redevelopment, it replaced existing housing with large, often high-rise superblocks and changed the physical form of entire neighborhoods. This was the urban intervention at the center of Jane Jacobs's critique of modernist planning (Jacobs, 1961).

In this paper, I examine how this mid-century construction program reshaped neighborhood composition and the private housing market, and whether its effects depended on how projects were built. I combine archival records for NYCHA developments with de-

[1] Projects completed between 1951 and 1970 account for almost two thirds of all public-housing units in the project records (see Appendix B2). Median project size exceeded 1,000 units in every cohort completed through 1960, median project height reached 18 stories in the 1961–70 cohort, and public housing represented almost one quarter of all housing constructed in New York City in that period. See Table B2.1 and Figure B2.1.

cennial U.S. Census data from 1930 to 2010 and focus on the 166 census tracts that first received public housing before 1970. This tract-level panel allows me to distinguish the direct addition of public-housing residents and apartments from changes in the surrounding private market. Because construction slowed and the developments built after 1970 differed in scale, design, location, and administration, I interpret the estimates as the effects of the mid-century construction regime rather than of public housing in every period.

I complement tract-level median contract rents with a new dataset of 19,463 geocoded rental listings from the *New York Times* real-estate section, digitized at ten-year intervals between 1930 and 2010. Median contract rent provides consistent coverage across census tracts, but it may include subsidized public-housing rents. The newspaper listings measure advertised private-market rents and therefore provide a separate indicator of private-market demand, although their spatial coverage is more limited.

To identify the causal effects of public housing, I exploit the staggered construction of NYCHA projects across New York City. Site selection was not random: NYCHA targeted areas that differed systematically from the rest of the city. I therefore compare changes in tracts that received public housing, and in immediately adjacent tracts, with changes in slightly more distant tracts around the same project site. Following Blanco and Neri (2025), I implement this comparison in a stacked difference-in-differences design centered on each tract's first project completion. This design allows me to trace the effects on population, the housing stock, and market rents for up to six decades after construction.

I find that public housing did not simply add subsidized units to an otherwise unchanged neighborhood. In the project tract, total population rises by about 1,500 residents at completion, with a point estimate roughly 820 above the counterfactual after six decades. The private-population estimates move in the opposite direction, falling by about 410 residents at completion and nearly 960 in the long run. The housing stock shows the same replacement pattern. Total housing rises by roughly 460 units at completion and remains about 340 units higher after six decades, while the private housing stock declines by about 350 units immediately and 720 units in the long run. The completion-period gap implies that roughly 810 public-housing units were added. Thus, the increase in total housing masked a persistent contraction of the private residential market.

Separating public-housing residents also changes the interpretation of racial composition. The privately housed White population falls by roughly 810 residents after six decades. The privately housed Black population declines by about 600 residents after three decades, although its longer-run estimate is smaller and imprecise. In contrast, total Black and Hispanic populations remain substantially above the counterfactual in the long run, while the corresponding private-population estimates are smaller. Much of the growth in total Black and Hispanic populations is therefore accounted for by public-housing occupancy

rather than comparable inflows through the private market.

The decline in the private housing stock is accompanied by lower, rather than higher, private-market rents. Median and average asking rent per room in the newspaper listings fall by about 0.75 and 0.70 log points after six decades, corresponding to declines of roughly 53 and 50 percent. Median contract rent falls by about 0.55 log points, or roughly 42 percent. A contraction of private housing supply alone would ordinarily place upward pressure on rents. The fact that both private housing and advertised rents decline is therefore consistent with a reduction in private housing demand beyond supply removal alone. The listing results further show that the rent decline is not solely a mechanical consequence of subsidized apartments entering the tract median.

I next examine whether the private-market response depended on the characteristics of the project. I use a static neighborhood-choice model, following Bayer et al. (2007) and Almagro et al. (2023), to translate the joint population and rent responses into the rent change that would offset the estimated neighborhood effect. This places the population and price estimates in a common metric. For a pre-1970 project with mean characteristics, the model implies a large compensating rent discount for White and Black private-market residents; the Hispanic estimate is smaller and less precise. These model-implied valuations are more negative in the project tract, while the corresponding long-run estimates in adjacent areas are smaller and statistically indistinguishable from zero.

Greater construction investment is associated with a less negative valuation. At the baseline calibration, a $1,000 increase in construction cost per room corresponds to a model-implied willingness to pay about $105 more per month for White and Black residents and $66 for Hispanic residents, in 2010 dollars. In contrast, taller projects are associated with model-implied rent discounts of approximately $149 per additional story for White residents and $127 for Black residents; the corresponding Hispanic estimate is smaller and imprecise. Ground coverage is less systematically related to neighborhood valuation once height and the other project attributes are held fixed. Initial tenant composition also produces less consistent estimates and is especially difficult to interpret because opening allocations reflected administrative rules and local conditions. I therefore view the construction-investment and height estimates as the more systematic findings: greater recorded investment is associated with smaller private-market discounts, while taller projects are associated with larger discounts.

The paper is most closely related to the literature on the neighborhood effects of public and subsidized housing construction. At the national level, Shester (2013) studies public-housing construction between 1940 and 1970 and finds that adverse neighborhood effects emerged only over the long run. In New York City, Schwartz et al. (2006) and Ellen et al. (2007) estimate the effects of more recent subsidized housing developments on nearby

property values, while Diamond and McQuade (2019) show that the capitalization of Low-Income Housing Tax Credit developments depends on initial neighborhood characteristics. This paper complements that literature by examining the original construction of the large public housing projects that shaped mid-century cities. I follow neighborhoods for up to six decades after project completion and distinguish the direct addition of public housing residents and units from equilibrium changes in the surrounding private population, housing stock, and advertised rents. This distinction is important because aggregate tract outcomes can change mechanically following project construction even when the surrounding private neighborhood evolves in the opposite direction.[2]

A second closely related literature examines the demolition and regeneration of distressed public housing. Aliprantis and Hartley (2015) and Sandler (2017) study crime following Chicago demolitions; Blanco (2022) studies house prices, rents, and residential composition; and Almagro et al. (2023) evaluate the equilibrium and distributional consequences. Blanco and Neri (2025) and Staiger et al. (2024) study regeneration programs that combine demolition with rebuilding, relocation, and changes in tenure mix. These papers estimate what happens when mature projects are removed or remade. I instead estimate what happened when the projects were initially built. The two interventions need not be mirror images: demolition follows decades of neighborhood adjustment and is often bundled with vouchers or mixed-income redevelopment.[3]

Finally, the paper relates to the literature on housing externalities, residential sorting, and the role of the built environment in shaping neighborhood outcomes. Rossi-Hansberg et al. (2010) and Autor et al. (2014) show that housing investment and regulation generate important spatial spillovers, while Asquith et al. (2023) study how new apartment construction affects nearby rents, migration, and housing supply. More recently, Redding and Sturm (2024) show that reconstruction quality can have persistent effects on neighborhood development. Direct evidence on architecture itself as a neighborhood amenity remains relatively limited, although Ahlfeldt and Holman (2018) document price premiums in architecturally distinctive neighborhoods. This paper contributes by linking variation in public housing design—including construction expenditure, building height, site coverage, and initial ten-

[2]Related historical work studies the broader spatial consequences of social housing. Dalmazzo et al. (2022) show that Italy's postwar INA-Casa program increased local population and industrialization, while Bromhead and Lyons (2023) document persistent effects of rural social housing on population distribution in Ireland. A complementary literature asks whether subsidized construction displaces private supply. Baum-Snow and Marion (2009) find crowd-out of nearby rental construction in gentrifying neighborhoods, while Eriksen and Rosenthal (2010) estimate substantial crowd-out at broader spatial scales.

[3]An adjacent literature follows residents affected by demolition or offered moves to lower-poverty neighborhoods. Jacob (2004) finds no effect of Chicago demolitions on student achievement, whereas Chyn (2018) finds higher adult employment and earnings and fewer violent-crime arrests among displaced children. Evidence from the Moving to Opportunity experiment shows gains for children who moved at young ages but not for adolescents (Chetty et al., 2016). These studies estimate effects on exposed households rather than on the neighborhoods in which projects were located.

ant composition—to long-run neighborhood responses. Following Bayer et al. (2007) and Almagro et al. (2023), I map the model's comparative statics to the reduced-form estimates. This allows me to express each project attribute as the rent change that would offset its estimated effect on neighborhood attractiveness and to quantify the sampling uncertainty in that measure.

The paper proceeds as follows. Section 2 describes the historical setting. Section 3 presents the data, and Section 4 develops the empirical strategy. Section 5 reports the long-run effects of public-housing construction. Section 6 studies project attributes and reports willingness-to-pay estimates, and Section 7 concludes.

2 Background

Public housing emerged as a key response to urban poverty during the mid-20th century, aiming to replace slums, stabilize communities, and provide affordable housing to working-class families. These efforts also reshaped neighborhood racial and socioeconomic dynamics in lasting ways. In response to the Great Depression's economic devastation and housing crisis, federal programs like the Public Works Administration and the 1937 Wagner-Steagall Act launched a wave of slum clearance and new construction to foster health and safety through modern housing (Allen and Van Riper, 2020; Radford, 2008; Fogelson, 2003).

In New York City, the newly formed Housing Authority (NYCHA) pioneered high-quality, professionally managed public housing. Its first project, the First Houses (1935), exemplified the era's optimism: replacing overcrowded tenements with clean, modern apartments for working families (Roosevelt House Public Policy Institute at Hunter College, 2025). Early residents often recalled public housing as "paradise," illustrating the broader optimism and strong community cohesion that marked the early era of federally supported housing (Williams, 2014; Bloom, 2008; Marcuse, 1986; Hunt, 2009).[4]

Authorities like NYCHA and the Chicago Housing Authority implemented strict tenant screening, emphasizing stable employment, nuclear families, and proximity to transit and schools. Early projects often reinforced patterns of racial segregation, with tenant selection policies and housing placements reflecting broader societal prejudices. In New York City, some developments were designated for White families, while others primarily housed

[4]Maude Davis, a retired public school principal, described Altgeld Gardens in Chicago - opened in 1945 - as "paradise," noting, "We felt this was just the greatest housing that we could live in! There was pride in it." So did Vonsell Ashford when moving into the 1955 completed Harold L. Ickes Homes in Chicago: "The building was new, and they had a beautiful playground for the children. You couldn't ask for a better location, and the place was just marvelous. I had three bedrooms, a nice storage area, and a linen closet ... I thought I was moving to paradise" (Hunt, 2009). Though expressed by Chicago residents, these sentiments reflect the broader optimism that characterized early public housing, including NYCHA developments.

Black or Hispanic households.[5] These developments were operated like private apartment complexes, with routine maintenance and community-building efforts (Hunt, 2009; Bloom, 2008).

Demand for public housing surged in the postwar period amid a housing shortage and returning servicemen. The 1949 Housing Act spurred expansion while reaffirming its dual goals of slum clearance and affordability. In NYC alone, 72,499 units were built in the 1950s and 42,721 more in the 1960s (Plunz, 2016). Construction sites were typically selected based on land costs and surrounding population densities, leading to a concentration of developments in lower-cost neighborhoods.[6] By 1970, public housing made up 25% of all new residential construction. Designs shifted from low-rise blocks to high-density towers in the modernist *Tower in the Park* model, emphasizing open space around high-rise buildings.[7]

These projects were not isolated housing interventions. Especially during the Robert Moses era, public housing construction formed part of a broader program of slum clearance, urban renewal, arterial-road construction, and large-site redevelopment that jointly changed accessibility, parcel structure, and land use (Caro, 1975). Large projects often replaced sections of the existing street grid and many small parcels with superblocks, making them durable changes to the physical organization of their neighborhoods.[8]

By 1968, NYCHA eased its admission criteria, declaring that it would no longer evaluate applicants based on moral considerations. Landmark cases such as *Escalera v. NYCHA* (1970) led to the relaxation of strict tenant screening practices, including the elimination of race-based admissions policies that had long characterized early projects. While these changes reflected broader efforts to promote equity, they also contributed to increasing concentrations of poverty within developments. This period coincided with a decline in federal funding and growing maintenance backlogs, straining NYCHA's resources and infrastructure (Bloom, 2008).

The trajectory of public housing in New York City diverged sharply from its early promise. The optimism embodied in the initial construction boom gradually gave way to

[5]Moreover, early public housing projects exhibited explicit segregation, with a predominant allocation to White residents (see Figure B2.6). Against this backdrop, tracts with public housing projects followed different demographic trajectories than trends in the rest of New York City. Specifically, tracts designated for public housing had significantly higher White population in 1930. In 2010, the same tracts had a lower White and higher Black population than the average tract in the rest of the city (see Figure B1.1).

[6]Federal regulations capped land acquisition costs at $1.50 per square foot, making many centrally located slum areas financially infeasible. NYCHA faced considerable challenges assembling large parcels in built-up areas due to fragmented ownership, political opposition, and zoning constraints. As a result, the Authority often settled on sites that were affordable but marginal, including deindustrialized zones and peripheries of established neighborhoods (Genevro, 1986).

[7]Figure B2.1 shows the evolution of public housing construction by construction cohort.

[8]This transformation coincided with New York City's 1961 Zoning Resolution, which comprehensively revised the regulation of land use and building density. Subsequent redevelopment around public-housing sites therefore took place under a different zoning regime from the one that had shaped much of the displaced prewar housing stock (Plunz, 2016).

concerns about crime, neglect, and physical deterioration. The public housing design, called the *Tower in the Park*, drew criticism and lost public support. Famously, urban sociologist Jane Jacobs and architect Oscar Newman criticized the *Tower in the Park* as a utopian design that created places marked by crime and social isolation—spaces that were difficult to monitor and maintain. Rising opposition resulted in a policy shift at the local level in favor of low-density public housing (Jacobs, 1961; Newman, 1997).

Despite these challenges, NYCHA maintained a degree of resilience that distinguished it from other public housing systems, largely due to its commitment to management and maintenance (Bloom, 2008). However, NYCHA's developments were not immune to broader urban challenges. As Marilyn Jones, who moved into the Queensbridge housing project in 1970, puts it: "In the beginning for about the first 2 or 3 years it was fine, but then all of a sudden, crime started, people running around with guns, shooting everybody, people throwing people off the rooftops, police all over the place everywhere" (Petrus and Rosner, 2019).

By the 1980s, chronic underfunding and demographic shifts exacerbated social and economic isolation within public housing communities. Federal programs like HOPE VI aimed to modernize or demolish aging public housing and introduce mixed-income communities (Goetz, 2012; Fernandez, 2010). By the early 2000s, survey evidence showed that concerns about maintenance, crime, property values, and design extended to low-income housing more broadly (Belden Russonello & Stewart, 2003). The survey documents the persistence of these perceptions, rather than the effects of public housing itself.

Today, NYCHA remains emblematic of both the early promise and enduring challenges of public housing. Its developments house over 400,000 residents, making it the largest public housing authority in North America. However, it faces a $40 billion capital repair backlog, with aging infrastructure and funding constraints threatening its future. Nationally, public housing continues to grapple with questions of sustainability, equity, and integration (NYCHA, 2021).

3 Data

This section describes the NYCHA development records, the demographic, housing-stock, and rent outcomes, and the construction of the tract panel and estimation sample.

Public Housing Characteristics. I obtain information from three sources. First, I use the New York City Housing Authority (NYCHA) Development Data Book, available annually from 1948 onward. It provides information on construction date, height, number

of apartments, construction costs,[9] and ground coverage—the total ground-floor area of a development's building footprints divided by its site area. I recover corresponding information for 1940 from the LaGuardia and Wagner Archives. The administrative records cover 299 NYCHA-operated developments.[10] Second, I supplement these records with the racial composition of public-housing residents, including the number of White and Black residents at initial occupancy and in each census year. For developments constructed through 1970, I digitize records from the LaGuardia and Wagner Archives; Figure B2.5 shows an example. For later years, I use the NYCHA Resident Data Book. I spatially match each development to 2010 census tracts. When a development spans several tracts, I allocate additive quantities, such as residents and apartments, in proportion to the share of the development's site in each tract. I then construct the tract-level design measures from the matched development records. For height, I use the maximum reported number of stories among new-construction developments intersecting the tract.

Rents. I primarily use the median contract rent reported in the decennial census, which is available from 1930 to 2010 at the census tract level. However, this variable has several limitations. First, it is top-coded, meaning that rents above a certain threshold are reported as a single value, which can distort the true distribution of rents. Second, it might not reflect true market rents, as it includes public housing rents, which are often subsidized.

Therefore, I complement the census data with private rental market data from the *New York Times* real estate section. I digitize rent prices from the *New York Times* real estate section for each decennial census year from 1930 to 2010 to examine the impact of public housing on rents, resulting in 19,463 rental listings. Only properties with an exact address or cross-street information were included to ensure accurate geolocation, and the Google Maps API was used to geocode the rental data. Using property-level rent data presents both advantages and limitations. On the one hand, it circumvents issues inherent to median contract rent reported in the United States census; on the other hand, relying on newspaper data, particularly from the *New York Times*, introduces two key limitations. First, as an upper-middle-class newspaper, the *New York Times* may not provide comprehensive coverage across all market segments and is biased toward the higher end of the market. However, no newspaper systematically covers the lower end of the rental market. Table B5.1 compares both rent measures and shows that listing prices are considerably higher than census rents. However, when comparing the indexed median

[9] Construction costs exclude land acquisition expenses, which are accounted for separately as part of development costs.

[10] Not all NYCHA developments were built by public entities. Under the *Turnkey* program, introduced in 1969, private developers acquired sites and constructed buildings that NYCHA subsequently purchased. This method accounts for roughly one quarter of the developments in the administrative records.

contract rent to both the average real rent per room from NYT listings and a hedonic rent index constructed from listing characteristics, all three measures exhibit similar trend growth over time (see Figure B5.3). Moreover, the rent-per-room measures from the listings align closely with the level trend of the census rent series (Appendix B5 compares the pairings in detail). Since the empirical strategy relies on a difference-in-differences design, it is the evolution of rents over time — rather than differences in levels — that matters for identification. Second, the listing data have limited spatial coverage, as not all tracts are consistently represented in each year.[11] The census measure therefore provides broader and more consistent tract coverage. A full description of the data collection procedure, summary statistics, and an example of the source material are provided in Appendix B5.

Neither source perfectly measures market-clearing private rents. I therefore treat them as indicators of private-market price movements rather than frictionless prices faced by every tenant.[12]

Census Population and Housing. I obtain White, Black, Hispanic, and total population, total housing units, and census tract boundaries from IPUMS and IPUMS-NHGIS (Manson et al., 2024). Population and tract boundaries are available from 1930 through 2010, while housing-unit counts begin in 1940. A challenge when building a geographically consistent panel dataset is the presence of boundary changes over time. Census tract boundaries underwent substantial changes throughout most of the 20th century, especially in Brooklyn (Kings County) and Queens County.

Therefore, I harmonize the population and housing-unit counts in each available census year to 2010 census tract definitions using overlapping area weights, allowing for the construction of a geographically consistent panel. For 1930, I rely on full-count population census data, aggregated at the enumeration district level, which is geographically much smaller than census tracts. Mapping these enumeration districts to 2010 census tracts helps reduce potential measurement error. A potential limitation of this approach, however, is the implicit assumption that tract-level population characteristics are uniformly distributed across space. Further details on the harmonization procedure and data sources are provided in Appendix B4 and Appendix B1, respectively.[13] Finally, I use the administrative records to construct outcomes net of public housing. In each tract-year, I subtract

[11] Figure B5.2 displays the location of the geocoded listings by year.

[12] Census contract rents combine regulated and unregulated units, while the newspaper data report asking rents and do not consistently identify whether a listing was rent controlled or rent stabilized. The listings are closer to prices faced by marginal movers but disproportionately cover the upper part of the market. Rent regulation may also attenuate or delay measured adjustment.

[13] I also harmonize the median contract rent and the median household income in every census year to 2010 census tract boundaries. Because income is not a count variable, I calculate tract-level estimates as weighted averages across historical geographies using overlapping area weights.

public-housing residents from the corresponding population counts and the public-housing apartment stock in that census year from total housing units, setting negative residuals to zero. The resulting series measure population and housing stock outside public housing; I refer to them as private population and private housing stock.

Sample. I combine the harmonized census panel with the spatially matched NYCHA records at the 2010 census-tract level. The resulting data cover 2,164 non-water tracts at decennial intervals from 1930 through 2010. Of these, 2,161 are present in all nine census waves, although coverage of individual outcomes varies across tract-years. Because a NYCHA development can cross tract boundaries and a tract can contain several developments, I aggregate the public-housing records to the tract level. After aggregation, 232 tracts ever contain a new-construction development. I assign each of these tracts the completion year of its first such development; Section 4 explains how this date enters the empirical design.

The baseline cohort comprises 166 public-housing tracts whose first new-construction development was completed before 1970. The remaining 66 tracts, whose first development was completed in 1970 or later, enter only the construction-cohort comparison in Appendix C2. Population outcomes are available for all 166 baseline tracts. Because housing-unit counts begin in 1940, the housing sample omits tracts whose first development predates that year. The NYT sample contains at least one rental listing in the area surrounding 155 of the 166 tracts, with coverage varying substantially across tracts and years. All prices and costs are deflated by the CPI and expressed in 2010 dollars. Appendix B2 reports descriptive statistics for the public-housing records, and Appendix B5 describes the rental-listing sample.

4 Empirical Strategy

This section describes the empirical strategy for estimating the causal effects of public housing on population, housing stock, and rents. The main specification is a stacked difference-in-differences design that leverages variation in the timing of a tract's first exposure to public housing and in proximity to the treated tract. Treatment timing is defined by the completion date of the first new-construction NYCHA development in the tract, as in related studies of housing construction (Asquith et al., 2023; Pennington, 2021). Subsequent developments in the same tract do not define new treatment events.[14] Throughout the paper, the main treatment cohort consists of the 166 public-housing tracts first treated before 1970. Section 5.5 and Appendix C2 document that the 66 tracts first treated later were ex-

[14]Because I use decennial census years, a tract first exposed within a given decade appears as treated at the end of that period. For example, a first development completed between 1961 and 1970 is observed as treated in 1970. Treatment effects for a census cohort therefore average across tracts first exposed during the corresponding decade.

posed to smaller and institutionally distinct developments and produced muted responses, motivating this restriction.

The main challenge in the empirical analysis is selecting a suitable comparison group that accurately reflects how nearby tracts would have evolved in the absence of public housing. This concern is important because public housing sites were not randomly assigned across New York City. Construction sites were often chosen based on pre-construction characteristics such as land prices and population density, so public housing tracts may have differed from the average city tract before construction.

To address this challenge, I construct treated-tract-specific stacks following Blanco and Neri (2025). For each public-housing tract, I define rings of census tracts around it. The public-housing tract and adjacent tracts form the treated and inner-ring groups, while the outer ring serves as the comparison group. Because tract boundaries are fixed, proximity is defined by adjacency to the public-housing tract. Treated tracts are excluded from the first and second rings of other stacks, ensuring that each comparison group consists only of never-treated tracts. Repeating this procedure for each treatment site creates a stacked dataset in which some control tracts may appear in several comparisons. Figure 1 illustrates the spatial layout of fixed tract rings and overlapping tracts, while Figure C6.1 maps the ring classification across the city.

The identifying assumption is conditional parallel trends within each treated-tract stack: absent the arrival of public housing, outcomes in the treated and inner-ring tracts would have evolved similarly to outcomes in the outer-ring comparison tracts, conditional on the controls and fixed effects described below. This assumption does not require public housing sites to be randomly assigned or for treated and comparison tracts to have identical pre-treatment levels. Instead, the event-study coefficients before construction provide a test for differential pre-trends, while comparisons of pre-treatment characteristics are used as supporting diagnostics (Appendix C1). Balance between treated and comparison rings would strengthen the design; differences between the project-area tracts and the rest of New York City would also be consistent with the concern that public housing sites were selected rather than randomly assigned. Table C1.1 shows both patterns: project areas were denser, poorer, and less White than the rest of the city, while within treatment-site stacks the remaining differences between treated and comparison rings are three to nine times smaller, though not zero. Let $\mathcal{R} = \{T, 1\}$ denote the public-housing tract and adjacent first ring, and let $\mathcal{T} = \{-40, -30, \ldots, 60\} \setminus \{-10\}$ denote event time in years, excluding the census decade immediately before completion. For census outcomes, I estimate the following event-study equation at the census tract k, neighborhood (NTA) m, treatment-site stack p, and year t level:

$$y_{kmpt} = \sum_{r \in \mathcal{R}} \sum_{\tau \in \mathcal{T}} \beta_{\tau,r} \mathbb{1}\{t - Y_p = \tau\} \mathbb{1}\{R(k,p) = r\} + \boldsymbol{\delta}_p' \mathbf{Z}_{kt} + \lambda_p y_{kp}^0 + \rho_{pmt} + u_{kmpt}. \tag{1}$$

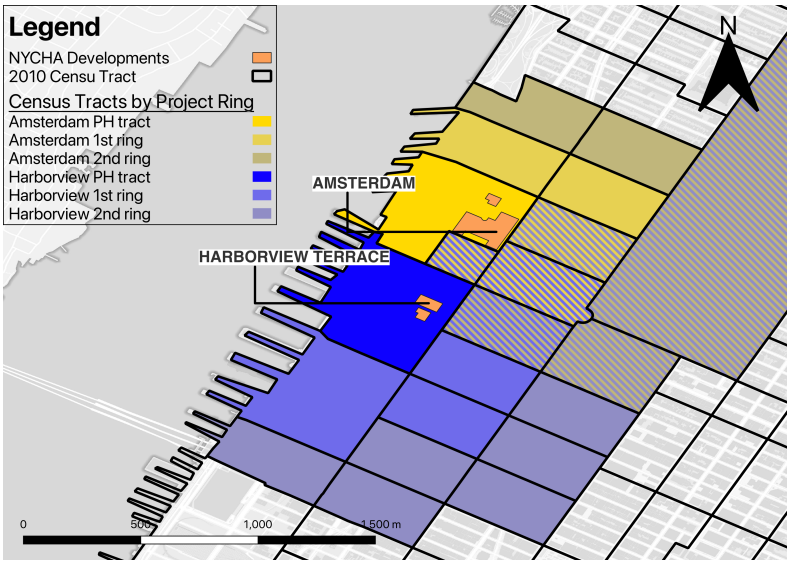
The coefficient $\beta_{\tau,r}$ captures the effect of a tract's first public-housing completion at event time τ in ring r, relative to second-ring tracts and to event time $\tau = -10$. The function $R(k,p)$ assigns tract k to a ring around treatment site p, and Y_p is the completion year of the first new-construction development in that site.

The vector $\mathbf{Z}_{kt}$ records whether tract k has been exposed by year t to urban renewal, an arterial road expansion, or a new subway connection; $\boldsymbol{\delta}_p$ lets these associations differ by stack. I also control for the pre-completion baseline outcome, y_{kp}^0, with stack-specific coefficient λ_p.

The fixed effects ρ_{pmt} absorb shocks common to tracts in the same treatment-site stack, neighborhood, and census year.[15] Identification therefore comes from ring comparisons among tracts in the same stack, neighborhood, and census year, conditional on controls whose coefficients are estimated within stacks.

[15] Each stack-by-neighborhood-by-year cell lies within a single stack-year cell, so every stack-year indicator is a linear combination of the finer ρ_{pmt} effects. Separate stack-year fixed effects are therefore absorbed and cannot be identified jointly.

Figure 1: Treatment construction



Note. Figure 1 provides an illustrative example of overlapping neighborhood/distance rings for two public housing projects: Harborview Terrace and Amsterdam Houses. The concept of neighborhood rings is depicted, with blue and yellow hatched census tracts representing the areas that belong to the respective public housing tract and are located within their respective rings. It is important to note that this tract may appear multiple times in the dataset. If a public housing tract was lying within a neighborhood ring to another public housing tract, it was excluded from the respective ring such that no treated tract appears in the control group.

This approach is equivalent to estimating Equation 1 separately for each treatment-site stack, then aggregating the coefficients using regression weights.[16] Since the number of treated and control observations varies across sub-experiments, estimates are weighted by their relative tract frequency. Baseline standard errors are clustered by stack, allowing arbitrary dependence within a treatment site and its rings. A control tract can, however, enter several stacks, and nearby stacks may share local shocks.[17]

The ring design identifies local relative effects rather than the aggregate effect of public housing on New York City. Citywide equilibrium responses may therefore cause the estimates to understate the full effect.[18]

[16] A stacked difference-in-differences design effectively accounts for heterogeneous treatment effects, a known limitation of traditional DiD estimators (Wing et al., 2024).

[17] In Section 5.4, I assess estimator and comparison-group sensitivity using synthetic difference-in-differences and the de Chaisemartin and D'Haultfoeuille estimator (De Chaisemartin and D'Haultfœuille, 2020). Appendix C7 addresses inference directly: two-way clustering by stack and tract and Conley spatial-HAC errors at 1, 2, and 5 kilometre cutoffs yield similar intervals and leave the conclusions unchanged.

[18] Figure B1.2 in Appendix B shows that treated tracts diverge substantially from the citywide trend, whereas comparison tracts track it closely; sorting into comparison areas should cause those areas to diverge as well. In addition, rents may respond when a project is announced. Because treatment is observed only at decennial intervals, the event-time estimates combine anticipation and post-completion effects.

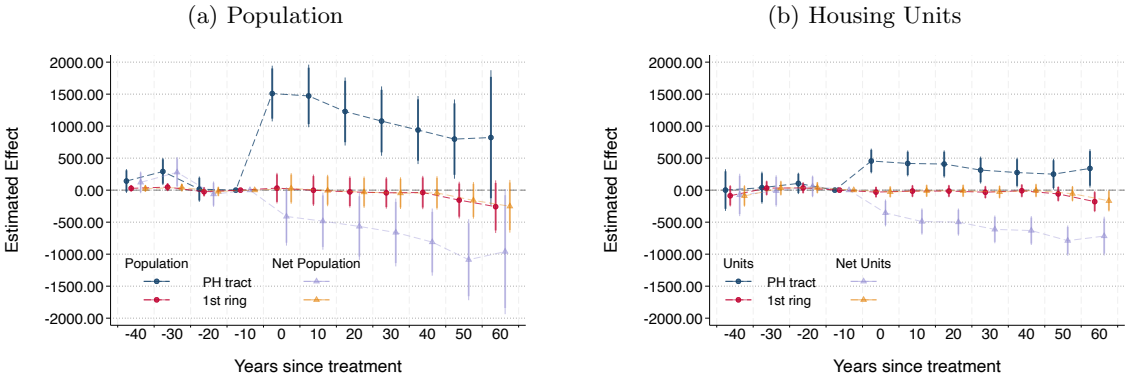
5 Reduced Form Results

5.1 Population and Housing Stock

Figure 2 separates the direct addition of public housing from changes in the private residential market, estimated on the 166 public-housing tracts first treated before 1970. Because developments were often built through clearance and large-site redevelopment, construction could remove private buildings and residents while adding public units and tenants. I therefore report both total outcomes and outcomes net of public-housing residents or units. The gap between the series is the public-housing component, while the net series measures the remaining private population or housing stock.[19]

Panel (a) shows that total population in the public-housing tract rises by about 1,500 residents at completion and remains roughly 820 above the counterfactual after six decades. Private population moves in the opposite direction, falling by about 410 residents at completion and nearly 960 in the long run. The initial gap implies that approximately 1,910 public-housing residents were added while the tract contained about 410 fewer private residents than its counterfactual. First-ring estimates remain small, indicating that the population response was concentrated on the project footprint rather than appearing as a measurable inflow into immediately adjacent tracts.

Figure 2: Long-Run Effects on Population and Housing Stock



Note: The figure plots level coefficients from Equation 1. Panel (a) shows total population and population net of public-housing residents; Panel (b) shows total housing units and housing units net of the public-housing stock. Circles denote total outcomes and triangles net outcomes. Blue and purple denote the public-housing tract; red and orange denote the first ring. The second ring and the decade before completion are the reference categories. Estimates use tract-frequency weights and stack-clustered standard errors; bars show 90% and 95% confidence intervals. The sample is restricted to public-housing tracts first treated before 1970.

[19]These repeated aggregate counts do not follow individuals and therefore cannot determine where displaced residents moved, whether they entered the new development, or whether they later returned.

Panel (b) provides the corresponding building-stock evidence.[20] Total housing rises by roughly 460 units at completion and about 340 units after six decades, whereas the private housing stock falls by about 350 and roughly 720 units, respectively. Public housing therefore did not simply add subsidized units to an unchanged neighborhood. It replaced part of the private stock and reorganized the residential footprint of the tract. First-ring housing estimates are smaller than the project-tract effects but not zero, at roughly 180 fewer total and 170 fewer private units in the long run.

5.2 Race

Figure 3 decomposes the population response by race. Total counts combine residents placed in public housing with those in the private market, whereas net counts subtract public-housing residents of the corresponding group. The comparison therefore distinguishes changes in tract composition from changes among privately housed residents, although it does not follow individual households.

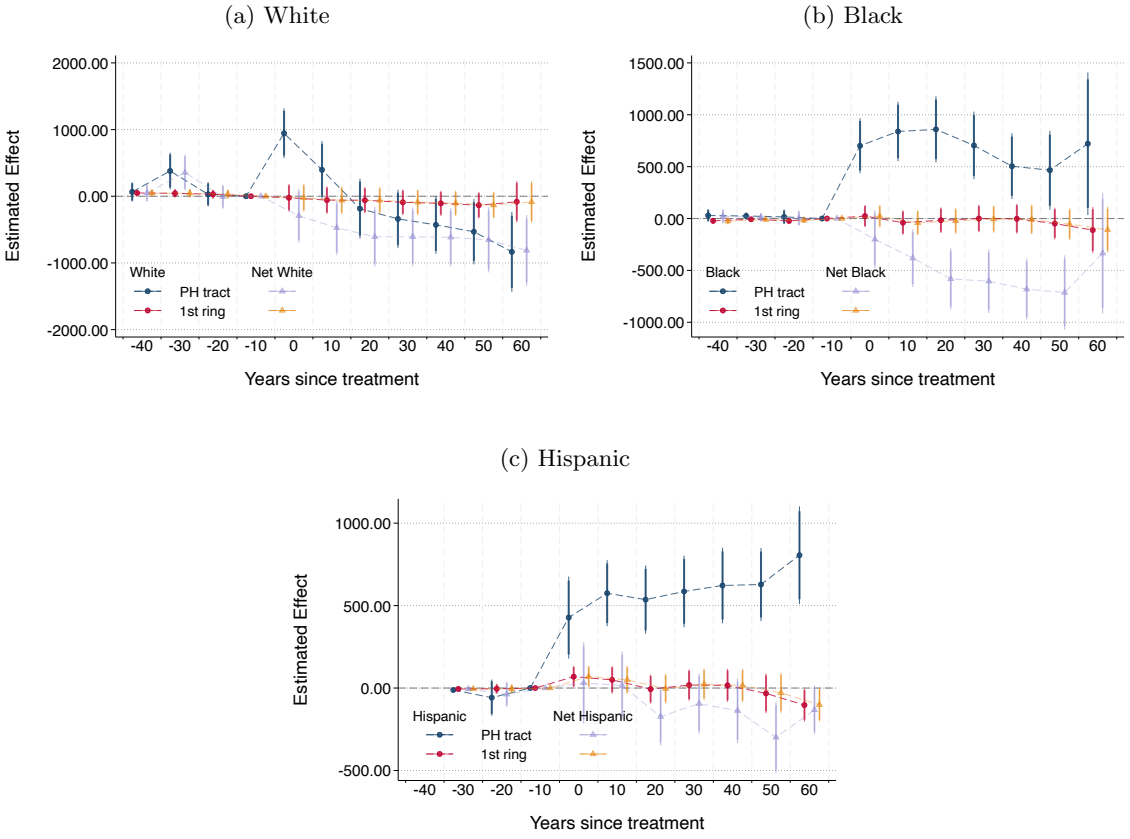
Panel (a) shows that total White population rises by about 950 at completion, reflecting the placement of White tenants in early projects, while private White population falls by about 290. The private decline grows to roughly 810 residents after six decades, by which point the temporary gross increase has disappeared. Panel (b) shows a different gross trajectory for Black residents but a similar private-market response. Total Black population rises by about 700 at completion and remains approximately 720 above the counterfactual after six decades. Private Black population instead declines by about 200 at completion and 600 after three decades, although the long-run estimate is smaller and imprecise. The sustained gross increase thus reflects Black occupancy of public housing rather than growth in the privately housed Black population.

Panel (c) shows that total Hispanic population rises from about 430 residents at completion to approximately 810 after six decades, while the private Hispanic estimates remain much closer to zero.[21] Taken together, the panels show that public housing did not produce the same population pattern across racial groups. Among residents living outside public housing, both White and Black populations declined. When public-housing residents are included, however, total Black and Hispanic populations increased, while the total White population increased only temporarily. First-ring estimates are generally small, indicating no detectable aggregate population change in immediately adjacent tracts.

[20] Housing-unit counts begin in 1940, when the first Census of Housing was conducted, so the housing-stock outcomes exclude 1930. For the same reason, the housing-stock regressions exclude treatment sites first treated before 1940, which have no pre-treatment housing observation.

[21] The White, Black, and Hispanic categories do not exhaust the tract population, so the group-specific estimates do not constitute a complete decomposition of population change. Net racial outcomes also combine harmonized tract-level Census counts with project-level public-housing resident counts. Appendix B4 discusses the residual racial categories and the associated measurement issues.

Figure 3: Long-Run Effects on Racial Composition



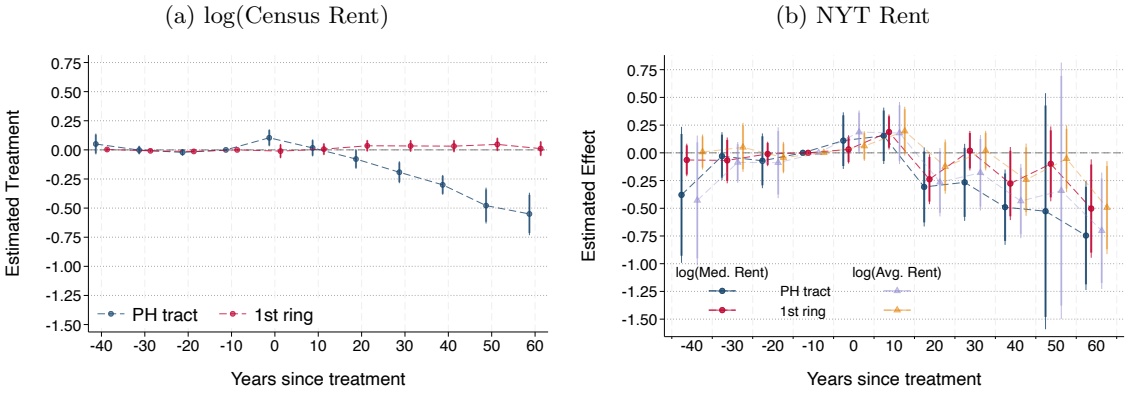
Note: The figure plots level coefficients from Equation 1. Panels (a), (b), and (c) show White, Black, and Hispanic population, respectively. Within each panel, circles denote total group population and triangles population net of public-housing residents of that group. Blue and purple denote the public-housing tract; red and orange denote the first ring. The second ring and the decade before completion are the reference categories. Estimates use tract-frequency weights and stack-clustered standard errors; bars show 90% and 95% confidence intervals. The sample is restricted to public-housing tracts first treated before 1970.

5.3 Rents

Rent effects provide a separate test of whether public housing changed private-market demand beyond the mechanical replacement documented above. The two available measures have different limitations. Census contract rents cover all tracts and years, but they include subsidized public-housing units, so part of their decline in treated tracts can be mechanical: as a project comes to dominate a tract's rental stock, the tract median shifts toward below-market public-housing rents. New York Times listings measure advertised private rents directly and are therefore the preferred market signal, although they are noisier and cover fewer tracts.

Panel (b) reports the listing-based estimates. After six decades, median and average rent per room are approximately 0.75 and 0.70 log points lower—roughly 53 and 50 percent—respectively; the corresponding first-ring estimates are also negative, at about 0.50 log points. Panel (a) shows that Census contract rents follow the same trajectory: they initially rise but subsequently reverse, reaching about 0.15 log points below the counterfactual after three decades and 0.55 log points—roughly 42%—after six, with first-ring estimates close to zero. The agreement across sources indicates that the long-run decline is not an artifact of the subsidized-rent component of the Census measure. Because only the listings isolate private-market prices, they serve as the rent outcome in the attribute and willingness-to-pay analyses in Section 6.

Figure 4: Long-Run Effects on Rents



Note: The figure plots coefficients from Equation 1. Panel (a) reports log median Census rent; Panel (b) reports log median and average NYT rent per room. Blue and purple denote the public-housing tract; red and orange denote the first ring. The second ring and the decade before completion are the reference categories. Estimates use tract-frequency weights and stack-clustered standard errors; bars show 90% and 95% confidence intervals. The sample is restricted to public-housing tracts first treated before 1970.

Together with the population results, the rent decline distinguishes the long-run response from the direct placement of tenants alone. Public housing changed the private residential

stock and was followed by a persistent reduction in the market valuation of the tract where it was built, with more limited changes in adjacent tracts in the Census-based outcomes, although the sparser listing series also declines in the first ring.[22]

5.4 Robustness of the Research Design

I assess whether the baseline findings are driven by repeated control-tract use or by the choice of estimator and comparison group. To this end, I use synthetic difference-in-differences (Arkhangelsky et al., 2021) and the estimator of De Chaisemartin and D'Haultfœuille (2020), comparing public-housing tracts first treated before 1970 with never-treated tracts and excluding adjacent tracts. Because each control tract enters these designs only once, the exercise also addresses dependence from repeated use across stacks. Both estimators produce small pre-treatment coefficients and the same qualitative post-treatment dynamics: declining private population and housing stock, losses among privately housed White and Black residents, little change among privately housed Hispanic residents, and lower long-run Census rents (Appendix C6). Because the specifications use different comparison groups, the important question is whether they show similar patterns over time, not whether their coefficient estimates are identical. I separately assess inference under repeated tract use and spatial correlation. Two-way stack-and-tract clustering and Conley spatial-HAC errors at 1, 2, and 5 kilometre cutoffs yield similar confidence intervals and leave the substantive conclusions unchanged (Appendix C7). These robustness checks retain the baseline sample of 166 tracts first treated before 1970. In the next section I compare this early cohort with the 66 tracts first treated from 1970 onward to explain the sample restriction and show how public-housing construction changed across the two periods.

5.5 Construction Regimes

The estimates above use the 166 public-housing tracts first treated before 1970. Appendix C2 compares them with the 66 tracts first treated in 1970 or later in a single pooled specification that tests the cohort difference formally and asks whether it is explained by observed development characteristics (Table C2.1). The earlier-treated tracts show the replacement pattern documented above—large additions of public-housing residents and units alongside contractions of private population and housing—while the implied effects in later-treated tracts on the population and housing margins are small and mostly indis-

[22] The persistence of the private-stock decline should also be read in light of New York City's 1961 Zoning Resolution. Because subsequent rebuilding occurred under different land-use and density rules from those governing the displaced prewar stock, zoning may have limited or redirected the recovery of private housing. This supply-side path dependence cannot by itself explain the simultaneous decline in private asking rents: a binding constraint on rebuilding would tend to raise rents, so the joint fall in private quantities and rents continues to point to lower neighborhood demand.

tinguishable from zero. Conditioning on initial tenant composition and maximum building height absorbs much of the demographic contrast, indicating that later sites differed in part because the developments built there were different; the rent contrast is the exception, with census rents in later-treated tracts declining regardless. The clearest differences therefore concern development scale, physical replacement, and demographic accounting.

This split should not be interpreted as the causal effect of crossing a 1970 cutoff. The cohorts differed in scale, clearance practices, physical form, tenant composition, location, and administrative context. I interpret them as distinct historical interventions rather than as the same treatment delivered on opposite sides of a date threshold. This concentration of the private-market response in the mid-century regime motivates restricting the paper's remaining estimates to tracts first treated before 1970.

Within that sample, Appendix C3 shows that Census-rent declines are largest in the lowest-income tracts, while first-ring outcomes display no systematic income gradient. This pattern differs from evidence that amenity changes or subsidized housing are capitalized differently across neighborhood income levels (Gaigné et al., 2022; Diamond and McQuade, 2019). Appendix C4 likewise finds larger effects where the pre-existing housing stock was less dense. Both results point to the size of the project relative to the existing neighborhood stock, a treatment-intensity margin examined directly next.

5.6 Treatment Intensity

I next examine whether public-housing effects vary with local treatment scale, measured by the number of public-housing apartments in the treated tract. A site with more units adds more apartments and residents and therefore represents a larger local intervention. This margin differs from both baseline neighborhood density and the design attributes considered below. Because treatment scale was not randomly assigned, the estimates describe heterogeneity by treatment intensity, not the causal effect of adding one public-housing unit.

Let U_p denote the number of public-housing apartments aggregated within treatment site p. I rank treatment sites by U_p and let $B(U_p) \in \{1, 2, 3\}$ denote the resulting tertile: sites with the fewest units, a middle group, and sites with the most units. Let $\mathcal{H} = \{\text{pre}, \text{med}, \text{long}\}$ denote the three pooled event-time horizons. I estimate a stacked event-study specification that interacts ring, horizon, and public-housing-unit tertile:

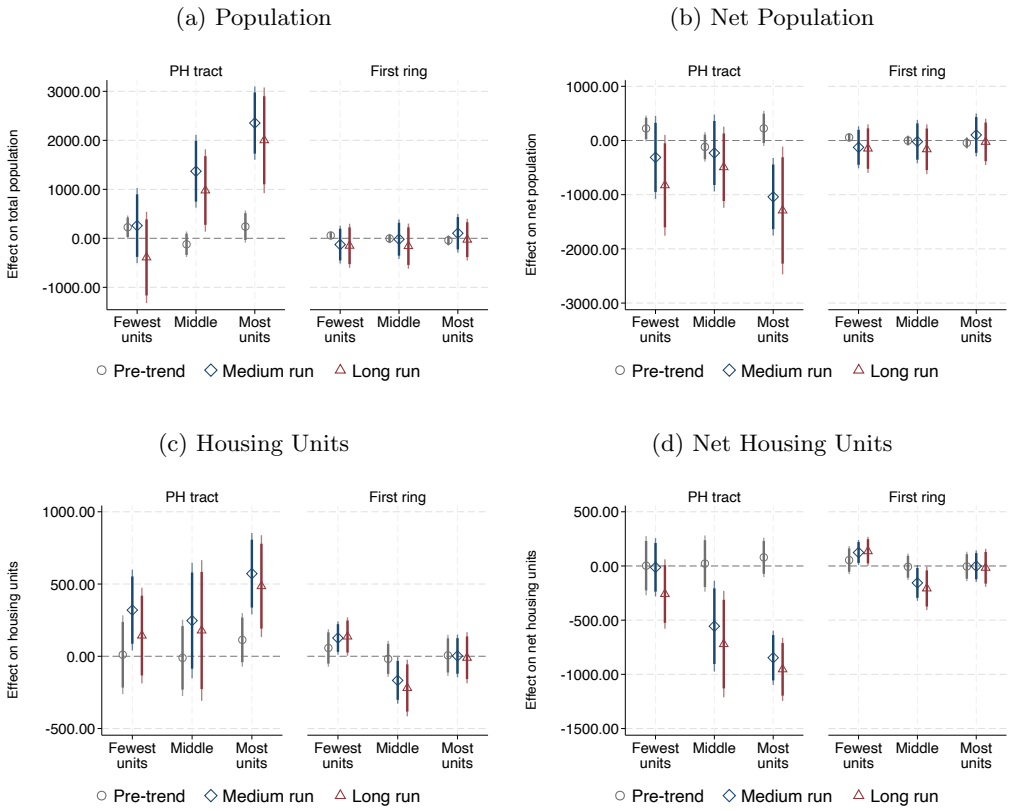
$$y_{kmpt} = \sum_{b=1}^{3} \sum_{r \in \mathcal{R}} \sum_{h \in \mathcal{H}} \beta_{r,h}^{(b)} \mathbb{1}\{B(U_p) = b\} \mathbb{1}\{R(k,p) = r\} \mathbb{1}\{H(t - Y_p) = h\} + \boldsymbol{\delta}_p' \mathbf{Z}_{kt} + \lambda_p y_{kp}^0 + \rho_{pmt} + u_{kmpt}. \tag{2}$$

Event time $\tau = t - Y_p$ is pooled into the pre-period ($\tau \leq -20$), medium run (0–30), and long run ($\tau \geq 40$). The reference category is the second ring at $\tau = -10$, so $\beta_{r,h}^{(b)}$ measures the response for unit tertile b in ring r and horizon h. Controls, baseline adjustment, and stack-by-neighborhood-by-year fixed effects follow the main specification. Standard errors are clustered by stack, observations are weighted by tract frequency within ring and unit tertile, and the sample, as throughout the paper, consists of public-housing tracts first treated before 1970.

Figure 5 compares total and private outcomes, separating the public-housing component from changes in the remaining housing market. The total-population response becomes increasingly positive across the three tertiles, as expected when larger developments house more tenants. During the first 30 years, projects in the highest unit-count tertile raise total population by about 2,350 residents and total housing by roughly 570 units, while private population falls by about 1,040 residents and private housing by approximately 850 units. All four estimates are statistically significant. The gap between the total- and private-housing coefficients implies a public-housing component of roughly 1,420 units. On this basis, the estimated reduction in private stock amounts to about 60% of that public-housing component, although this coefficient ratio is descriptive rather than a separately identified causal parameter. This decline in non-public-housing stock is consistent with clearance or replacement, although the design cannot distinguish direct removal from foregone or later construction. The middle tertile loses about 560 private units, whereas the estimate for the smallest projects is close to zero and imprecise. The pattern persists in the long run: the largest projects are associated with about 2,000 additional total residents and 485 additional total units, but with approximately 1,290 fewer private residents and 950 fewer private units. Thus, the estimated loss of non-public-housing stock is concentrated among medium and large developments. Population effects in the first ring remain close to zero, and first-ring private-housing estimates do not become more negative with project size.

Figure 6 shows how tenant placement masks the response of the privately housed population. For all three racial groups, the total-population coefficients become more positive with project size, consistent with the placement of more tenants at larger developments. During the first 30 years, the largest projects add about 1,300 Black, 600 Hispanic, and 810 White residents in total, while the corresponding private counts fall by roughly 760, 320, and 530 residents. Gross gains at these sites therefore reflect tenant placement rather than private inflows. The private White point estimates are negative in every tertile in both horizons. During the first 30 years, they are similar across the three tertiles—about −516, −454, and −531 residents—although only the largest-site estimate is significant at 10%. The corresponding private Black estimates are about −247, −323, and −761. In the long run, private Black population falls by 583, 536, and 772 residents across the three tertiles,

Figure 5: Intensity: Population and Housing Stock



Note: The figure plots the horizon-by-ring coefficients from Equation 2, separately by treatment-site public-housing-unit tertile. Panels (a) and (c) report effects on total population and housing units. Panels (b) and (d) subtract public-housing residents and units, respectively, and therefore report effects on the private population and housing stock. Estimates are reported for the public-housing tract and adjacent first ring relative to the second ring in the census period immediately preceding completion. Markers distinguish the pooled pre-period ($\tau \leq -20$), medium run (0–30 years), and long run (40 to 60 years). Standard errors are clustered by stack; vertical bars show 90% and 95% confidence intervals. Estimates are weighted by tract frequency within ring and public-housing-unit tertile. The sample is restricted to public-housing tracts first treated before 1970.

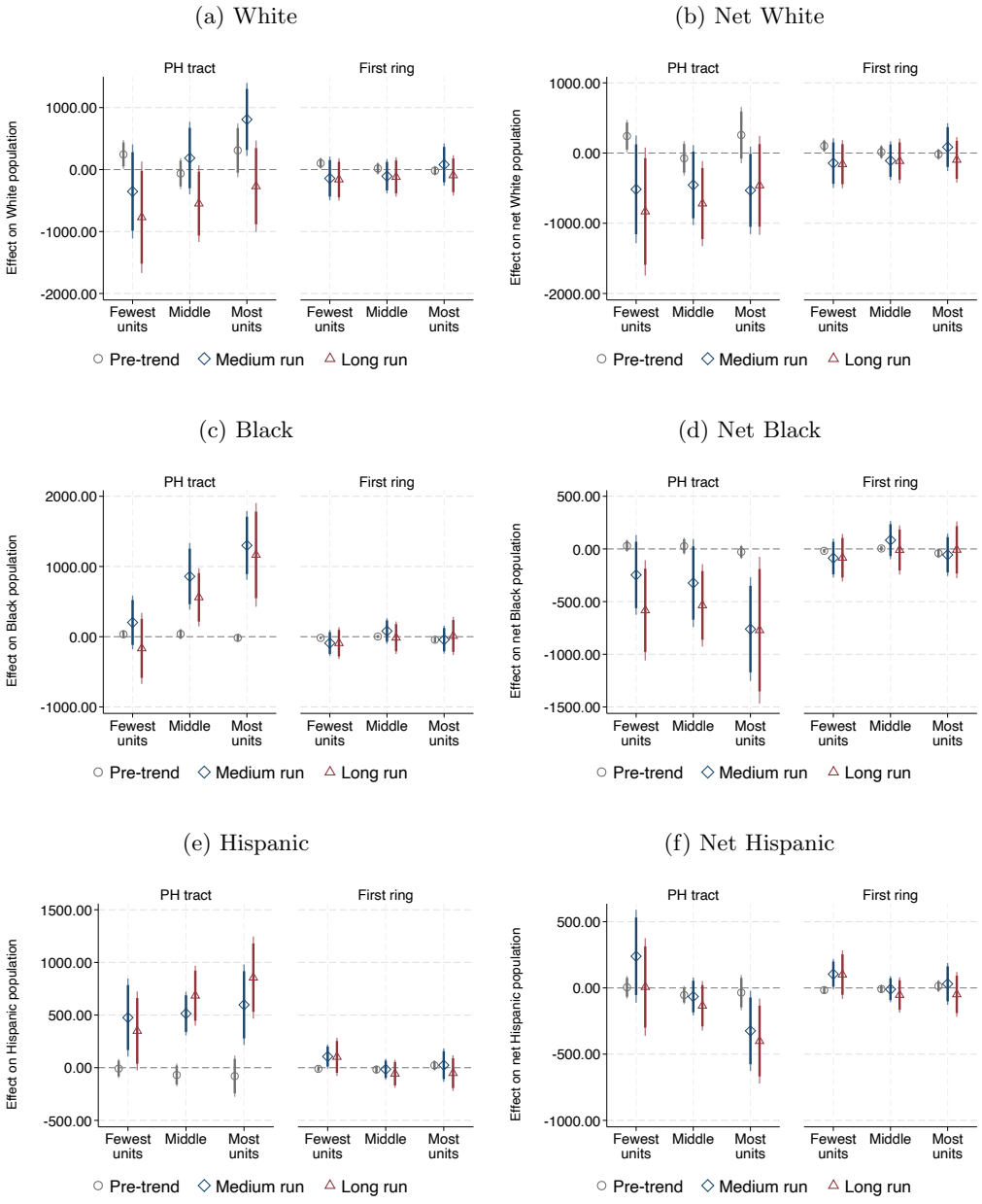
with all three estimates significant at 5%.[23] Private White estimates remain negative at −832, −719, and −460, and are significant for the smallest and middle groups only. These population gradients do not track the estimated private-stock loss one-for-one. In particular, the smallest projects have negative private White and Black point estimates despite no detectable medium-run decline in private housing, while long-run private Black losses are similar in the smallest and middle tertiles even though the housing-stock loss is substantially larger in the middle tertile. This divergence is consistent with residential sorting in addition to the housing-capacity channel, but aggregate counts cannot distinguish residents displaced by redevelopment from subsequent sorting. The estimates for privately housed White residents in the smallest-site tertile also warrant caution: the pooled pre-treatment coefficient is positive (about 245 residents, $p = 0.039$).

Figure 7 provides complementary evidence on private-market adjustment, but it does not support a smooth behavioral dose response. Census rents show little differentiation by project size during the first 30 years. In the long run, the public-housing-tract coefficients decline monotonically from −0.15 for the smallest projects to −0.28 for the middle tertile and −0.68 for the largest projects; all three are statistically significant, while first-ring estimates remain near zero. Because Census rents include subsidized public-housing units, however, this gradient may partly reflect the mechanical change in the composition of the housing stock and does not isolate private demand.

The New York Times series measures rents in the private market. Its project-tract coefficients are not monotonic: during the first 30 years they are −0.02, −0.28, and −0.04 across the three tertiles, and in the long run they are −0.22, −0.83, and −0.13. The long-run decline for the middle tertile is precisely estimated, whereas the project-tract estimate for the largest sites rests on very limited private-listing coverage. The adjacent first ring provides the most suggestive evidence of a behavioral response. Around the largest sites, the long-run New York Times coefficient is −0.98 ($p = 0.002$), while the private-housing response in that ring is essentially zero (−17 units, $p = 0.850$). Although Census rents do not show a corresponding first-ring decline, the New York Times pattern—private rents falling without a contraction in the adjacent private stock—is more consistent with a negative demand spillover than with the mechanical addition of public units. Taken together, treatment intensity is most clearly reflected in the housing stock: larger sites add more public housing, while medium and large sites are associated with greater losses of non-public-housing units. The racial-population and first-ring private-rent estimates also provide evidence consistent with sorting and demand responses beyond this accounting

[23] These long-run estimates pool the 40–60-year window and are therefore more precise than the decade-specific coefficient at $\tau = 60$ in Figure 3, which is smaller and imprecisely estimated for the private Black population. The two are consistent: the pooled window draws mainly on the better-supported 40- and 50-year horizons.

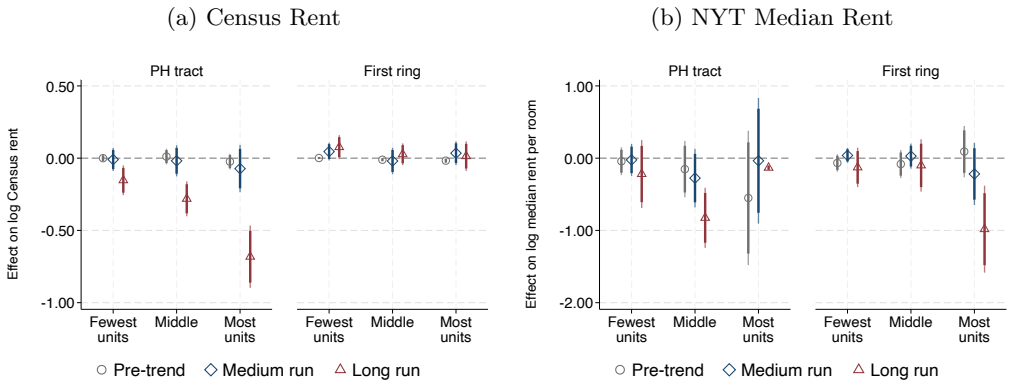
Figure 6: Intensity: Racial Composition



Note: The figure plots the horizon-by-ring coefficients from Equation 2, separately by treatment-site public-housing-unit tertile. Panels (a), (c), and (e) report effects on the total White, Black, and Hispanic populations. Panels (b), (d), and (f) subtract public-housing residents of the corresponding group and therefore report effects on the privately housed population. Estimates are reported for the public-housing tract and adjacent first ring relative to the second ring in the census period immediately preceding completion. Markers distinguish the pooled pre-period ($\tau \leq -20$), medium run (0–30 years), and long run (40 to 60 years). Standard errors are clustered by stack; vertical bars show 90% and 95% confidence intervals. Estimates are weighted by tract frequency within ring and public-housing-unit tertile. The sample is restricted to public-housing tracts first treated before 1970.

channel, although the behavioral component does not increase smoothly with project size and is not separately identified.

Figure 7: Intensity: Rents



Note: The figure plots the horizon-by-ring coefficients from Equation 2, separately by treatment-site public-housing-unit tertile. Panel (a) reports effects on the log of median contract rent from the U.S. Census, and Panel (b) reports effects on the log of median rent per room from New York Times rental listings. Estimates are reported for the public-housing tract and adjacent first ring relative to the second ring in the census period immediately preceding completion. Markers distinguish the pooled pre-period ($\tau \leq -20$), medium run (0–30 years), and long run (40 to 60 years). Standard errors are clustered by stack; vertical bars show 90% and 95% confidence intervals. Estimates are weighted by tract frequency within ring and public-housing-unit tertile. The sample is restricted to public-housing tracts first treated before 1970. New York Times coverage is sparse within some tertile–ring–horizon cells: the long-run public-housing-tract estimate for the largest-site tertile is based on five tract–year observations from three treatment sites, whereas the corresponding first-ring estimate is based on 39 tract–year observations from 18 treatment sites. Sites in the largest tertile occupy about 43% of the treated tract at the median (49% on average), consistent with limited private-market coverage within the public-housing tract.

6 Project Design and Willingness to Pay

The heterogeneity results show that public housing was not a uniform neighborhood intervention. The largest private-market changes occurred under the pre-1970 construction regime and where projects represented a large shock to the existing housing stock. This motivates a more specific question: conditional on a project being built, did its construction investment, physical form, and initial tenant composition help determine the response of the surrounding private market? I answer this question in two steps. I first estimate how private population and rents vary with project attributes. I then combine these responses into a rent-equivalent valuation—the rent change that would leave residents equally well off after a change in the attribute.

The valuation exercise concerns residents in the private housing market, not households administratively assigned to public housing. Accordingly, the population outcomes subtract

public-housing tenants, and the rent outcome is the advertised private-market rent from New York Times listings; Census contract rents are not used here because they include subsidized public-housing units. For group g, attribute a, ring r, and horizon h, let $\widehat{\gamma}^{g,pop}_{a,r,h}$ denote the attribute coefficient for log private population and $\widehat{\gamma}^{rent}_{a,r,h}$ the corresponding coefficient for log rent. The location-choice model implies the key decomposition

$$\widehat{\beta}^{g}_{a,r,h} = \widehat{\gamma}^{g,pop}_{a,r,h} + \alpha\widehat{\gamma}^{rent}_{a,r,h}, \qquad \widehat{MWTP}^{g}_{a,r,h} = \frac{\widehat{\beta}^{g}_{a,r,h}}{\alpha} = \frac{\widehat{\gamma}^{g,pop}_{a,r,h}}{\alpha} + \widehat{\gamma}^{rent}_{a,r,h}. \tag{3}$$

Here $\widehat{\beta}^{g}_{a,r,h}$ is the recovered preference index and $\widehat{MWTP}^{g}_{a,r,h}$ is MWTP in log-rent units. The decomposition adds back the indirect population response caused by the change in rent: population changes alone combine the direct utility effect of an attribute with the sorting response to the equilibrium price change. I calibrate the housing-expenditure share to $\alpha = 0.30$.[24] Appendix A derives the mapping and states its maintained assumptions, while Appendix C8 reports sensitivity to alternative housing-expenditure shares.

6.1 Attribute Heterogeneity

I begin with attribute heterogeneity. Conditional on public housing being built in a tract, I examine whether private-market responses differed with construction investment, building form, and the tenant composition assigned at opening. This exercise separates treatment scale from design. The number of apartments is treated as treatment intensity and studied in Section 5.6; the attribute analysis below focuses on characteristics that describe how the developments in a treatment site were built and initially occupied.

I estimate the attribute specification on public-housing tracts first treated before 1970, the period in which the reduced-form evidence shows the clearest private-market response. Later-treated sites produced more muted effects and were more strongly shaped by administrative placement, so they are less informative about market valuation of development design. The outcomes are the log privately housed White, Black, and Hispanic populations and the log median rent per room from New York Times rental listings. The population outcomes subtract public-housing tenants from total tract population, so the estimates capture sorting among residents outside the public housing system rather than the mechanical placement of tenants into public housing.

Let A_{ap} denote attribute a of treatment site p, constructed from the development records assigned to its public-housing tract. The vector $\mathcal{A}$ contains six variables: construction cost

[24] Published benchmarks support this value: the median gross rent-to-income ratio of New York City renter households moves between 20 and 34 percent over 1965–2023 in the New York City Housing and Vacancy Survey (Gaumer, 2024), and tenement-era budget studies place working-class rent shares near 25 percent around 1900–1910 (Chapin, 1909). Figure C8.1 in Appendix C8 collects these series.

per rental room,[25] maximum building height,[26] ground coverage, and the White, Black, and Hispanic tenant shares at initial occupancy. Construction cost is expressed in \$5,000 units, height in stories, ground coverage in 10-percentage-point units, and each tenant share in 5-percentage-point units. I define $\widetilde{A}_{ap} = A_{ap} - \bar{A}_a$, where $\bar{A}_a$ is the unweighted mean of attribute a across one observation per treatment site in the pre-1970 estimation sample. For $r \in \mathcal{R} = \{T, 1\}$ and $h \in \mathcal{H} = \{\text{pre}, \text{med}, \text{long}\}$, let $D_{r,h,kpt} = \mathbb{1}\{R(k,p) = r\}\mathbb{1}\{H(t - Y_p) = h\}$. I estimate:

$$\begin{aligned} y_{kmpt} = & \sum_{r\in\mathcal{R}}\sum_{h\in\mathcal{H}} \theta_{r,h} D_{r,h,kpt} \\ & + \sum_{a\in\mathcal{A}}\sum_{r\in\mathcal{R}}\sum_{h\in\mathcal{H}} \gamma_{a,r,h}\widetilde{A}_{ap} D_{r,h,kpt} \\ & + \boldsymbol{\eta}_t'\mathbf{B}_{kp} + \boldsymbol{\delta}_p'\mathbf{Z}_{kt} + \lambda_p y^0_{kp} + \rho_{pmt} + u_{kmpt}. \end{aligned} \tag{4}$$

Event time $\tau = t - Y_p$ is pooled into the pre-period ($\tau \leq -20$), the medium run ($0 \leq \tau \leq 30$), and the long run ($\tau \geq 40$). The second ring and the decade immediately preceding completion are the spatial and temporal reference categories, respectively. Because the attributes are demeaned, $\theta_{r,h}$ gives the response for a treatment site with mean observed attributes, while $\gamma_{a,r,h}$ describes how that response varies with attribute a.

The vector $\mathbf{B}_{kp}$ contains baseline White, Black, and Hispanic population shares and log population density, each interacted with census-year fixed effects.[27] The remaining controls, baseline-outcome adjustment, and stack-by-neighborhood-by-year fixed effects follow Equation 1; for the New York Times listing-rent outcome, the baseline adjustment uses Census rent. Observations receive stack-year ring-frequency weights, and standard errors are clustered by stack.

Project attributes were not randomly assigned across otherwise comparable neighborhoods. I therefore interpret $\gamma_{a,r,h}$ as conditional heterogeneity across treatment sites, rather than as the causal effect of changing one attribute while holding other determinants fixed. The baseline-characteristic interactions allow outcomes to evolve differently with observed pre-construction composition and density, while the pooled pre-period coefficients provide a diagnostic for differential trends. Appendix B3 documents correlations among project

[25] Construction cost per rental room are converted to 2010 dollars. It covers the structure, installed systems, and appliances, but excludes land and separately reported site improvements. For sites with multiple records, I average within completion decades and then across decades. The measure captures reported construction investment rather than total development cost or housing quality.

[26] Height is the maximum reported number of stories among new-construction developments intersecting the treated tract, including later developments in the same tract. When a record lists several heights or a range, I use the largest. The measure therefore captures the tallest recorded building at the site, not average height.

[27] Zero-filled values and year-specific missing-value indicators retain observations with unavailable baselines.

attributes and their relationship with pre-construction neighborhood characteristics. These adjustments cannot rule out selection on unobserved conditions, and the MWTP estimates below inherit the same limitation.

Figure 8 shows the clearest patterns for construction investment and height. In the treated tract, a $5,000 increase in construction cost per rental room is associated with private-population responses that are roughly 5–15 log points larger, with the largest coefficients for White and Black residents. The listing-rent estimates are less precise, although the long-run coefficient is positive. Higher reported construction investment is therefore associated with a smaller contraction of the private market.

Height is associated with outcomes in the opposite direction. In the long run, a one-story increase in the site's maximum recorded height is associated with a 5–10 log-point lower private White and Black population response in the treated tract and with lower New York Times rents; the private Hispanic-population coefficient is close to zero. This pattern is consistent with historical concerns about high-rise, *Tower in the Park* developments, but it does not isolate a causal effect of height.[28] Ground-coverage coefficients are imprecise and show no consistent relationship with private population or rents once height and the other attributes are held fixed.

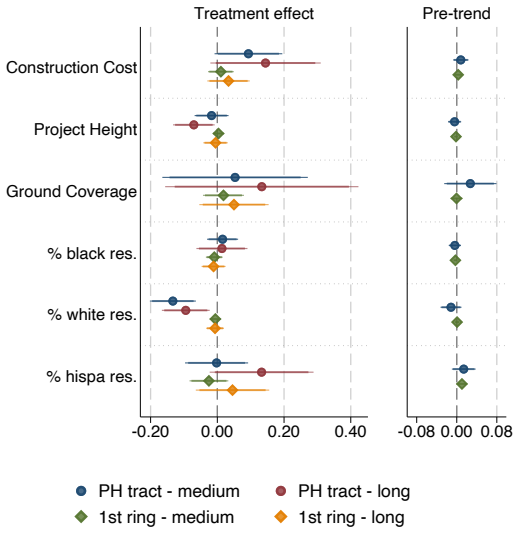
Initial tenant composition yields no comparably stable pattern. A higher White tenant share at opening is associated with lower private White and Black population responses in the treated tract, whereas the coefficients on initial Black and Hispanic shares are smaller and less consistent. Because opening composition reflected administrative assignment and differed systematically across neighborhoods, these coefficients should be read as cross-project associations; they do not support a simple racial-sorting interpretation.

The most consistent associations are therefore with construction investment and height. The next section expresses the corresponding population and rent responses on a common rent-equivalent scale.

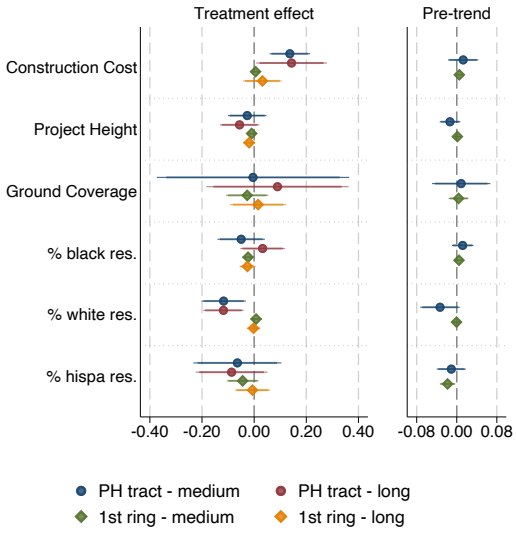
[28] Appendix C5 shows that the association with height differs by pre-treatment neighborhood income. In above-median-income neighborhoods, it appears primarily in Census rents; in below-median-income neighborhoods, it appears primarily through population sorting. This pattern is consistent with greater sensitivity to local amenities in higher-income neighborhoods (Gaigné et al., 2022).

Figure 8: Public Housing Attributes and Private-Market Responses

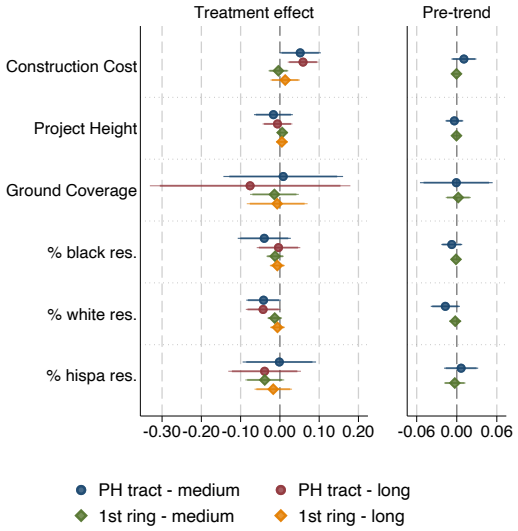
(a) Log private White population



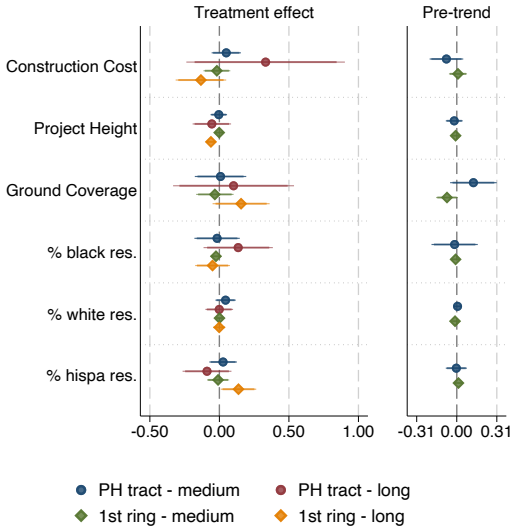
(b) Log private Black population



(c) Log private Hispanic population



(d) Log NYT rent per room



Note: The figure plots the attribute-interaction coefficients $\hat{\gamma}_{a,r,h}$ from Equation 4. Panels (a), (b), and (c) report log privately housed White, Black, and Hispanic population, respectively; Panel (d) reports log median rent per room from New York Times listings. The left side of each panel reports post-completion treatment effects for public-housing tracts and first-ring tracts in the medium run (0–30 years) and long run (40 to 60 years). The right side reports the corresponding attribute-specific coefficients from the pooled pre-period. The omitted period is the census decade immediately preceding first treatment, and the comparison group is the second ring. Attributes are measured at the treatment-site level and demeaned before estimation. Construction cost is CPI-deflated cost per rental room in 2010 dollars and is scaled by \$5,000; height is the maximum reported number of stories; ground coverage is scaled by 10 percentage points; and initial tenant shares are scaled by 5 percentage points. Initial tenant shares refer to the composition of public-housing residents at opening. Apartments are excluded from the attribute vector and analyzed separately as treatment intensity in Section 5.6. Standard errors are clustered by stack; horizontal bars show 90% and 95% confidence intervals. The sample is restricted to public-housing tracts first treated before 1970.

6.2 Marginal Willingness to Pay

The attribute estimates above show how projects with different characteristics generated different private-market responses. I now translate those coefficients into marginal willingness to pay (MWTP) using the decomposition in Equation 3. For each group, ring, horizon, and project attribute, the MWTP is the rent change that would offset the utility change implied by the attribute. Positive values therefore mean that private-market residents would be willing to pay higher rents to live in the neighborhood associated with that attribute. Negative values mean that they would require lower rents to remain indifferent.

I report the resulting log-rent MWTP in exact percentage and dollar equivalents:

$$\begin{aligned} \widehat{MWTP}^{g,\%}_{a,r,h} &= 100 \left[\exp\left(\widehat{MWTP}^{g}_{a,r,h} \right) - 1 \right], \\ \widehat{MWTP}^{g,\$}_{a,r,h} &= \bar{r}_{r,h} \left[\exp\left(\widehat{MWTP}^{g}_{a,r,h} \right) - 1 \right]. \end{aligned} \tag{5}$$

The population and listing-rent coefficients entering Equation 3 are estimated on their respective supports.[29] I use the housing-expenditure share $\alpha = 0.30$ introduced above and let $\bar{r}_{r,h}$ denote the average median rent per room in the relevant ring and horizon. The estimates should be interpreted conditional on this calibration; Appendix C8 reports results for values from 0.05 to 0.60.[30]

Three qualifications are important. First, the MWTP estimates inherit the identifying assumptions of the attribute specification. They should be read as valuations implied by conditional cross-project heterogeneity, not as the effects of independently changing a project's maximum recorded height, construction investment, or initial tenant composition. Second, the estimates describe the preferences of residents in the private housing market. They do not measure the value of public housing to tenants who received apartments, nor do they include the fiscal cost of construction. Third, New York's rent-control and rent-stabilization regimes mean that observed rents need not clear the market immediately or uniformly. Although the New York Times listings better approximate rents available to marginal movers than census contract rents, the data do not consistently identify regulatory status and cover the upper part of the market more intensively. The estimates therefore express the neighborhood effect as the change in rent that would leave private-market residents equally well off, under the model's assumption that rents adjust until housing demand equals supply. They are not complete welfare measures for tenants in either regulated or unregulated housing.

[29] The Census-based population regressions cover all 166 treatment sites, whereas the listing-rent regressions cover the 155 sites with geocoded listings and far fewer tract–year cells. Census rents cannot be used here because they include subsidised public-housing units, so this support difference is a direct consequence of measuring private-market prices with listings; the MWTP estimates should be read with it in mind.

[30] The event study does not identify α, and a constant share is unlikely to describe every demographic group and decade from 1930 to 2010. Standard errors are computed with the delta method using the sampling variance of the population and rent coefficients and treating α as fixed.

Table 1: Marginal willingness to pay for public-housing attributes, by race

	White				Black				Hispanic			
	0–30 yrs		40–60 yrs		0–30 yrs		40–60 yrs		0–30 yrs		40–60 yrs	
Attribute	%	$	%	$	%	$	%	$	%	$	%	$
PH Tract												
Public housing (mean attr.)	−97.68***	−453.07	−99.42***	−589.39	−98.19***	−455.41	−98.67***	−584.95	−31.32	−145.26	−44.27	−262.46
	(1.299)		(0.453)		(1.346)		(1.025)		(33.706)		(32.974)	
Construction cost	43.34**	201.03	125.72**	745.31	65.79***	305.14	124.94**	740.67	24.84**	115.19	69.72*	413.32
	(25.728)		(90.367)		(23.425)		(82.683)		(12.895)		(49.966)	
Max. stories	−5.97	−27.71	−25.08**	−148.69	−8.86	−41.09	−21.40*	−126.86	−5.75	−26.65	−7.19	−42.62
	(8.549)		(9.460)		(11.701)		(11.454)		(8.362)		(8.779)	
% Black res.	3.66	16.99	19.88	117.84	−16.65	−77.23	27.53	163.18	−13.86	−64.28	13.14	77.91
	(11.783)		(21.588)		(14.484)		(24.361)		(12.112)		(17.710)	
% Hisp. res.	1.95	9.06	42.39	251.28	−17.11	−79.35	−31.41	−186.20	2.23	10.35	−19.65	−116.49
	(17.053)		(39.482)		(23.970)		(17.080)		(16.922)		(14.417)	
% White res.	−32.95***	−152.81	−27.12**	−160.78	−29.24**	−135.64	−32.56***	−193.01	−8.98	−41.65	−13.36	−79.21
	(8.183)		(9.544)		(10.507)		(9.434)		(7.532)		(7.775)	
% Ground coverage	20.39	94.57	72.69	430.90	−0.62	−2.86	49.29	292.23	3.72	17.25	−14.00	−82.98
	(45.699)		(92.506)		(62.586)		(76.149)		(28.366)		(41.556)	
1st Ring												
Public housing (mean attr.)	0.31	1.78	−29.48	−272.44	127.86***	729.41	37.27	344.35	53.92**	307.62	11.69	108.02
	(19.344)		(22.852)		(52.933)		(45.537)		(30.998)		(30.320)	
Construction cost	1.78	10.17	−2.06	−19.03	0.03	0.14	−2.68	−24.78	−2.99	−17.08	−8.39	−77.54
	(8.271)		(13.809)		(5.640)		(15.080)		(6.260)		(10.203)	
Max. stories	1.06	6.05	−7.56	−69.87	−3.27	−18.63	−11.73***	−108.42	1.88	10.73	−4.28	−39.52
	(3.488)		(5.979)		(3.691)		(3.882)		(2.966)		(3.052)	
% Black res.	−5.35	−30.53	−8.31	−76.74	−9.74**	−55.55	−12.61*	−116.56	−6.32	−36.03	−6.86	−63.36
	(4.444)		(8.016)		(3.853)		(7.113)		(3.953)		(6.589)	
% Hisp. res.	−8.92	−50.87	33.51	309.67	−14.48	−82.61	12.32	113.81	−12.89	−73.52	8.45	78.07
	(9.666)		(26.246)		(9.494)		(14.709)		(8.081)		(11.183)	
% White res.	−1.87	−10.65	−2.35	−21.68	2.58	14.70	−0.89	−8.26	−4.23	−24.11	−2.13	−19.70
	(3.156)		(4.919)		(4.064)		(4.933)		(3.240)		(3.807)	
% Ground coverage	2.93	16.72	38.07	351.80	−11.41	−65.07	22.97	212.21	−7.60	−43.35	14.26	131.74
	(12.752)		(28.098)		(13.599)		(25.667)		(11.536)		(19.042)	

Notes: Each cell reports the marginal willingness to pay (MWTP) for the indicated public-housing attribute, in percent of rent (%) and in dollars ($), evaluated at a housing-expenditure share of $\alpha = 0.3$. MWTP ($) is computed at the average median rent per room in the relevant ring and horizon. Both columns are the exact compensating rent change, $100[\exp(m) - 1]$ percent and $\bar{r}[\exp(m) - 1]$ dollars, where m is the log-rent MWTP; this is preferred over the small-effect linear approximation ($100m$, $\bar{r}m$), which is poor for the discrete "Public housing (mean attr.)" row. "Public housing (mean attr.)" is the value of a project built with mean attributes, and "Max. stories" denotes maximum recorded building height. Apartments are excluded from the attribute vector and analyzed separately as treatment intensity. Delta-method standard errors in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

Table 1 shows that the average pre-1970 project was valued negatively by private-market residents in the tract where it was built. In the first 30 years, MWTP equals −3.77 log-rent units for White residents and −4.01 for Black residents; because the discrete mean-project effect is large, the exact compensating rent changes are −97.7 and −98.2 percent of rent. After 40–60 years, the corresponding estimates are −5.15 and −4.32 log-rent units (−99.4 and −98.7 percent). Hispanic estimates are smaller and imprecise. First-ring effects are much weaker: the mean project has little effect on White MWTP outside the treated tract, while the Black and Hispanic estimates turn positive in the medium run and are statistically indistinguishable from zero in the long run. The transformation therefore reinforces the reduced-form pattern that the main private-market response was concentrated in the tract physically changed by public housing.

Construction investment is the clearest positive attribute. In the public-housing tract, a $5,000 increase in construction cost per rental room[31] is associated with a long-run MWTP of about 126 percent of rent for White and Black residents and 70 percent for Hispanic residents, or roughly $745, $741, and $413 at the relevant average rents. Because the $5,000 increment is roughly 1.5 times the interquartile range, these magnitudes represent a relatively large cross-site comparison under the linear attribute specification. For a $1,000 increase, the exact equivalents are 17.7, 17.6, and 11.2 percent. These implied gains do not offset the negative valuation of the average project, but the estimated discount is smaller at sites with greater construction investment. In the first ring, construction cost has little systematic relationship with MWTP.

Height moves in the opposite direction. In the public-housing tract, a one-story increase in maximum recorded height is associated with long-run MWTP declines of about 25 percent of rent for White residents and 21 percent for Black residents, with dollar equivalents of about $149 and $127; Hispanic estimates are smaller and imprecise. First-ring effects are mostly near zero, apart from a negative long-run Black response. Ground coverage does not show a comparable pattern: conditional on height, construction investment, and tenant composition, estimates are imprecise and change sign. The historical critique of high-rise public housing therefore appears tied more to vertical scale than to built-up site coverage.

The MWTP estimates for initial tenant composition are less consistent. A higher initial White tenant share is associated with lower public-housing-tract MWTP for all groups, especially White and Black residents. A higher initial Black tenant share is more negative in the first ring than in the treated tract, while the initial Hispanic share has no consistent pattern. Because opening composition reflected administrative assignment and historical conditions, and because the shares are mechanically related, the coefficients capture differ-

[31] Across the 166 treatment sites first treated before 1970, the 25th and 75th percentiles of construction cost per rental room were $13,581 and $16,960 in 2010 dollars, yielding an interquartile range of $3,379.

ences across projects with different tenant mixes rather than the effect of changing any one group's share.

Private-market valuations differed across public-housing projects. The average pre-1970 project was valued negatively in the tract where it was built, but that discount was smaller for projects with greater construction investment and larger for projects with greater maximum recorded height. The relevant policy margin was therefore not only whether to build public housing, but also how projects were built and where they were placed.

Finally, in a back-of-envelope calculation, I assess whether the construction-investment margin paid for itself. Holding project size fixed and setting aside benefits to public-housing tenants—which higher construction investment would, if anything, increase—raising investment by $1,000 per room in the median pre-1970 project would have cost $3.0 million. In the treated tract, the calculation implies avoided capitalized private-rent losses of $1.1–3.4 million (at discount rates of 7 to 3 percent), between one third and roughly the full marginal construction dollar. The first-ring point estimate, however, has the opposite sign and is larger in dollar terms because the ring's rent base is about three times larger, so the combined externality is a small negative number that is estimated far too imprecisely to sign: the confidence bands on the combined ratio span values well below zero and well above one. The calculation also excludes tenant-side benefits of higher-quality buildings, which would raise the return. Appendix C9 details the calculation and its caveats. The exercise therefore cannot establish whether the historical investment level was too low or too high; it shows only that the sums at stake are of the same order as the construction outlay, making design quality a quantitatively serious margin of the program.

7 Conclusion

This paper provides new evidence on the long-run effects of public housing construction on neighborhood composition and housing markets in New York City. Leveraging archival data and distinguishing between public housing tenants and privately housed residents, I document persistent demographic and price changes unfolding over decades. Public housing replaced much of the private housing stock where it was built; privately housed White and Black residents left treated tracts for decades afterward; and long-run asking rents fell by roughly 50 percent, with Census contract rents down about 40 percent. The joint decline of private population and rents indicates a persistent reduction in private demand for these neighborhoods rather than the removal of private supply alone.

These responses are concentrated among the 166 tracts that received New York City's early public-housing developments. The results should therefore be understood as evidence on this early era of public-housing construction, when projects were generally larger and

built under a distinct planning and administrative regime, rather than as effects of public housing in every period. Later developments produced more muted neighborhood responses.

To move beyond average effects, I embed the reduced-form estimates in a discrete location choice model and recover marginal willingness to pay (MWTP) for key public housing attributes. Within the public housing tract, the MWTP estimates are expressed as monthly rent per room in 2010 dollars. A $1,000 increase in construction cost per rental room is associated with a long-run MWTP increase of about $105 for White and Black residents and $66 for Hispanic residents. A one-story increase in project height is instead associated with declines of about $149 for White residents and $127 for Black residents, consistent with historical concerns about high-rise *Tower in the Park* design. The corresponding Hispanic estimate is smaller and imprecise. Estimates for initial tenant composition are more selected and less systematic than the construction cost and height estimates.

These findings highlight a core policy trade-off. While large-scale developments expand the housing stock, they risk triggering neighborhood flight and rent declines when they represent large local interventions in the surrounding private market. In contrast, higher-quality construction appears to mitigate those discounts in the project tract but comes at higher public costs. Designing effective affordable housing policy thus requires not only more housing, but better housing, tailored to how physical design and resident composition shape long-run neighborhood dynamics.

The estimated private-market response likely combines several channels. The ring design nets out citywide suburbanization and the broad postwar decline in central-city White population, but public housing may nevertheless have accelerated departures among households already near the suburbanization margin (Boustan, 2010). The estimates also combine direct responses to project buildings and residents with indirect effects through schools, accessibility, and other neighborhood institutions. The location-choice model further abstracts from tenure, although replacing private ownership with public rental housing may have affected reinvestment and ownership opportunities. The intervention also unfolded alongside coordinated urban renewal, zoning constraints, and rent regulation. The recovered MWTP thus summarizes the private-market value of the historical intervention in equilibrium rather than a structural preference for public housing.

Future research should explore optimal public housing scale and placement by modeling resident preferences and planner trade-offs. Additional work linking building attributes to schools, crime, or renovation needs could help distinguish among the mechanisms behind the estimated private-market response.

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Online Appendix for "Public Housing Preferences: New York City, 1930–2010"

Maximilian Günnewig-Mönert

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A Model Details

This appendix develops the model used to interpret the reduced-form population and rent estimates. The purpose of the model is deliberately narrow. It maps changes in the private population and private-market rents into a rent-equivalent measure of how residents value exposure to public housing and its attributes. It is not a model of admission to public housing, the administrative assignment of public-housing tenants, or the value of a subsidized apartment to its recipient.

That distinction is important for the paper's empirical setting. Construction can change a tract through three conceptually different channels. First, slum clearance can remove private buildings and displace their occupants. Second, the completed project mechanically adds public-housing units and tenants. Third, the project can change the attractiveness of the surrounding private market through its presence, physical design, construction investment, and tenant composition. The data separate the second channel by subtracting public-housing tenants and units from the tract totals. The model below allows the first channel to operate through private housing supply and uses the joint response of private population and private-market rents to recover the third channel.

A1 Households, housing demand, and location choice

Let $g \in \{W, B, H\}$ index White, Black, and Hispanic households, k index census tracts, and t index census years. There are N_t^g private-market households of group g in the city. The empirical outcomes are resident counts rather than household counts; the model treats the two as proportional within group and year, so they have the same log changes.

Following Bayer et al. (2007), earnings do not vary across residential locations. Every household in group g and year t has the same citywide resources $\bar{y}_t^g$, regardless of tract choice. A household that lives in tract k consumes a numeraire good c and private housing services q at price r_{kt}, subject to

$$c + r_{kt} q = \bar{y}_t^g. \tag{A.1}$$

Under this assumption, no neighborhood-specific wage term appears in the model, which avoids introducing local labor demand, commuting, or workplace choice into a model designed only to describe residential sorting. Differences in resources across groups or years are permitted, but they are common to every tract in a group-year cell. Their contribution to utility is therefore a group-by-year fixed component, $\lambda_t^g \equiv \log \bar{y}_t^g$, rather than a neighborhood characteristic.

Preferences over the two goods are Cobb–Douglas. I normalize direct utility so that optimized utility has a particularly simple form:

$$U_{ikt}^g = \exp\left(Q_{kt}^g + \epsilon_{ikt}^g\right) \left(\frac{c}{1-\alpha}\right)^{1-\alpha} \left(\frac{q}{\alpha}\right)^{\alpha}, \qquad 0 < \alpha < 1. \tag{A.2}$$

Here α is the share of expenditure devoted to housing, Q_{kt}^g is the group-specific quality of tract k, and ϵ_{ikt}^g is an idiosyncratic location taste. The optimal demands implied by Equation A.1 and Equation A.2 are

$$c_{ikt}^{g*} = (1-\alpha)\bar{y}_t^g, \qquad q_{ikt}^{g*} = \frac{\alpha \bar{y}_t^g}{r_{kt}}. \tag{A.3}$$

Substitution into direct utility gives log indirect utility

$$v_{ikt}^g = Q_{kt}^g + \lambda_t^g - \alpha \log r_{kt} + \epsilon_{ikt}^g \equiv \delta_{kt}^g + \lambda_t^g + \epsilon_{ikt}^g, \tag{A.4}$$

where $\delta_{kt}^g \equiv Q_{kt}^g - \alpha \log r_{kt}$ is the component that varies across residential locations. Thus α has two related interpretations: it is the housing-expenditure share and the absolute elasticity of indirect utility with respect to rent. The common fixed component λ_t^g shifts utility for all locations equally and therefore does not affect relative tract choices.

Calibration and temporal variation in α. The event-study coefficients do not identify α. I therefore calibrate it to 0.30 in the baseline analysis. This is a round benchmark for the share of household resources devoted to gross housing expenses; it is not an estimate of a constant New York City housing share from 1930 through 2010. The maintained baseline imposes the same value across groups, rings, and horizons so that differences in MWTP reflect the estimated population and rent responses rather than different calibrations.

That restriction is substantive. Housing expenditure shares can change as rents, incomes, household size, tenure, and housing subsidies change, and these forces evolved considerably over the paper's eight-decade period. The appropriate share could therefore differ across demographic groups and between the medium and long run. The historical data do not provide a consistent group-by-tenure housing share for every census decade, so allowing α to vary freely would replace a transparent calibration with a series of incompletely measured inputs.

The consequences of the calibration are transparent. If the relevant share for horizon h were α_h rather than 0.30, the recovered preference and MWTP would be

$$b_{a,R,h}^g(\alpha_h) = \gamma_{a,R,h}^{g,pop} + \alpha_h \gamma_{a,R,h}^{rent}, \qquad m_{a,R,h}^g(\alpha_h) = \frac{\gamma_{a,R,h}^{g,pop}}{\alpha_h} + \gamma_{a,R,h}^{rent}. \tag{A.5}$$

A smaller housing share places more weight on the population response and magnifies its contribution to MWTP; a larger share moves MWTP toward the reduced-form rent effect. Temporal variation in α therefore changes magnitudes and can, when the population and rent coefficients have opposite signs, change signs. It does not alter the reduced-form estimates themselves. Appendix C8 recomputes the estimates for housing shares from 0.05 to 0.60. The exercise shows that the main signs and rankings are stable over this deliberately wide range, although the magnitudes—especially the dollar equivalents—remain conditional on the chosen share. For this reason, I interpret the baseline MWTP as a calibrated rent-equivalent preference index rather than a parameter identified independently of the housing-share assumption.

A2 Public housing in neighborhood quality

Consider a public-housing project p, completed in year Y_p. Let $R(k,p) \in \{T, 1, 2\}$ denote whether tract k contains the project (T), lies in the adjacent first ring (1), or belongs to

the second-ring comparison group (2). Let $H(t - Y_p)$ assign event time to the pooled pre-period, the medium run (0–30 years), or the long run (40 to 60 years). For $R \in \{T, 1\}$ and post-construction horizon h, define

$$D_{Rhpkt} = \mathbb{1}\{R(k,p) = R\} \times \mathbb{1}\{H(t - Y_p) = h\}. \tag{A.6}$$

The second ring is the unexposed reference category. Because the empirical design is stacked by project, the derivation considers one project comparison at a time; the influence of other projects is included in background neighborhood quality.

Let A_{ap} be attribute a of project p and $\widetilde{A}_{ap} = A_{ap} - \bar{A}_a$ its demeaned value in the estimation sample. I parameterize neighborhood quality as

$$Q^g_{kt} = \xi^g_k + \sum_{R \in \{T,1\}} \sum_h D_{Rhpkt} \left(b^g_{0,R,h} + \sum_{a \in \mathcal{A}} b^g_{a,R,h} \widetilde{A}_{ap} \right). \tag{A.7}$$

The term ξ^g_k is a group-by-tract amenity fixed effect. It captures persistent features of tract k that make it more or less attractive to group g, including fixed differences in access to employment. It is not a local wage: earnings remain $\bar{y}^g_t$ in every tract. Time-varying infrastructure and neighborhood shocks are handled by the controls and fixed effects in the stacked empirical specification.

Because the attributes are demeaned, $b^g_{0,R,h}$ is the utility effect of exposure to a project with mean attributes. The coefficient $b^g_{a,R,h}$ is the change in mean utility associated with a one-unit increase in attribute a, conditional on the other included attributes, for group g, ring R, and horizon h.

The empirical attribute set $\mathcal{A}$ contains construction cost per room, maximum recorded building height, ground coverage, and the White, Black, and Hispanic tenant shares at initial occupancy. The units follow the estimating equation: construction cost is scaled by \$5,000 per room, height by one story, ground coverage by 10 percentage points, and tenant shares by 5 percentage points. The number of public-housing apartments is excluded because the paper treats it as treatment intensity rather than a design attribute. Since the tenant shares are compositional and project attributes were not randomly assigned, each $b^g_{a,R,h}$ is a conditional preference index, not the effect of an independently randomized design feature. In particular, a change in one tenant share is meaningful only relative to an explicit offsetting change in the remaining shares.

Assume that ϵ^g_{ikt} is independently distributed type-I extreme value. Because λ^g_t is common to all tracts, it cancels from the location probabilities:

$$\pi^g_{kt} = \frac{\exp(\delta^g_{kt})}{\mathcal{I}^g_t}, \qquad \mathcal{I}^g_t = \sum_j \exp(\delta^g_{jt}). \tag{A.8}$$

The equilibrium private population of group g in tract k is

$$N^{g*}_{kt} = N^g_t \pi^g_{kt}, \tag{A.9}$$

or, in logs,

$$\log N^{g*}_{kt} = \log N^g_t + Q^g_{kt} - \alpha \log r^*_{kt} - \log \mathcal{I}^g_t. \tag{A.10}$$

There is consequently no neighborhood wage or income term in the population equation. Group-year resources affect conditional housing consumption and aggregate housing demand, but not the relative probability of choosing one tract rather than another.

The inclusive value need not be assumed constant. It is common to every tract for a given group and year, and the population regressions are run separately by group. Consequently, its change is absorbed by the project-by-year comparisons in the stacked design. The empirical mapping therefore uses differences between an exposed tract and the second-ring comparison tract, not a claim that a project literally has no effect on citywide expected utility.

A3 Private housing supply and market clearing

Public-housing construction frequently demolished private dwellings before adding subsidized units. To represent this part of the paper's story, let the supply of *private* housing services be isoelastic:

$$S_{kt} = \kappa_{kt} r_{kt}^{\phi_k}, \qquad \phi_k \geq 0. \tag{A.11}$$

The shifter κ_{kt} captures the amount and productivity of the private housing stock. Clearance, conversion, or the loss of developable private land can lower κ_{kt}. Public-housing apartments are not included in S_{kt} because their occupants and rents are outside the choice problem being modeled.

Using individual housing demand from Equation A.3, aggregate private housing demand is

$$\begin{aligned} D_{kt} &= \sum_g N_{kt}^{g*} q_{ikt}^{g*} \\ &= \alpha r_{kt}^{-(1+\alpha)} \sum_g \frac{N_t^g \bar{y}_t^g}{\mathcal{I}_t^g} \exp(Q_{kt}^g). \end{aligned} \tag{A.12}$$

Equating Equation A.11 and Equation A.12 gives the private-market rent:

$$\log r_{kt}^* = \frac{1}{\phi_k + 1 + \alpha} \left[\log \alpha + \log \left(\sum_g \frac{N_t^g \bar{y}_t^g}{\mathcal{I}_t^g} \exp(Q_{kt}^g) \right) - \log \kappa_{kt} \right]. \tag{A.13}$$

This expression makes the two private-market channels explicit. A deterioration in neighborhood quality lowers housing demand and rent, while a reduction in the private housing stock raises rent, holding demand fixed. To see the distinction, let z be a project exposure or attribute, let $s_{z,k} = \partial \log \kappa_{kt} / \partial z$, and let ω_{kt}^g denote group g's share of the demand aggregate inside parentheses in Equation A.13. Under the local-market approximation that a single project has a negligible effect on the citywide inclusive values,

$$\frac{\partial \log r_{kt}^*}{\partial z} = \frac{\sum_g \omega_{kt}^g b_z^g - s_{z,k}}{\phi_k + 1 + \alpha}. \tag{A.14}$$

The citywide resources $\bar{y}_t^g$ affect the demand weights ω_{kt}^g, but they do not vary across tracts and are not interpreted as a neighborhood mechanism. Hence a rent coefficient alone

does not identify preferences: the observed rent change combines group-weighted demand effects, private-stock changes, and the local supply elasticity. The population equation supplies the additional information needed for the preference mapping.

A4 From equilibrium responses to preference parameters

Differentiate Equation A.10 with respect to z:

$$\frac{\partial \log N_{kt}^{g*}}{\partial z} = b_z^g - \alpha \frac{\partial \log r_{kt}^*}{\partial z} - \frac{\partial \log \mathcal{I}_t^g}{\partial z}. \tag{A.15}$$

When the exposed tract is compared with the second-ring tract in the same project-year, the common inclusive-value term cancels. Let $\theta_{R,h}^{g,pop}$ and $\theta_{R,h}^{rent}$ denote the reduced-form effects of exposure to a mean-attribute project on log private group-g population and log private-market rent. The structural mean-project preference index is therefore

$$b_{0,R,h}^g = \theta_{R,h}^{g,pop} + \alpha \theta_{R,h}^{rent}. \tag{A.16}$$

Similarly, let $\gamma_{a,R,h}^{g,pop}$ and $\gamma_{a,R,h}^{rent}$ be the attribute-interaction coefficients from Equation 4. Then

$$b_{a,R,h}^g = \gamma_{a,R,h}^{g,pop} + \alpha \gamma_{a,R,h}^{rent}. \tag{A.17}$$

Equations A.16 and A.17 are the central estimating relationships. They do not require knowledge of ϕ_k, the group demand weights, or the private-stock response $s_{z,k}$. Those objects determine how the preference and stock shocks are divided between quantities and rents in equilibrium, but the joint population–rent response recovers the change in mean utility directly.

This result also clarifies how the model addresses the mechanical effects emphasized in the main text. Adding public-housing tenants affects total population but not the private-population outcome used here. Removing private housing changes κ_{kt}; in a frictionless equilibrium its population effect is mediated by the rent response and therefore nets out of Equation A.17. A simultaneous decline in private population and private-market rent cannot be generated by a negative private-supply shock alone: it requires a decline in private-market demand. The mapping is least secure during the demolition and construction transition, when households may be forcibly displaced before rents clear. Accordingly, the long-run estimates have the cleanest equilibrium interpretation; the 0–30-year estimates should be read as equilibrium-equivalent preference indices under the maintained market-clearing assumption.

A5 Marginal willingness to pay and units

Consider a marginal change in attribute a, holding the remaining project attributes fixed. From Equation A.4 and Equation A.7, the compensating change in log rent solves

$$b_{a,R,h}^g \, dA_a - \alpha \, d \log r = 0.$$

Marginal willingness to pay in log-rent units is therefore

$$m^g_{a,R,h} \equiv \left.\frac{d\log r}{dA_a}\right|_{dv=0} = -\frac{\partial v/\partial A_a}{\partial v/\partial \log r} = \frac{b^g_{a,R,h}}{\alpha} = \frac{\gamma^{g,pop}_{a,R,h}}{\alpha} + \gamma^{rent}_{a,R,h}. \tag{A.18}$$

The same expression with $b^g_{0,R,h}$ gives the log-rent equivalent of exposure to a mean-attribute project. Because that exposure is discrete rather than marginal, it is more accurately described as the compensating rent change for a mean project, not the MWTP for a marginal attribute.

Three units must be kept distinct. For a one-unit change in an attribute:

$$\text{log-rent MWTP} = m^g_{a,R,h}, \tag{A.19}$$

$$\text{exact percentage rent change} = 100\left[\exp\left(m^g_{a,R,h}\right) - 1\right], \tag{A.20}$$

$$\text{exact dollar rent change} = \bar{r}_{R,h}\left[\exp\left(m^g_{a,R,h}\right) - 1\right]. \tag{A.21}$$

For a sufficiently small $m^g_{a,R,h}$, these become $100m^g_{a,R,h}$ percent and $\bar{r}_{R,h}m^g_{a,R,h}$ dollars. Thus a value of $m = 0.40$ means 0.40 log-rent units, approximately 40 percent, not 0.40 percent. The linear dollar conversion $\bar{r}m$ is a first-order approximation and can be poor for the discrete mean-project effect or any other large estimate. For a change of ΔA_a units, replace m by $m\Delta A_a$ in Equations A.20 and A.21.

The baseline application evaluates dollar equivalents at the average median rent per room for the relevant ring and horizon. Equation A.5 and Appendix C8 describe how these valuations change with the calibrated housing share.

A6 Sampling uncertainty

Treat α as fixed and collect the estimated population and rent coefficients in

$$\widehat{\boldsymbol{\gamma}} = \left(\widehat{\gamma}^{g,pop}_{a,R,h}, \widehat{\gamma}^{rent}_{a,R,h}\right)'.$$

Equation A.18 is a linear transformation with gradient $\mathbf{d} = (1/\alpha, 1)'$. Its sampling variance is

$$\begin{aligned}\operatorname{Var}(\widehat{m}^g_{a,R,h}) &= \mathbf{d}'\operatorname{Var}(\widehat{\boldsymbol{\gamma}})\mathbf{d} \\ &= \frac{1}{\alpha^2}\operatorname{Var}(\widehat{\gamma}^{g,pop}_{a,R,h}) + \operatorname{Var}(\widehat{\gamma}^{rent}_{a,R,h}) \\ &\quad + \frac{2}{\alpha}\operatorname{Cov}\left(\widehat{\gamma}^{g,pop}_{a,R,h}, \widehat{\gamma}^{rent}_{a,R,h}\right).\end{aligned} \tag{A.22}$$

Equivalently, the variance of the preference parameter $\widehat{b} = \widehat{\gamma}^{g,pop} + \alpha\widehat{\gamma}^{rent}$ is

$$\begin{aligned}\operatorname{Var}(\widehat{b}^g_{a,R,h}) &= \operatorname{Var}(\widehat{\gamma}^{g,pop}_{a,R,h}) + \alpha^2\operatorname{Var}(\widehat{\gamma}^{rent}_{a,R,h}) \\ &\quad + 2\alpha\operatorname{Cov}\left(\widehat{\gamma}^{g,pop}_{a,R,h}, \widehat{\gamma}^{rent}_{a,R,h}\right).\end{aligned} \tag{A.23}$$

The reported MWTP variance assumes that the estimated population and rent coefficients have zero covariance. This is a maintained assumption, not a consequence of estimating the two outcomes separately: because both outcomes are observed around the same projects, shared project-level shocks could generate nonzero covariance. Equation A.22 makes this assumption explicit.

For the exact percentage transformation $P = 100[\exp(m) - 1]$, the delta method gives

$$\operatorname{Var}(\widehat{P}) \simeq [100 \exp(\widehat{m})]^2 \operatorname{Var}(\widehat{m}). \tag{A.24}$$

If $\bar{r}_{R,h}$ is treated as fixed, replace 100 with $\bar{r}_{R,h}$ for the variance of the exact dollar transformation. Under the linear dollar approximation, $\operatorname{Var}(\bar{r}\widehat{m}) = \bar{r}^2 \operatorname{Var}(\widehat{m})$. If the uncertainty in $\bar{r}$ is to be included, its variance and its covariance with $\widehat{m}$ must also enter the delta-method calculation.

For the log-rent statistic, the reported two-sided normal-approximation p-value is

$$p = 2\Phi\left(- \left| \frac{\widehat{m}^g_{a,R,h}}{\operatorname{SE}(\widehat{m}^g_{a,R,h})} \right| \right), \tag{A.25}$$

where Φ is the standard normal cumulative distribution function. The exact percentage and dollar transformations are monotone in m, so a test of zero based on m has the same p-value.

Because α is calibrated rather than estimated, the baseline standard errors condition on $\alpha = 0.30$ and do not include uncertainty about the housing share. The sensitivity analysis in Appendix C8 varies the calibration deterministically; it is not a confidence interval for α. If a time- or group-specific housing share were estimated, its sampling variance and covariance with the reduced-form coefficients would also need to enter the delta method.

A7 Interpretation and maintained assumptions

The recovered object is a private-market, equilibrium-equivalent valuation. Its interpretation requires the following conditions. First, after conditioning on the stacked design, exposed and comparison tracts must provide valid counterfactual changes in private population and rent. Second, the private-population measure must remove public-housing tenants, and the rent outcome must measure the private rental market. Third, local private housing and location choices must have adjusted enough for the static market-clearing condition to be informative. Fourth, project effects outside the modeled rings must either be negligible or common to the within-stack comparison. Fifth, earnings are common across residential locations within a group-year cell. Sixth, the baseline imposes a common housing-expenditure share of $\alpha = 0.30$ across groups and census years. This calibration requires that the housing share be approximately stable over the study period, an assumption that cannot be verified decade by decade. Appendix C8 provides supporting evidence: across the published New York household-level series assembled in Figure C8.1, rent-share estimates vary between roughly 20 and 34 percent, so the benchmarks bracket the baseline value throughout the study period without establishing exact constancy. The same appendix also relaxes the calibration by re-evaluating the MWTP estimates over a grid of α from 0.05 to 0.60: the signs and ranking of the headline valuations are stable across this range, while their magnitudes vary with α and should be interpreted conditional on the maintained share.

Taken together, these six conditions define the environment in which the recovered valuations carry their intended interpretation. Even when all of them hold, the attribute estimates inherit the selection described in Appendix B3. They measure valuations implied by conditional cross-project heterogeneity, not causal responses to randomly assigning height, construction investment, or tenant composition. Nor are they a complete welfare calculation. They exclude the benefits received by public-housing tenants, admission and displacement costs, fiscal construction and operating costs, and any citywide general-equilibrium effects. Within those limits, the model gives a transparent answer to the paper's narrower question: how much of the observed private-market population response remains after accounting for the accompanying change in the price of housing?

B Data

This appendix describes the data underlying the empirical analysis and provides additional descriptive statistics. Table B.1 lists all variables used in the analysis, together with their temporal coverage, geographic level, definitions, and sources. Appendix B1 presents long-run trends of the main outcome variables—White, Black, and Hispanic population and rents—in public housing neighborhoods relative to the rest of the city. Appendix B2 documents the construction of the public housing dataset and provides descriptive statistics on project characteristics, siting, and the racial composition of residents. Appendix B3 examines the dependence among project attributes and tests whether pre-construction neighborhood conditions predict the characteristics of subsequently built projects. Appendix B4 details the harmonization of historical census geographies to 2010 census tract boundaries, and Appendix B5 describes the collection of the rental listings data from the *New York Times*.

Table B.1: Definition of Variables and Data Sources

	Years	Level	Description	Source
Asking rents	1930–2010	Property	Asking rent and property characteristics.	New York Times Real Estate Section; `proquest.com`
Med. contract rent	1930–2010	Tract	Median contract rent (2010 $) at the census-tract level.	Manson et al. (2024)
White population	1930–2010	Tract	White population at the census-tract level.	Manson et al. (2024)
Black population	1930–2010	Tract	Black population at the census-tract level.	Manson et al. (2024)
Total housing units	1940–2010	Tract	Total number of housing units at the census-tract level.	Manson et al. (2024)
Private housing units	1940–2010	Tract	Total housing units less the public-housing apartment stock in that census year, bounded below at zero.	Author's calculations using Census and NYCHA records
Black public-housing residents	1930–2010	Project	Black residents at project opening date.	NYCHA Resident Data Book; LaGuardia & Wagner Archives
White public-housing residents	1930–2010	Project	White residents at project opening date.	NYCHA Resident Data Book; LaGuardia & Wagner Archives
Maximum building height	1930–2010	Tract	Maximum reported number of storeys among new-construction developments intersecting the tract.	NYCHA Development Data Book
Ground coverage	1930–2010	Project	Ground-floor area / project area (%).	NYCHA Development Data Book
Number of public-housing units	1930–2010	Project	Number of apartments in each public-housing project.	NYCHA Development Data Book
Arterial street	1930–2010	Segment	NYC arterial streets with opening dates.	NYC Open Data; `nycroads.com`
Urban-renewal programmes	1930–2010	Area	Areas designated for urban renewal and start date.	Digital Scholarship Lab (2018)
Subway stations	1930–2010	Station	NYC subway stations with opening dates.	NYC Open Data; Wikipedia

Note. Table B.1 lists all variables used in the analysis, along with their temporal coverage, geographic level, definitions, and data sources. Because the Census and NYCHA measures come from separate sources and are harmonized independently, the public-housing apartment count exceeds total Census housing units in 176 tract-year observations across 60 public-housing tracts; the implied private-housing count is set to zero in these cases. Information on public housing residents at the time of project opening is drawn from the LaGuardia & Wagner Archives for projects constructed up to 1970, and from the NYCHA Resident Data Book for projects completed afterward. Subway and arterial road shapefiles are obtained from NYC Open Data, while opening dates are sourced from Wikipedia and `nycroads.com`.

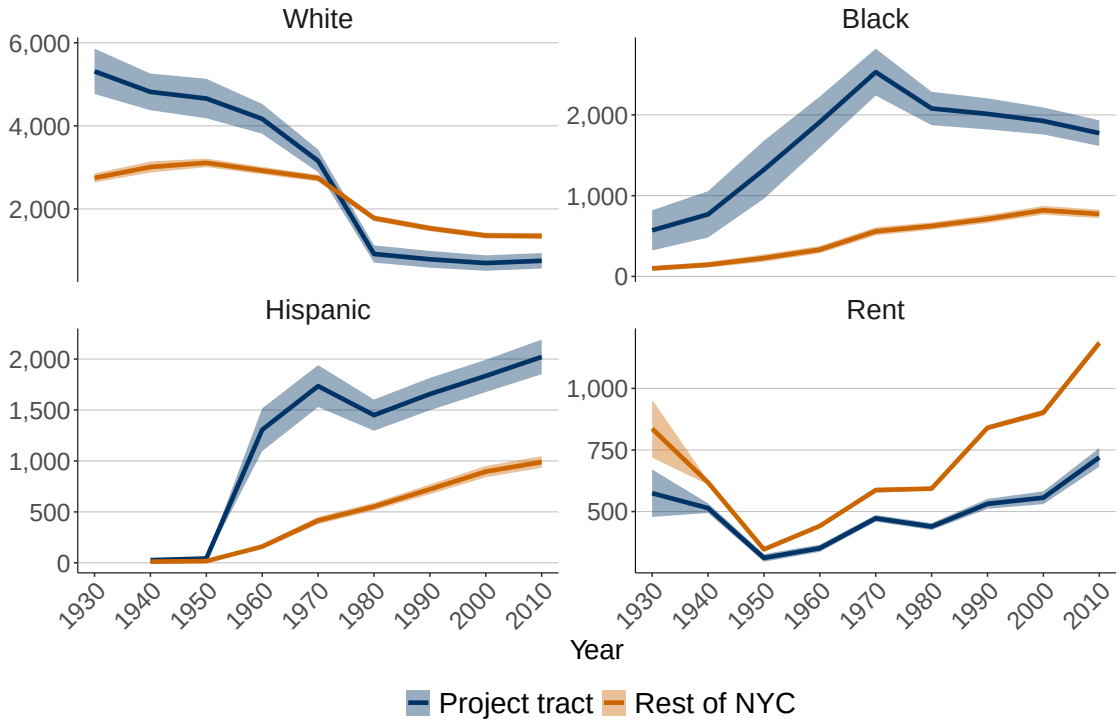
B1 Census Statistics

This subsection reports descriptive statistics for the census data. Population counts by race and median contract rents are drawn from the decennial census between 1930 and 2010, obtained from IPUMS and IPUMS-NHGIS and harmonized to 2010 census tract boundaries

as described in Appendix B4. All rents are deflated using the CPI and expressed in 2010 dollars. The two figures below compare the long-run evolution of the main outcome variables in public housing neighborhoods with the rest of the city.

Figure B1.1 plots census-year averages of the White, Black, and Hispanic population and of real median contract rent, separately for project tracts—tracts that ever contain a public housing unit—and all remaining tracts in New York City. Project tracts are more populous than the average tract: in 1930, before public housing construction began, they housed almost twice as many White residents as the average remaining tract. From 1950 onward, however, the White population of project tracts declines steeply and falls below that of the rest of the city by 1980. The Black and Hispanic populations grow substantially faster in project tracts than elsewhere, with the Black population peaking around 1970 and the Hispanic population continuing to grow through 2010. Rents in project tracts are already below the city average in 1930. Rents in both groups decline until 1950 and rise thereafter; from the 1970s onward, however, rents in the rest of the city grow considerably faster, opening a substantial gap to project tracts by 2010.

Figure B1.1: Demographic Trends

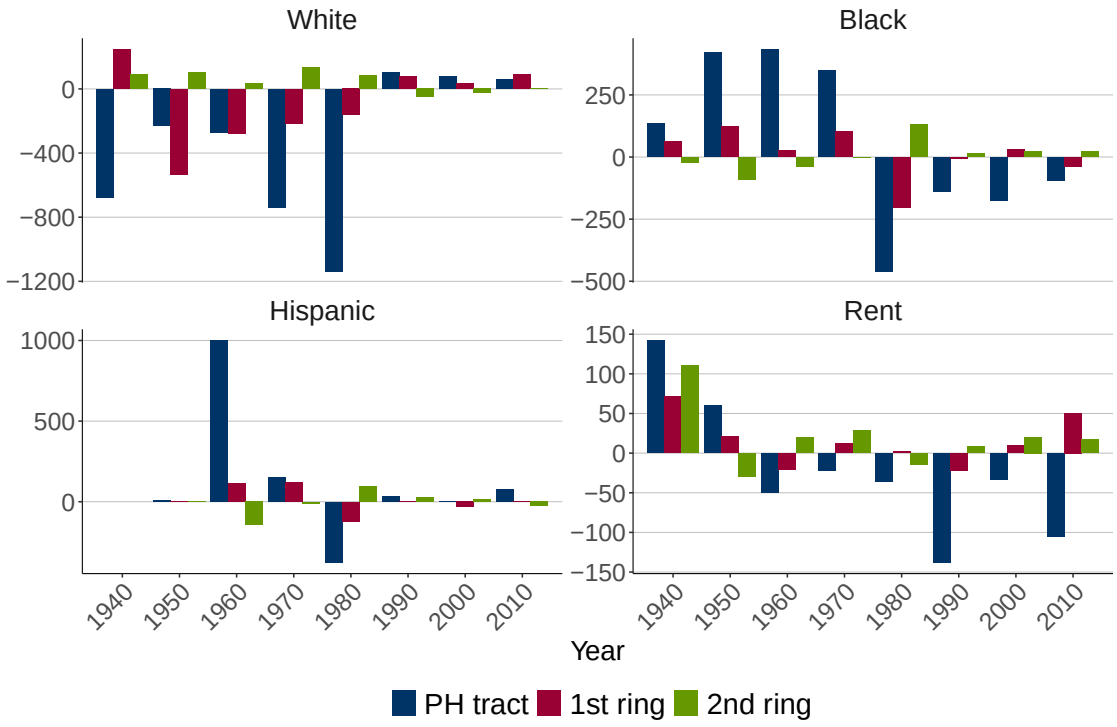


Note. Figure B1.1 reports trends of the main outcome variables. It shows census-year averages of the White, Black, and Hispanic population and of real median contract rent. I compute averages for all treated tracts, or, in other words, those which ever had a public housing unit within their boundaries (Project tract) and all remaining tracts in New York City (Rest of NYC). Shaded areas indicate 95 percent confidence intervals.

Source. US Decennial Census. Details on construction of the data set can be found in section 3.

Figure B1.2 translates these trends into deviations from the average city tract for the three groups of tracts used in the empirical analysis, as defined in Section 4: public housing tracts (PH tract), adjacent tracts (1st ring), and control tracts (2nd ring). Each bar reports the change over the preceding decade in the difference between the group average and the average census tract in New York City. Public housing tracts lose White population relative to the average tract in every decade between 1940 and 1980, with the largest relative losses occurring between 1970 and 1980. They gain Black population relative to the city between 1940 and 1970, a pattern that reverses in the 1970s. The Hispanic population in public housing tracts rises sharply relative to the average tract between 1950 and 1960 and declines relative to the city in the 1970s. Relative rents in public housing tracts decline in most decades after 1950. The deviations of first- and, in particular, second-ring tracts are generally smaller, indicating that the surrounding neighborhoods track the average city tract more closely than the public housing tracts themselves.

Figure B1.2: Deviation from Average City Tract



Note. Figure B1.2 reports, for each census year, the decadal change in the deviation of the average treated (PH tract), adjacent (1st ring), and control tract (2nd ring), as defined in Section 4, from the average census tract in New York City.
Source. US Decennial Census. Details on construction of the data set can be found in section 3.

B2 Public Housing Statistics

This subsection documents three connected dimensions of New York City's public housing program: the scale and physical form of construction, the neighborhoods in which projects were placed, and the residents initially allocated to them. These dimensions are important for the empirical analysis because a public housing project is not a homogeneous treatment. The large, high-rise developments of the mid-century program differ from later projects in size and design, while both siting and tenant allocation changed substantially across construction cohorts. The descriptive evidence therefore provides the basis for the construction-period comparison in subsection C2 and for the project-attribute analysis in section 6.

I construct the project-level dataset by combining the NYCHA Development Data Books, the NYCHA Resident Data Books, and archival records from the LaGuardia and Wagner Archives. The Development Data Books report each project's completion year, number of units, number of stories, ground coverage, and acquisition method. They also separate development expenditure into land, construction, site-improvement, and other costs, allowing construction costs per room to be compared in real terms. Table B2.1 summarizes these measures by construction cohort and establishes how the physical intervention changed over time. Of the 299 NYCHA-operated developments in the administrative records, the table covers the 273 unique new-construction developments with a recorded completion year; the remainder are rehabilitation or acquisition developments or lack the information required here.

Table B2.1: Public Housing Characteristics by Construction Decade

	1931–40	1941–50	1951–60	1961–70	1971–80	1981–90	1991–
	Total counts						
Projects	9	28	58	80	63	26	9
Units	10244	30382	70601	41933	14911	5715	626
	Median characteristics						
Max. stories	6	10	14	18	12	7	6
Units	1531	1156	1246	441	221	189	51
Ground coverage	29.74%	18.86%	14.50%	17.62%	33.47%	28.01%	42.68%
Construction cost	$12,854.93	$14,712.01	$13,963.55	$15,409.28	$17,287.91	$25,965.57	$26,962.55
	Average characteristics						
Max. stories	5.33	9.43	12.78	16.95	12.27	8.12	5.44
Units	1138.22	1085.07	1217.26	524.16	236.68	219.81	69.56
Ground coverage	27.87%	18.29%	14.92%	21.61%	31.86%	28.26%	46.71%
Construction cost	$18,622.04	$16,744.14	$16,626.30	$17,682.21	$17,123.95	$25,097.79	$29,298.38

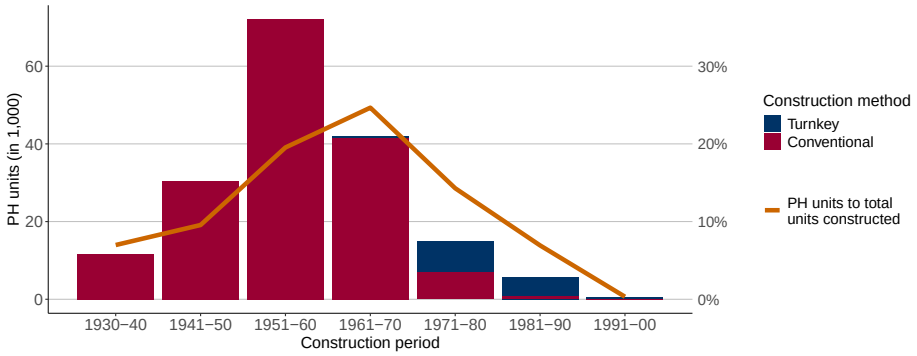
Note. Projects are assigned to construction cohorts using their opening dates; each column covers completions in the indicated decade, and the final cohort pools all projects completed after 1990. The first two rows report the number of projects and units; the next four report median project characteristics; and the final four report project-level means. Maximum stories is the largest number of stories reported for a development. Construction cost is measured per room, deflated by the CPI, and expressed in 2010 dollars. *Source.* NYCHA Development Data Book. Details on construction of the dataset appear in section 3.

The table reveals a sharp historical break. Projects completed between 1951 and 1970 account for almost two thirds of all units. These developments embody the characteristic tower-in-the-park form: median project size exceeds 1,000 units in every cohort completed through 1960, median height rises from six stories in the earliest cohort to 18 stories among

projects completed in 1961–70, and median ground coverage falls from about 30 percent to below 15 percent by 1951–60. Median project size drops sharply already among projects completed in 1961–70, and after 1970 projects become progressively smaller and lower, ground coverage rises, and real construction cost per room increases substantially. Figure B2.1 complements the project-level table by showing the aggregate construction cycle and its importance relative to all new housing in the city.

In Figure B2.1, the bars report completed public housing units and distinguish the conventional and Turnkey acquisition methods. The Turnkey program, introduced in 1969, appears only among the later cohorts and accounts for a substantial share of their production. The orange line scales public housing construction by all housing units built in New York City during the same period. Public housing production peaks at roughly 70,000 units in the 1950s, whereas its share of citywide construction peaks at almost one quarter in the 1960s as private construction slows. Both measures subsequently contract, reaching virtually zero by the 1990s.

Figure B2.1: Evolution of Public Housing by Construction Period



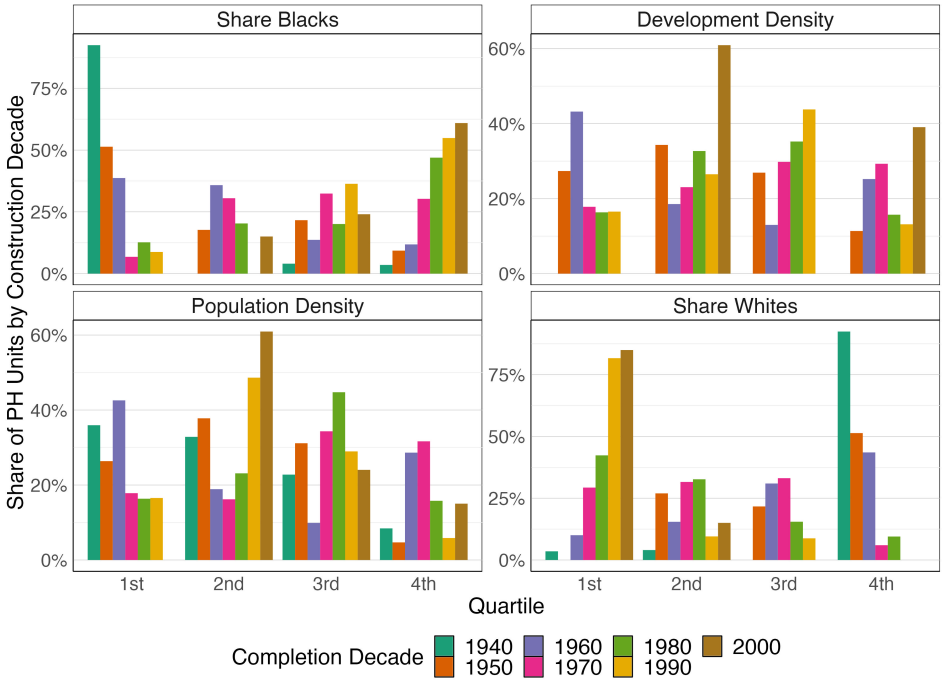
Note. Bars report public housing units by construction cohort and acquisition method. Under the conventional method, NYCHA acquires the land and contracts separately for construction work. Under the Turnkey method, a developer acquires the land, constructs the development, and sells it to NYCHA under a pre-agreed contract. The orange line reports public housing units as a share of all housing units constructed in New York City during the same period.
Source. NYCHA Development Data Book. Details on construction of the dataset appear in section 3.

Together, Table B2.1 and Figure B2.1 show that the pre- and post-1970 cohorts represent different forms of intervention, not merely projects completed at different dates. The early program combined large individual projects with a sizable role in citywide housing production; the later program used smaller developments and a different acquisition model at a much lower construction intensity. This distinction motivates treating construction period and project attributes as related sources of heterogeneity rather than pooling all projects into a single average effect.

Project form also changed together with project location. Figure B2.2 distributes the units in each construction cohort across quartiles of four tract characteristics measured in the census decade before completion: Black population share, White population share, population density, and development density. The earliest units were placed almost exclusively in tracts in the lowest quartile of the Black share and, equivalently, the highest quartile

of the White share. Later cohorts shift progressively toward tracts in the highest quartile of the Black share. The distribution across population- and development-density quartiles is more balanced, indicating that racial composition changed more systematically across siting regimes than baseline density did.

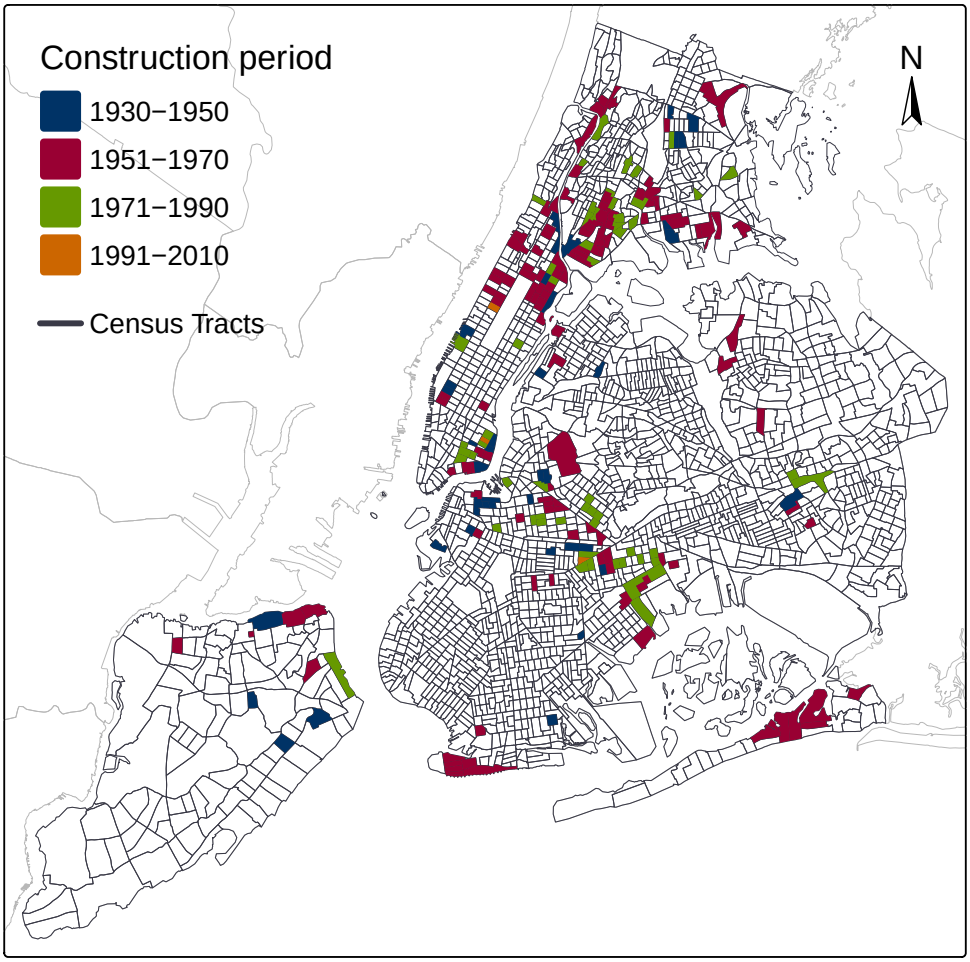
Figure B2.2: Public Housing Units by Construction Cohort and Quartile of Baseline Tract Characteristics



Note. For each construction cohort, the figure reports the share of public housing units located in each quartile of the indicated tract characteristic. Characteristics are measured in the census decade before project completion, and quartile shares sum to one within each cohort and characteristic.
Source. NYCHA Development Data Book and US Decennial Census. Details on construction of the dataset appear in section 3.

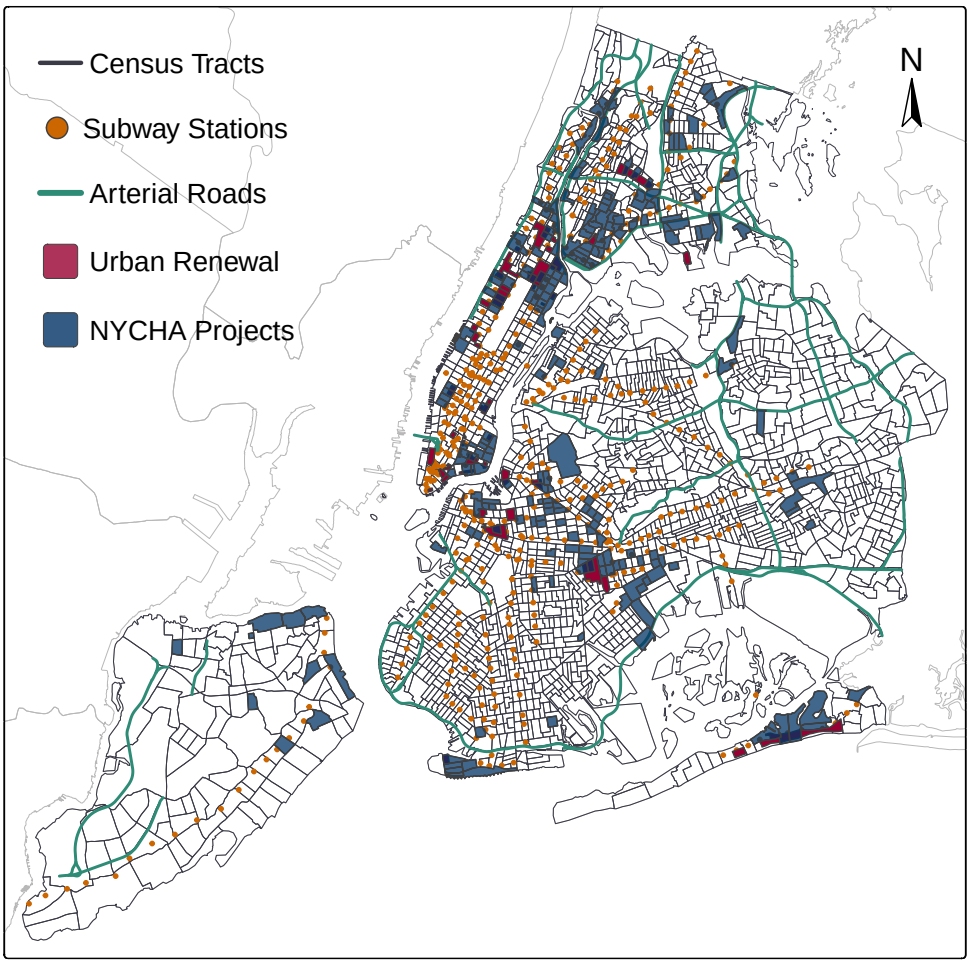
Figure B2.3 and Figure B2.4 translate this cohort-level siting evidence into geography. The first map classifies each public housing tract by the construction period of its first project, recording when public housing initially entered a neighborhood rather than every subsequent addition. The second overlays all project locations with subway stations, arterial roads, and urban renewal areas. Viewed together, the maps show why siting cannot be treated as independent of neighborhood history: projects arrived in places with different racial composition, accessibility, and planning status, and these relationships changed across cohorts. The main specification therefore includes infrastructure and planning controls (section 4), while its project-specific stacks and pretrend tests address remaining differences in neighborhood trajectories.

Figure B2.3: Spatial Expansion of Public Housing by Construction Period



Note. Colored tracts contain at least one public housing project and are classified by the construction period of their first completed project: 1930-1950 (blue), 1951-1970 (red), 1971-1990 (green), or 1991-2010 (orange). Tracts containing projects from multiple periods are classified using the earliest completion.
Source. LaGuardia and Wagner Archives and NYCHA Development Data Book. Details on construction of the dataset appear in section 3.

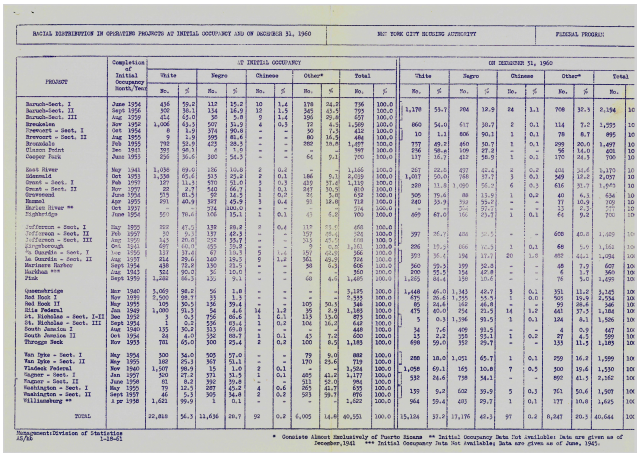
Figure B2.4: Public Housing Locations and Urban Infrastructure in New York City



Note. The map displays NYCHA public housing projects together with census tract boundaries, subway stations (orange), arterial roads (emerald), and designated urban renewal areas (red).
Source. NYCHA Development Data Book and New York City spatial data. Details on construction of the dataset appear in section 3.

The final component of the dataset measures who initially occupied each project and how NYCHA's resident population evolved. I digitize the LaGuardia and Wagner Archives series *Racial Distribution in Operating Projects at Initial Occupancy*, available for 1946, 1950, 1960, and 1970. Because no earlier report is available, the 1946 observation is assigned to the earliest cohort. The 1946 and 1950 reports distinguish Black and White residents, while the 1960 and 1970 reports additionally identify Hispanic and Asian residents. Figure B2.5 reproduces a page from the 1960 report and illustrates the project-level records from which the initial-allocation measures are constructed.

Figure B2.5: Archival Source for Project-Level Racial Data



RACIAL DISTRIBUTION IN OPERATING PROJECTS AT INITIAL OCCUPANCY AND ON DECEMBER 31, 1960 — NEW YORK CITY HOUSING AUTHORITY — FEDERAL PROGRAMS

PROJECT	Completion of Initial Occupancy Month/Year	AT INITIAL OCCUPANCY: White No.	White %	Negro No.	Negro %	Chinese No.	Chinese %	Other* No.	Other* %	Total No.	Total %	ON DECEMBER 31, 1960: White No.	White %	Negro No.	Negro %	Chinese No.	Chinese %	Other* No.	Other* %	Total No.	Total %	
Baruch-Sect. I	June 1956	436	59.2	112	15.2	10	1.4	178	24.2	736	100.0											
Baruch-Sect. II	Sept 1956	302	38.1	134	16.9	12	1.5	345	43.5	793	100.0	1,178	59.7	704	19.9	24	1.1	708	19.3	2,194	100.0	
Baruch-Sect. III	Aug 1959	214	65.0	30	5.8	9	1.4	196	25.8	607	100.0											
Brownsville	Nov 1952	1,096	43.5	507	31.9	4	0.3	19	4.3	1,589	100.0	860	54.0	617	30.7	2	0.1	114	7.2	1,593	100.0	
Claremont - Sect. I	Oct 1954	6	1.9	374	90.8	-	-	30	7.3	412	100.0	10	1.3	806	90.1	3	0.1	78	8.7	895	100.0	
Claremont - Sect. II	Aug 1955	8	1.9	395	91.6	-	-	28	6.5	431	100.0											
Glenmore	Jan 1952	798	53.8	472	31.5	-	-	208	14.6	1,478	100.0	577	49.2	460	39.1	1	0.1	136	11.6	1,174	100.0	
Harlem River Park	Dec 1951	291	54.7	241	45.3	-	-	-	-	532	100.0	292	46.2	259	41.0	1	0.1	80	12.7	632	100.0	
Jasper Park	June 1953	256	36.6	380	54.3	-	-	64	9.1	700	100.0	137	16.2	412	48.9	1	0.1	293	34.8	843	100.0	
East River	May 1941	1,038	89.6	106	10.8	2	0.2	-	-	1,146	100.0	257	22.2	837	72.4	4	0.2	58	5.2	1,156	100.0	
Idlewild	Oct 1953	1,158	65.6	513	29.2	2	0.1	186	5.1	2,029	100.0	1,017	50.0	766	37.7	3	0.1	249	12.2	2,037	100.0	
Lincoln - Sect. I	Feb 1959	197	11.5	970	51.0	3	0.3	439	37.2	1,119	100.0	109	11.8	1,080	58.3	4	0.3	616	33.7	1,673	100.0	
Lincoln - Sect. II	Nov 1957	22	2.7	540	66.2	1	0.1	247	30.5	810	100.0											
Stuyvesant	June 1954	255	81.5	58	18.5	1	0.3	51	5.8	639	100.0	305	75.6	88	15.9	1	0.2	40	8.3	634	100.0	
Wagner	Apr 1955	291	40.9	387	45.9	3	0.4	91	12.8	718	100.0	240	33.9	392	55.2	-	-	77	10.9	709	100.0	
Harlem River **	Oct 1957	-	-	274	100.0	-	-	-	-	274	100.0	-	-	254	97.3	-	-	7	2.7	261	100.0	
Highbridge	June 1954	569	78.6	106	15.1	1	0.1	43	6.2	700	100.0	669	67.0	166	25.1	1	0.1	66	9.2	700	100.0	
Jefferson - Sect. I	Dec 1959	302	47.5	139	20.2	2	0.4	133	31.9	686	100.0											
Jefferson - Sect. II	Feb 1957	30	9.3	137	42.3	-	-	107	33.4	324	100.0	397	35.7	486	30.5	-	-	608	33.8	1,489	100.0	
Jefferson - Sect. III	Jan 1957	145	21.8	258	38.7	-	-	263	39.5	666	100.0											
Gompers	Oct 1941	687	46.0	655	43.8	-	-	9	0.6	1,351	100.0	286	19.2	866	58.3	1	0.1	66	5.9	1,166	100.0	
Wm Harold - Sect. I	Jan 1959	137	37.6	87	23.9	5	1.4	130	35.8	366	100.0	203	36.1	196	34.8	20	3.6	143	25.5	562	100.0	
La Guardia - Sect. II	Aug 1957	214	29.6	140	19.3	9	1.2	361	49.9	724	100.0											
Marcus Garvey	Sept 1954	438	72.9	130	21.5	-	-	33	5.5	606	100.0	360	59.5	199	32.8	-	-	46	7.7	607	100.0	
Markham **	Jan 1943	324	90.0	36	10.0	-	-	-	-	360	100.0	200	55.5	154	42.8	-	-	6	1.7	360	100.0	
Polo	Sept 1959	1,488	86.2	151	9.1	-	-	68	4.6	1,685	100.0	1,265	84.4	156	10.4	-	-	78	5.2	1,499	100.0	
Queensbridge	Nov 1940	5,069	98.2	56	1.8	-	-	-	-	5,125	100.0	1,448	46.0	1,343	42.7	3	0.1	351	11.2	3,145	100.0	
Red Hook I	Dec 1949	7,500	98.7	95	1.3	-	-	-	-	7,595	100.0	675	24.4	1,956	70.5	1	0.0	505	18.2	2,774	100.0	
Red Hook II	May 1955	105	50.5	136	39.4	-	-	105	30.5	346	100.0	85	24.6	168	48.8	-	-	93	26.6	346	100.0	
Rita Federal	Jan 1949	1,080	91.5	54	4.6	14	1.2	35	2.9	1,183	100.0	475	40.0	374	31.5	14	1.2	461	37.3	1,324	100.0	
St. Nicholas - Sect. I-II	Dec 1952	3	0.3	756	84.6	1	0.1	113	15.0	873	100.0	5	0.3	1,396	91.5	1	0.1	124	8.1	1,526	100.0	
St. Nicholas - Sect. III	Sept 1954	1	0.2	536	83.4	1	0.2	106	16.5	642	100.0											
South Jamaica I	Aug 1940	135	30.2	313	69.8	-	-	-	-	448	100.0	34	7.6	409	91.5	-	-	4	0.9	447	100.0	
South Jamaica II	Oct 1954	24	4.0	556	89.7	1	0.1	43	7.0	620	100.0	17	2.2	755	93.1	1	0.2	37	4.5	599	100.0	
Throggs Neck	Nov 1953	701	65.0	300	25.4	2	0.2	100	8.5	1,183	100.0	698	59.0	352	29.7	-	-	133	11.5	1,183	100.0	
Van Dyke - Sect. I	May 1954	300	34.0	505	57.0	-	-	79	9.0	882	100.0											
Van Dyke - Sect. II	May 1955	160	25.3	367	51.1	-	-	170	23.6	719	100.0	288	18.0	1,051	65.7	1	0.1	259	16.2	1,599	100.0	
Vladeck Federal	Nov 1940	1,507	98.9	15	1.0	2	0.1	-	-	1,524	100.0	1,056	69.1	165	10.8	7	0.5	300	19.6	1,530	100.0	
Wagner - Sect. I	Dec 1959	561	78.2	156	21.8	1	0.1	695	43.2	1,613	100.0	538	24.8	798	54.1	-	-	899	41.5	2,162	100.0	
Wagner - Sect. II	June 1959	70	8.2	392	91.8	-	-	215	51.0	1,172	100.0											
Washington - Sect. I	May 1956	70	12.5	287	45.2	4	0.6	365	41.7	835	100.0	133	9.2	602	39.9	5	0.3	361	50.6	1,501	100.0	
Washington - Sect. II	Sept 1957	46	5.5	305	36.8	2	0.2	503	59.7	836	100.0											
Williamsburg **	2 pr 1958	1,621	99.9	1	0.1	-	-	-	-	1,622	100.0	964	59.4	483	29.7	1	0.1	177	10.8	1,625	100.0	
TOTAL		22,818	56.3	11,636	28.7	92	0.2	6,005	14.8	40,551	100.0	15,324	37.9	17,176	42.5	97	0.2	8,247	20.5	40,644	100.0	

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* Consists Almost Exclusively of Puerto Ricans ** Initial Occupancy Data Not Available; Data are given as of December, 1941 *** Initial Occupancy Data Not Available; Data are given as of June, 1945.

Note. The figure reproduces page two of NYCHA's December 31, 1960 report *Racial Distribution in Operating Projects*. The report records the racial composition of each project at initial occupancy and at the reporting date.
Source. LaGuardia and Wagner Archives, NYCHA Collection, MS Collection IV B.

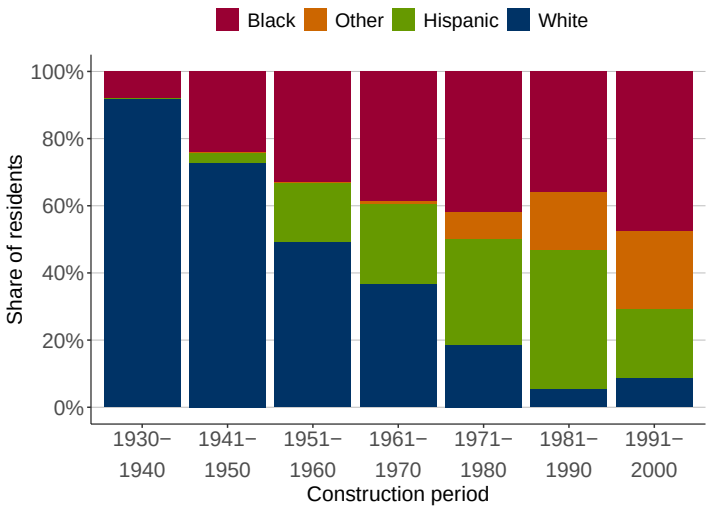
I supplement the archival reports with the NYCHA Resident Data Books, which provide annual project-level racial statistics. Matching these records to project completion dates recovers resident composition at opening and extends the aggregate series over time. White, Black, and Hispanic residents can be identified consistently across the combined sources. Because Asian and Other categories are not reported consistently, I combine them into a single category.

Figure B2.6 summarizes both margins. Panel (a) shows that the White share at initial occupancy falls from approximately 90 percent in the earliest cohort to below 10 percent after 1980, when Black and Hispanic residents account for most initial occupants. Panel (b) follows NYCHA's population over calendar time: the White population peaks around 1960 and then contracts, while the Black population rises rapidly and is the largest group by 1970. Hispanic and Other residents appear separately only from 1960 onward because earlier reports do not identify them.

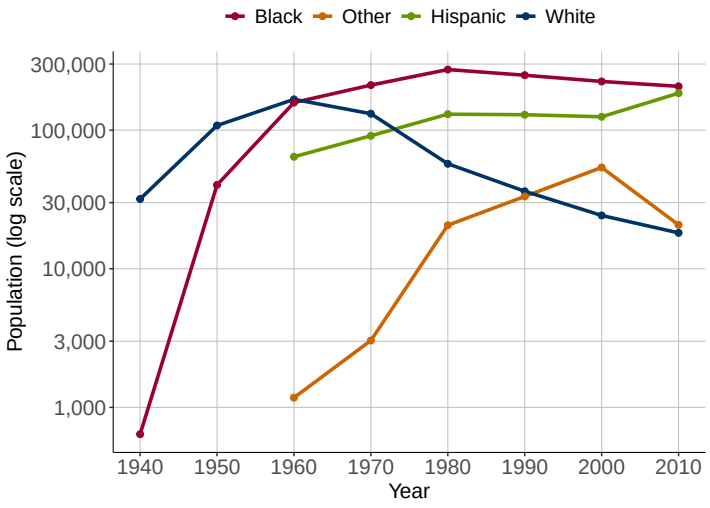
Together, these exhibits show that construction cohorts bundle changes in project form, siting, and tenant allocation. Early projects were larger, placed in whiter neighborhoods, and housed mostly White residents; later projects were smaller, located in neighborhoods with larger Black populations, and housed mostly Black and Hispanic residents. The heterogeneity analysis therefore conditions on neighborhood and project characteristics.

Figure B2.6: Racial Composition by Construction Decade

(a) Initial Allocation of Residents



(b) Number of Residents by Race



Note. Panel (a) reports resident shares at initial occupancy for projects grouped by construction cohort. Panel (b) reports the total number of NYCHA residents by race in each census year.
Source. LaGuardia and Wagner Archives and NYCHA Resident Data Books. Details on construction of the dataset appear in section 3.

B3 Project Attributes and Pre-Construction Neighborhood Selection

The project-level analysis asks whether developments that differed in physical form, construction expenditure, and initial tenant composition produced different neighborhood responses. Project scale is analyzed separately as treatment intensity in subsection 5.6; the attribute vector in subsection 6.1 contains maximum building height, ground coverage, construction cost per room, and White, Black, and Hispanic tenant shares at initial occupancy. Two concerns shape the interpretation of those estimates. First, project attributes may be correlated with one another, so the coefficient on any one attribute is a partial association conditional on the others. Second, NYCHA may have adapted project design and tenant allocation to pre-existing site conditions. The resulting estimates would then compare projects built in systematically different neighborhoods rather than isolate attributes assigned to otherwise identical locations.

The diagnostics below use average building height at the development level rather than the maximum-height measure in the main specification. The two measures capture related but distinct aspects of vertical form, so the appendix provides complementary evidence on design selection rather than an exact diagnostic for the headline height measure. I first document dependence among project attributes and then test whether pre-construction neighborhood conditions predict the attributes of the project subsequently built. Each diagnostic is reported separately for projects completed before 1970 and those completed in 1970 or later. Because the attribute and baseline measures are not available for every development, the regressions below use the 118 pre-1970 and 56 post-1970 projects with complete information. This cohort split is important because the descriptive evidence in subsection B2 shows that project scale, design, siting, and tenant composition changed together after the mid-century construction program.

Table B3.1 begins with the pre-1970 cohort and reports pairwise correlations among project scale, physical form, construction expenditure, and initial tenant composition. Apartment count is included only as a diagnostic for its relationship with the design variables; it does not enter the attribute vector in Equation 4. The table therefore indicates both how much independent variation the included attributes contain and how their coefficients must be interpreted when entered jointly.

Table B3.1: Pairwise Correlations among Public Housing Attributes: Pre-1970 Cohort

	Units	White	Black	Hisp.	Cost	Stories	Coverage
Units							
White	0.04						
Black	-0.08	-0.87***					
Hisp.	0.04	-0.68***	0.28***				
Cost	-0.08	0.18*	-0.13	0.10			
Stories	-0.11	-0.45***	0.29***	0.43***	0.01		
Coverage	-0.37***	0.14	-0.16*	-0.02	0.07	-0.20**	

Note. The table reports pairwise Pearson correlations among project-level attributes for projects completed before 1970, using all available nonmissing observations for each pair. Units is the number of apartments; White, Black, and Hisp. are tenant shares at initial occupancy; Cost is CPI-deflated construction cost per room in 2010 dollars; Stories is average building height; and Coverage is building-footprint area divided by total project area. Stars test whether the pairwise correlation equals zero. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The most pronounced pre-1970 relationships involve initial tenant composition. The White share is strongly negatively correlated with the Black share (-0.87) and the Hispanic share (-0.68), as expected for variables constrained to sum to one. Tenant-share coefficients must consequently be interpreted as conditional reallocations rather than independent increases in each group. The design variables are related but not redundant. Apartment count is negatively correlated with coverage (-0.37), while stories and coverage have a smaller correlation of -0.20. Using the entries in the table, the partial correlation between stories and coverage conditional on apartment count is approximately -0.26. Height and site coverage thus describe distinct, although related, margins of physical form.

The pre-1970 matrix also provides little support for treating construction cost per room as a proxy for height. Cost has a correlation of only 0.01 with stories and 0.07 with coverage. It contributes variation that is empirically distinct from the basic form measures, although expenditure remains an imperfect proxy for quality because it may also capture materials, labor costs, procurement rules, and site-specific constraints. I therefore interpret cost per room as construction investment rather than a clean structural measure of latent quality.

Table B3.2 asks whether the same relationships hold for projects completed in 1970 or later. Keeping this table separate is useful because later projects were smaller and built under a different institutional and design regime; pooling the correlations would obscure whether the dependence among attributes itself changed across cohorts.

Table B3.2: Pairwise Correlations among Public Housing Attributes: Post-1970 Cohort

	Units	White	Black	Hisp.	Cost	Stories	Coverage
Units							
White	0.29***						
Black	-0.18*	-0.52***					
Hisp.	0.13	-0.18*	-0.02				
Cost	0.04	0.05	-0.07	0.02			
Stories	0.47***	0.44***	-0.30***	0.08	0.03		
Coverage	-0.56***	-0.40***	0.26**	-0.22**	0.10	-0.47***	

Note. The table reports pairwise Pearson correlations among project-level attributes for projects completed in 1970 or later, using all available nonmissing observations for each pair. Units is the number of apartments; White, Black, and Hisp. are tenant shares at initial occupancy; Cost is CPI-deflated construction cost per room in 2010 dollars; Stories is average building height; and Coverage is building-footprint area divided by total project area. Stars test whether the pairwise correlation equals zero. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The later cohort displays stronger dependence among scale and physical form. Apartment count is positively correlated with stories (0.47) and negatively correlated with coverage (-0.56), while stories and coverage have a correlation of -0.47. Even here, however, the partial correlation between stories and coverage conditional on apartment count is only about -0.28. The two design measures remain sufficiently distinct to enter the joint specification, but their coefficients compare height holding coverage fixed and coverage holding height fixed. The White–Black tenant-share correlation remains sizeable at -0.52, reinforcing the conditional interpretation of the composition coefficients. As in the early cohort, cost is nearly uncorrelated with stories (0.03) and coverage (0.10). Across both tables, the central issue is therefore not mechanical collinearity, but the substantive meaning of holding related project features fixed.

Pairwise dependence among attributes does not reveal whether projects with different attributes were placed in different kinds of neighborhoods. To examine that selection margin, I estimate the following project-level regression separately by construction cohort c and project attribute j:

$$\widetilde{A}_{jp} = \widetilde{\mathbf{X}}_p^{0\prime}\boldsymbol{\beta}_{jc} + \mu_{g(p),j} + u_{jp}. \tag{B.1}$$

The dependent variable $\widetilde{A}_{jp}$ is a standardized project attribute. The standardized baseline vector $\widetilde{\mathbf{X}}_p^0$ contains the White and Black population shares, population density, housing-unit density, and real median rent in the project's census tract immediately before completion. Thus, each element of $\boldsymbol{\beta}_{jc}$ gives the standard-deviation difference in attribute j associated with a one-standard-deviation difference in the baseline characteristic. For pre-1970 projects, $\mu_{g(p),j}$ denotes borough-by-completion-decade fixed effects; for the smaller post-1970 cohort, it denotes completion-decade fixed effects. I report heteroskedasticity-robust standard errors and an F-test of the joint significance of the five baseline variables. Because baseline White and Black population shares are highly negatively correlated, the joint test is more informative than either coefficient in isolation.

Table B3.3 reports Equation (B.1) for the cohort used in the paper's main attribute analysis. It asks whether two pre-1970 projects completed in the same borough and construction

period nevertheless received systematically different designs, expenditures, or tenant allocations depending on conditions in their respective tracts before construction.

Table B3.3: Pre-Construction Neighborhood Predictors of Project Attributes: Pre-1970 Cohort

Dependent Variables:	Apartments	Avg. stories	Bldg. coverage share	ln(Cost / room)	Init. % hispanic tenants	Init. % black tenants	Init. % white tenants
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
% white	0.0053	-4.915**	-3.848	4.726	2.567	0.9588	0.9540
	(2.691)	(2.187)	(5.704)	(3.035)	(3.562)	(5.742)	(2.401)
% black	0.0349	-4.808**	-4.028	4.459	2.528	1.295	0.7063
	(2.689)	(2.163)	(5.677)	(2.992)	(3.566)	(5.727)	(2.381)
Pop. density	-0.1920	0.0415	-0.2041	0.1799	0.8842	0.3990	-0.0643
	(0.1818)	(0.1231)	(0.2124)	(0.1613)	(0.6831)	(0.4284)	(0.2087)
Housing-unit density	-0.0278	-0.0765	0.5420*	-0.0964	-0.6682	-0.3783	-0.4151*
	(0.2525)	(0.1753)	(0.3016)	(0.2529)	(0.6477)	(0.4434)	(0.2310)
Median rent	-0.2252***	-0.0639	0.2125*	0.1753**	-0.0307	-0.0571	-0.0696
	(0.0617)	(0.0735)	(0.1152)	(0.0807)	(0.1158)	(0.1066)	(0.0987)
Borough × Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	118	118	118	118	118	118	118
R^2	0.48901	0.66457	0.36002	0.38820	0.31078	0.24254	0.46212
F-test	4.9359	10.219	2.9015	3.2728	2.3258	1.6516	4.4313

Note. The table reports project-level estimates of Equation B.1. The dependent variables are total apartments, average stories, ground-coverage share, log CPI-deflated construction cost per room, and Hispanic, Black, and White tenant shares at initial occupancy. The sample contains one observation per project completed before 1970 ($n = 118$). Dependent variables and baseline regressors are standardized within the cohort, so coefficients are measured in standard-deviation units. Baseline regressors are measured in the project's census tract immediately before completion; housing-unit density uses the average pre-construction housing stock. All specifications absorb borough-by-completion-decade fixed effects. Heteroskedasticity-robust standard errors are in parentheses. The reported R^2 includes the fixed effects, and the F-statistic tests the joint significance of the five baseline regressors. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The table shows substantial selection on observed pre-construction conditions. Except for the initial Black tenant share ($F = 1.65$), the joint F-statistics range from 2.33 to 10.22, and the full specifications explain between 24% and 66% of the cross-project variation in the seven attributes. Baseline median rent is the most recurrent individual predictor: a one-standard-deviation increase predicts 0.23 standard deviations fewer apartments, 0.21 greater ground coverage, and 0.18 higher construction cost per room. Housing-unit density predicts greater coverage (0.54) and a lower initial White tenant share (-0.42), consistent with project form and allocation adapting to already built-up sites.

The coefficients on baseline racial composition are individually significant only for average building height, where the White and Black shares enter with large coefficients of the same sign. Because those baseline shares are almost perfectly negatively correlated in this sample, the coefficients largely offset one another; the joint F-test is the more stable summary. The broader conclusion does not depend on any one coefficient: pre-1970 project attributes cannot be described as independent of prior neighborhood conditions.

Table B3.4 applies the same diagnostic to the later cohort. The comparison distinguishes selection patterns that persist across policy regimes from relationships specific to the large mid-century projects, while the smaller sample of 56 later projects necessarily limits statistical precision.

Table B3.4: Pre-Construction Neighborhood Predictors of Project Attributes: Post-1970 Cohort

Dependent Variables:	Apartments	Avg. stories	Bldg. coverage share	ln(Cost / room)	Init. % hispanic tenants	Init. % black tenants	Init. % white tenants
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
% white	0.1298	0.3934	0.3108	0.0152	-0.7351**	-0.8763***	0.7202***
	(0.2611)	(0.2940)	(0.2398)	(0.3272)	(0.3218)	(0.2224)	(0.2461)
% black	0.1625	0.0681	0.2250	-0.2336	-0.8567***	-0.4475*	0.0071
	(0.1913)	(0.2006)	(0.1556)	(0.3172)	(0.2567)	(0.2255)	(0.1460)
Pop. density	-0.3059	-0.2245	0.2164	0.1475	0.3868	0.2744	-0.0829
	(0.2503)	(0.2794)	(0.2838)	(0.1809)	(0.2432)	(0.3906)	(0.1745)
Housing-unit density	-0.0308	0.2266	0.1788	-0.1668	-0.3327	-0.5561	-0.3605*
	(0.2738)	(0.2991)	(0.2409)	(0.2040)	(0.2420)	(0.4001)	(0.1801)
Median rent	-0.0669	0.1221	0.2612***	-0.0322	-0.2790**	0.1045	0.0601
	(0.1456)	(0.1280)	(0.0963)	(0.0773)	(0.1215)	(0.1068)	(0.0936)
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	56	56	56	56	56	56	56
R^2	0.17355	0.16990	0.30245	0.10228	0.27988	0.16239	0.56052
F-test	1.4400	1.4035	2.9732	0.78128	2.6651	1.3294	8.7457

Note. The table reports project-level estimates of Equation B.1. The dependent variables are total apartments, average stories, ground-coverage share, log CPI-deflated construction cost per room, and Hispanic, Black, and White tenant shares at initial occupancy. The sample contains one observation per project completed in 1970 or later ($n = 56$). Dependent variables and baseline regressors are standardized within the cohort. Baseline regressors are measured in the project's census tract immediately before completion; housing-unit density uses the average pre-construction housing stock. All specifications absorb completion-decade fixed effects. Borough-by-completion-decade effects are omitted because the small sample does not support saturated borough-cohort cells. Heteroskedasticity-robust standard errors are in parentheses. The reported R^2 includes the fixed effects, and the F-statistic tests the joint significance of the five baseline regressors. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Selection is less uniform after 1970 but remains visible. Baseline conditions have limited joint predictive power for apartments ($F = 1.44$), stories ($F = 1.40$), construction cost ($F = 0.78$), and the initial Black share ($F = 1.33$). The joint statistics are larger for coverage (2.97), the initial Hispanic share (2.67), and especially the initial White share (8.75). Initial tenant composition remains closely related to neighborhood racial composition: a one-standard-deviation higher baseline White share predicts lower initial Hispanic (-0.74) and Black (-0.88) shares and a higher initial White share (0.72). Median rent predicts greater coverage (0.26) and a lower initial Hispanic share (-0.28), while housing-unit density predicts a lower initial White share (-0.36). The R^2 of 0.56 for initial White composition therefore reflects systematic sorting rather than random variation. Conversely, failure to reject the joint null for several other attributes is not evidence of random assignment in a sample of only 56 projects.

These diagnostics reinforce the conditional interpretation of the attribute estimates in subsection 6.1. Because they use average rather than maximum height, they do not test selection in the headline height measure exactly. They nevertheless show that a closely related dimension of vertical form covaries with other project attributes and earlier neighborhood conditions. In the main specification, the maximum-height coefficient compares treatment sites of different height holding coverage, cost, and tenant composition fixed, and the coverage coefficient has the analogous interpretation. Construction cost per room is empirically distinct from the physical-form measures, although it remains an imperfect proxy for quality. More broadly, Equation (B.1) shows that observed project characteristics–especially in the pre-1970 cohort used for the main analysis–systematically covary with earlier neighbor-

hood conditions. Attribute interactions should therefore be read as conditional cross-project heterogeneity, not as effects generated by random assignment of project design.

The main specification addresses observed selection by interacting baseline racial composition and log population density with census-year fixed effects, while the pooled-pre coefficients in Figure 8 provide a pretrend diagnostic. These adjustments absorb observed differences and associated persistence but cannot rule out selection on unobserved neighborhood conditions. The qualification is particularly important for initial tenant composition: the shares are jointly constrained and closely related to baseline neighborhood composition, so they cannot be interpreted as independently or randomly assigned neighborhood attributes.

B4 Tract Harmonisation

A major challenge in using census tract-level data for longitudinal analysis is that tract boundaries change substantially over time, making it difficult to construct a time-consistent panel dataset. I address this issue by reweighting observations from every census year from 1930 through 2010 based on overlapping area weights, using 2010 census tract boundaries as the common target geography. Specifically, I aggregate count variables—such as White and Black population—using this method. For 1930, I use the full-count population census from IPUMS, which I aggregate at the enumeration-district (ED) level. Thus, instead of census tract, I use overlapping area weights for enumeration districts. Let i index 2010 census tracts and j index historical tracts from earlier census years. For each 2010 tract i, the weighted estimate $\hat{y}_i$ for a given outcome variable y is calculated as:

$$\hat{y}_i = \sum_{j \in \mathcal{J}_i} \frac{A_{ij}}{A_j} \cdot y_j,$$

where y_j is the value of the outcome variable in historical tract j, A_{ij} is the area of intersection between historical tract j and 2010 tract i, and A_j is the total area of tract j. The set $\mathcal{J}_i$ contains all historical tracts j that overlap with tract i.

This method assumes that the outcome variable is uniformly distributed within each historical tract, which may introduce error. For instance, if tract j in year t contains a spatially concentrated low-income population, but overlaps evenly with multiple 2010 tracts, the method would misallocate residents uniformly across those tracts—biasing spatial estimates.

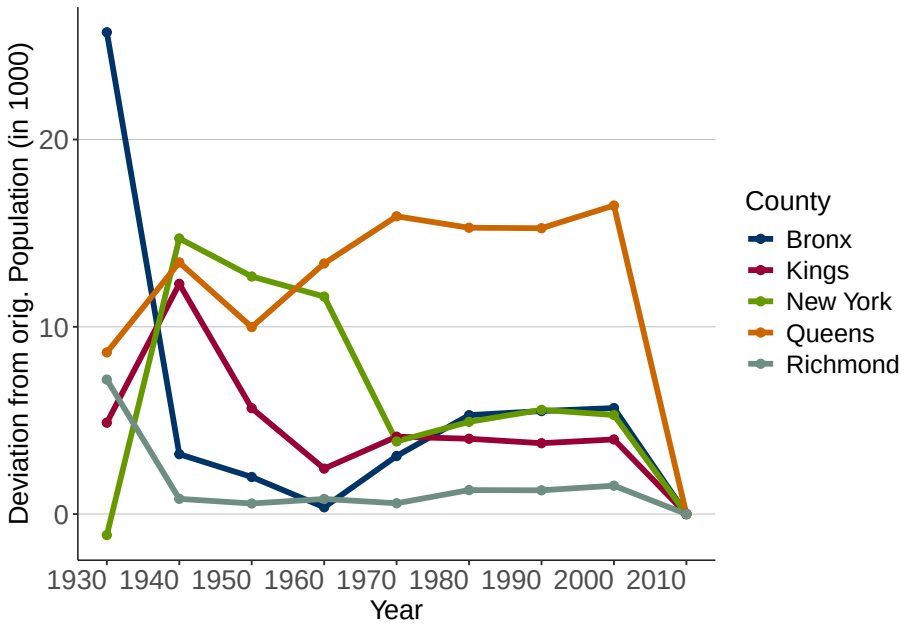
For variables such as median rent and household income, which are not count-based, I calculate tract-level estimates as weighted averages across historical geographies using overlapping area weights. Specifically, for each 2010 census tract i, I compute:

$$\hat{y}_i = \frac{\sum_{j \in \mathcal{J}_i} y_j \cdot w_{ij}}{\sum_{j \in \mathcal{J}_i} w_{ij}},$$

where w_{ij} is the proportion of area of historical tract j overlapping with 2010 tract i, conditional on $w_{ij} > 0.05$. This approach accommodates spatial uncertainty while preserving information from all overlapping historical tracts. A fallback value from the highest-overlap tract is also computed, but not used in the main analysis. Two limitations of this interpolation should be kept in mind. A weighted average of source-tract medians need not equal the

median of the combined population living in the target tract, and because the weights are shares of each source tract's own area, small source tracts lying mostly inside a target tract can receive substantial weight. I therefore treat the harmonized medians as interpolated neighborhood-level indicators rather than exact target-tract statistics.

Figure B4.1: Deviation due to Boundary Harmonization



Note. Figure B4.1 shows the difference between total population aggregated from original census tracts and total population estimated using area-weighted aggregation based on overlapping boundaries, by borough. *Source.* U.S. Decennial Census.

Figure B4.1 compares the reweighted population series at the borough level to the original series. While overall trends are preserved, deviations from the original data vary across boroughs and over time. The largest single deviation occurs for the Bronx in 1930, at roughly 26,000 residents, in the year that relies on the ED-based aggregation and the least standardized historical boundaries. Queens shows the most persistent deviations, of roughly 15,000–17,000 residents between 1970 and 2000, reflecting its more complex boundary changes, such as tract splits and consolidations. Kings (Brooklyn) and New York (Manhattan) show deviations of similar magnitude in the 1940s and 1950s that decline as boundary stability improves, while Richmond (Staten Island) exhibits minimal deviation throughout the entire period, consistent with relatively stable tract definitions. Two properties of these deviations matter for interpretation. First, they are almost uniformly positive: the area-weighted series slightly exceeds the original borough totals. Because area weighting with uniform within-tract allocation conserves totals over a complete and exactly aligned spatial partition, this systematic component cannot be explained by tract splits or the uniform-allocation assumption alone; it instead reflects imperfect geometric

alignment between digitized historical boundaries and 2010 tract boundaries—overlapping or non-nested polygons and shoreline differences—which can double-count small areas of overlap. Second, the deviations are small relative to borough populations in the millions, but they are most relevant when interpreting long-run neighborhood trends and estimating historical baseline conditions.

As discussed in Logan et al. (2021), spatial harmonization accuracy depends critically on the nature of boundary changes, with larger errors expected for non-nested, irregular transitions. While the area-weighting method provides a feasible approach to creating a tract-level panel, it may understate heterogeneity in neighborhoods with high internal variation or complex administrative histories.

Racial decomposition. The White, Black, and Hispanic series do not exhaust tract population. Asian residents and smaller racial categories are reported inconsistently across the historical Census and NYCHA sources and are grouped into a residual Asian/Other category. A comparable eight-decade net series cannot therefore be constructed for this group. Growth in the residual category, especially in later decades, can allow total population to recover even while privately housed White and Black populations decline. The group-specific estimates should not be interpreted as an exhaustive or symmetric decomposition of tract population.

The net racial outcomes combine harmonized tract-level Census counts with project-level public-housing resident counts. In a small number of post-treatment project–tract cells, the project count exceeds the harmonized tract count for the same racial group. This can make the total and net White series overlap or appear mechanically reversed in a few long-run cells. I therefore interpret the total–net gap as an accounting decomposition rather than as a separate causal estimate of changes among public-housing residents. This measurement issue does not alter the finding that privately housed White residents decline relative to the comparison group.

B5 Newspaper Data

The rental price data used in this paper were collected from the real estate sections of the New York Times (NYT) for each decennial year between 1930 and 2010. The goal was to identify approximately 1,500 valid rental listings per sampled year. Sampling focused on the last Sunday of each month—when real estate listings were most abundant—across all 12 months. I sampled across all columns in the real estate section so that all areas covered by the newspaper would be included in the sample. Then, I randomly selected listings that satisfied three criteria: (1) the listing included a rental price, (2) it specified size (in rooms or bedrooms), and (3) it provided either the exact street address (with housenumber) or an intersection. If a sufficient number of qualifying listings could not be found for a given month, assistants consulted Sunday editions from adjacent months. An example of a typical NYT rental page is shown in Figure B5.1.

In the next step, addresses from NYT rental listings were geocoded using the Google Geocoding API. Listings were retained only if the geocoder successfully matched either an exact street address or a recognizable intersection (main and cross streets). Listings with incomplete or ambiguous location information were excluded. This procedure yielded a

Figure B5.1: Example of a Rental Listings



8 W APARTMENTS Sunday advertisements must be ordered before 2 P. M. Saturday. THE

Apartments Furnished—Manhattan

Apartments of Three, Four, Five Rooms

Continued From Preceding Page

70'S (Park-Madison)—Sublet most unusual studio apartment, 4 rooms, 2 baths, beautifully furnished in the modern manner; dining gallery overlooks spacious duplex living room; cool, quiet. RHinelander 4-6437.

70'S, EAST—Three rooms, southern exposure, cross-ventilation; to October; reasonable. Sholl.
Albert B. Ashforth, Inc., 12 E.44 MU.2-1100

70'S, Lower (off Central Park West)—4½ rooms; quiet; cool; beautifully furnished; 3 exposures; piano; May-October, longer; reasonable. ENdicott 2-2045.

70'S (lower, off Central Park West)—Sublet 4½ rooms till October, 1941; 3 exposures, spacious; $90. ENdicott 2-2045.

70'S, WEST—Unusual, beautiful, large 3 rooms; reasonable right party. TRafalgar 7-6848.

70'S, WEST—4 (2 Bedrooms), Elegantly Furnished! Outstanding Bargain; $80. TRafalgar 4-3156.

71ST (240 West End) (14B)—Three large, cool rooms, kitchen, exquisite, completely furnished; accommodating 2 or 4 persons.

71ST, 325 WEST—Living room, bedroom, complete kitchen, bath; elevator; telephone; reasonable.

72D, EAST—Sublet charmingly furnished 3½ (16th floor); outside dinette-kitchen; unobstructed; cross-ventilation; October-longer. RHinelander 4-8785.

72D, 405 EAST—3 large rooms; May 15 occupancy. RE. 7-4772, afternoons or evenings; also unfurnished.

72D, EAST—4½ rooms, 2 baths, June-Oct., $100. BUtterfield 8-8882, Monday.

72D, 124 WEST—Luxurious 3 rooms, 3 exposures, surrounding terrace, $125. Supt.

73D, 28 EAST—Summer bargain. Attractively furnished, 3 large room, bath, kitchen, dinette, extra toilet; cross draft; new elevator building. Waterman, ELdorado 5-6900.

73D, 210 EAST (9C)—Furnished 4 rooms, 2 baths; 9th floor; East Village; sublet Summer. Kahn, BUtterfield 8-3886.

73D, 105 WEST—Modern elevator, 3 rooms, large kitchen, unobstructed view; lease, $65. SU. 7-5554, 5A.

73D, 40 WEST—Airy 3-room apartment, attractively furnished, bath, kitchenette, frigidaire.

75TH ST., 167 EAST—Cooperative building; living room, bedroom, dressing room and bath; fireplace; well furnished; $100; excellent meals served if desired; selective tenantry. Apply Mrs. Brown, premises. RHinelander 4-0550.

Apartments Furnished—Manhattan

Apartments of Three, Four, Five Rooms

187TH, 825 WEST (overlooking Hudson River—Sublet 4 rooms, furnished, with antiques, unusual view, reasonable rental. See Supt.

189TH, 701 WEST (Broadway) (5E)—Sublet, 3½ modern rooms, cozy, elevator; $55.

BEEKMAN PLACE
Attractive Summer Rental
4½ rooms, master bedroom, paneled nursery or library, connecting baths, large living room, dining foyer, kitchen, cross-ventilation; East River view; immediate occupancy until October, by appointment. Telephone PLaza 8-0349. REASONABLE RENTAL.

BEEKMAN PLACE, 32—Sublease now until October, attractive 3 rooms, cool, smart, river view. Sacrifice. Call PLaza 5-3155.

CABRINI BLVD., 160 (Castle Village) (Apt. 36)—Artist's beautifully decorated 4½-room apartment facing Hudson; sublet furnished-unfurnished; immediate occupancy to October. WAdsworth 7-2816.

CENTRAL PARK WEST (1 West 85th)—Five rooms, 7 windows overlooking park; sacrifice; piano, radio; complete. TRafalgar 7-5321.

CENTRAL PARK WEST—Attractive 4 light rooms, 2 baths, cross ventilation; June-October; references. TRafalgar 7-4378.

CENTRAL PARK WEST, 25—5 rooms to sublet till October; attractive furnishings. Apply renting office, Century Apartments.

CLAREMONT AVE., 191 (Columbia University)—Delightful 4 rooms, elevator, piano; $38. Hinkins.

EAST END AVE.—Sublet to October; 3 rooms, bath, kitchen, cross-ventilation; top floor, walk-up; worth it for gorgeous view across Carl Schurz Park and miles of East River; also roof privileges; $50. REgent 7-4152.

GRAMERCY SECTION, modern building, 3½ rooms, beautifully furnished, roof garden, maid service; 15th floor; sacrifice. GRamercy 3-4667, or office, STuyvesant 9-3884.

GRAMERCY PARK—Unusually attractive apartment; beautifully furnished 5 rooms, magnificent view; 3 exposures; 12th floor; $150. GRamercy 5-4029.

GRAMERCY PARK, 3—3½ rooms, new; 3 exposures; all housekeeping appliances, radio; 4 months lease. Supt.

GRAMERCY PARK NORTH, 60—3-4, modern, twin beds, dinette. GRamercy 3-7399.

GREENWICH VILLAGE—Comfortable, attractive sublet, elevator building, 3 rooms, 2 exposures; $65.

Apartments Furnished—Manhattan

Apartments of Six Rooms and Over

RIVERSIDE DRIVE in 80's: 7, outside rooms, 3 baths, 10th floor, marvelous views, grand piano, 4 radios, television; rent for Summer or longer; $185 per month. TRafalgar 4-0208.

RIVERSIDE DRIVE—Sacrifice 6 rooms, new Steinway grand; river view. Algra Realty Corp., 70 Pine St.

FURNISHED beautiful Park Ave. 6-room apartment, real home, 3 baths, 35-ft. living room, den, dining, kitchen, 2 master bedrooms, maid's room; best transportation everywhere; for 3 to 5 months to responsible party; $300. W 486 Times.

EXCEPTIONALLY well-furnished apartment, 6 rooms, all 4 exposures; $150 monthly until October or longer. SUsquehanna 7-1745.

MUSICIAN sublet furnished 6 rooms, grand piano; beautiful river view. MOnument 2-3726.

Penthouse—Terrace Apartments

Furnished

18T TO 5TH AVES.—EAST-WEST.
Mrs. Ida Catlin, 1049 Lexington Ave.
PENTHOUSE SPECIALIST.
Unfurnished-furnished and apartments; many attractive offerings; excellent values; mail orders; state requirements; immediate or October. REgent 4-1400.

5TH AVE.—24TH FLOOR—DUPLEX
40' Salon, 4 Masters, 5 Baths. Landscaped Terraces. Also Unfurnished. BUtterfield 8-9430.

28TH, 137 EAST—Sublet approximately May 25, Sept. 25; unusually distinctive 3½ rooms, modern penthouse; large terrace; reasonable to suitable tenant. CAledonia 5-5366.

40TH (between Madison and 5th)—Deluxe duplex penthouse apartment; living room, dining room, kitchen and pantry; maid's room and bath; 2 master bedrooms, 3 baths, terrace; handsome antique furnishings; $600 month; occupancy April 30. Call for appointment. MUrray Hill 3-5245.

50'S, EAST—Unusual penthouse; studio living room, 2 master bedrooms, 3 baths, dining room, bar, kitchen, pantry, maid's room, fireplaces; terrace surrounds apartment; furnished by decorator; also unfurnished. PLaza 3-5058.

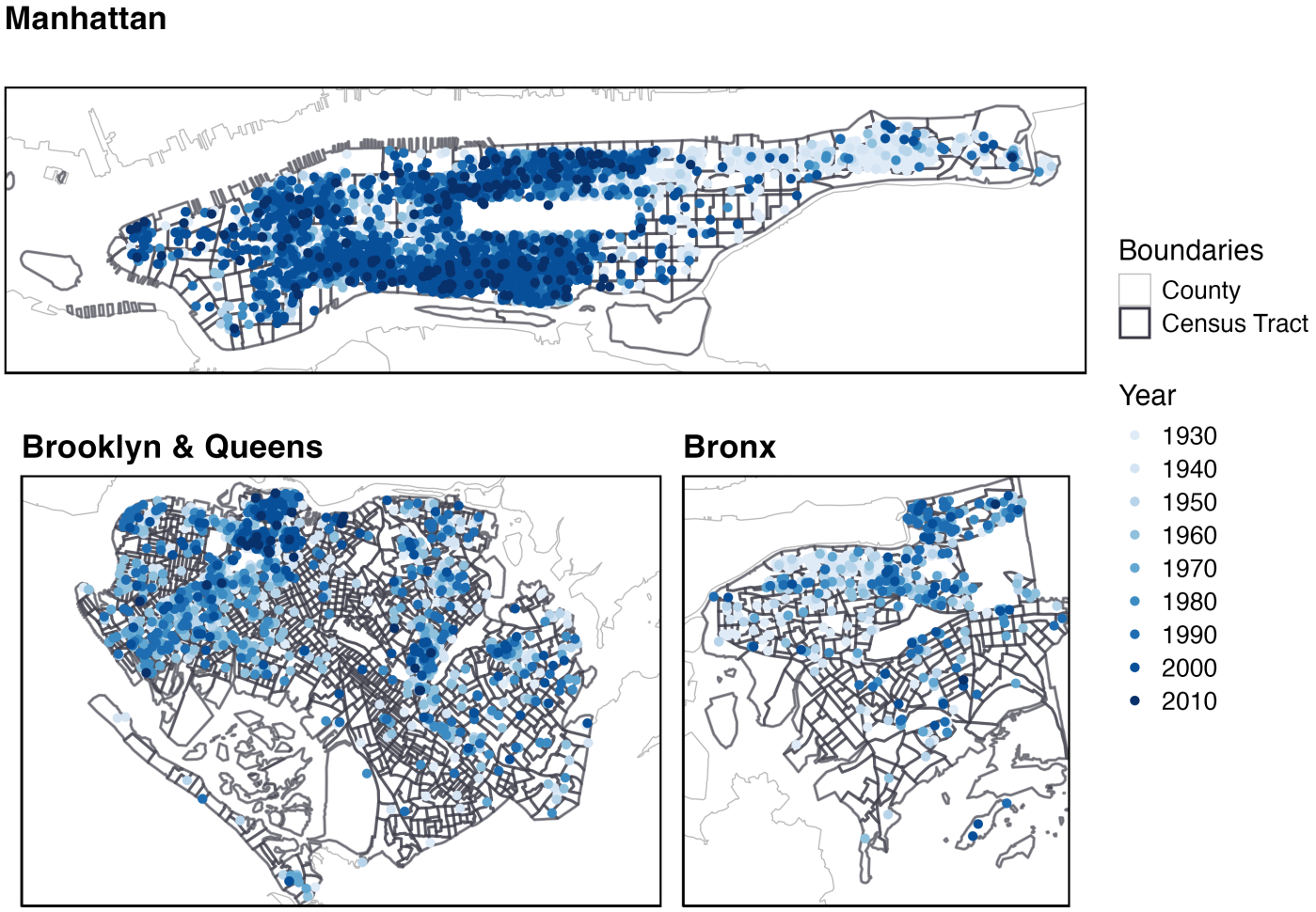
50'S, EAST—River view; 5-6 room duplex, with roof terrace, large living room, separated by plate glass partition from dining room; grand piano; furnished-unfurnished. Weekdays, Chace, PLaza 3-1700.

50'S (Sutton Place Vicinity)—River View, Garden Living Room; three rooms, bath; exquisite furnishings. $135. Immediate. PLaza 3-5572.

Note. A sample real estate listing from the NY Times in 1940.
Source. NYT 28.04.1940.

final sample of 19,463 rental listings across nine decennial years. Figure B5.2 documents the spatial coverage of the listings data across New York City by decade. Coverage is broad but not uniform across either tracts or years, which is why the tract-level analysis relies primarily on the census rent measure, while the listings provide a separate measure of private-market rents.

Figure B5.2: Spatial Distribution of Rental Properties



Note: Figure B5.2 shows the spatial distribution of geocoded rental listings from the *New York Times* real estate section across New York City's five boroughs. Each point represents a rental property, shaded by decade from 1930 to 2010. Boundaries correspond to census tracts (dark grey) and county (light grey) lines.
Source. NYT Real Estate Section. Details on the construction of the data set can be found in Section 3 and Appendix B5.

Table B5.1 summarizes rent levels from both U.S. Census tract data and NYT rental listings. Average rents increase steadily over time, with NYT-listed rents consistently exceeding census-based averages. This likely reflects the fact that NYT listings capture asking rents for advertised units — often skewed toward higher-end properties — whereas the census reports contract rents for existing tenants (i.e., sitting rents). Across all years, tracts with public housing exhibit lower average rents; this is a descriptive level comparison—public housing was placed in lower-rent areas—and does not by itself identify an effect of public housing on rents. While total rents rise sharply across decades, rent per room increases more gradually, suggesting improvements in unit size or quality. Among all measures, the strongest cross-source similarity is found between census average rents and NYT median rent per room. This suggests that per-room measures from NYT listings provide the most comparable counterpart to tract-level census rents. Finally, the number of listings is largest in 1930, reflecting intentional oversampling to ensure strong baseline coverage across tracts. Coverage is thinnest in 2010, when only 111 listings met the inclusion criteria, as classified rental advertising had largely migrated out of the printed real-estate section by then.

Table B5.1: Rent Price Statistics

	US census		NYT rental listings					
Year	Avg. rent	Avg. rent no p.h.	Med. Rent	Med rent p.r.	Avg. rent	Avg. rent p.r.	Avg rooms	Obs
1930	808.22	836.31	1358.23	507.88	1876.96	712.49	3.63	8845.00
	(2505.53)	(2637.84)			(1867.17)	(771.46)	(2.51)	
1940	605.45	618.01	1168.39	389.46	1840.45	615.18	3.64	1561.00
	(185.39)	(185.94)			(2386)	(1121.86)	(2.28)	
1950	342.83	346.55	1076.93	434.39	1332.28	511.38	2.99	1572.00
	(174.18)	(179.16)			(1408.2)	(357.42)	(1.51)	
1960	431.71	441.39	1053.66	324.20	1290.01	390.27	3.46	1629.00
	(164.9)	(167.46)			(756.67)	(187.88)	(1.34)	
1970	574.78	587.08	1826.87	615.51	2059.32	774.81	3.37	1428.00
	(226.33)	(233.7)			(1155.11)	(558.26)	(1.72)	
1980	576.6	593.11	1571.23	741.12	1881.81	1000.77	2.8	1498.00
	(174)	(173)			(1440.91)	(845.29)	(1.98)	
1990	806.78	839.85	1919.01	1251.53	2209.35	1298.46	2.48	1572.00
	(237.28)	(223.46)			(1220.93)	(795.81)	(2.12)	
2000	864.51	901.53	3229.70	1551.52	3741.84	1849.6	2.94	1247.00
	(304.1)	(293.09)			(2574.13)	(1226.25)	(2.1)	
2010	1132.89	1184.11	2700.00	1297.50	3483.18	1612.85	3.24	111.00
	(339.46)	(308.38)			(2584.45)	(1161.03)	(2.41)	

Note. Table B5.1 presents descriptive statistics for rent prices using data from the U.S. Census and New York Times (NYT) rental listings, with all prices deflated to 2010 dollars. Columns (2) and (3) report the average of census tract-level median contract rents; column (3) excludes tracts with public housing. Columns (4) and (5) show the median rent and median rent per room from NYT listings. Columns (6) and (7) report the average rent and average rent per room. Column (8) gives the average number of rooms per listing, and column (9) shows the number of listings per year. Standard deviations are reported in parentheses.

While Table B5.1 compares raw levels, I further assess the consistency of rental price trends across data sources by constructing rent indices from NYT listings and comparing them to an index based on U.S. Census tract-level median contract rents. To ensure comparability, all series are normalized to 1930 (=100). Because the NYT listings reflect advertised asking rents, which may vary by unit quality, location, and timing, I adjust for compositional changes over time including unit characteristics such as included utilities, furnishings, and parking availability, along with census tract and seasonal fixed effects. I

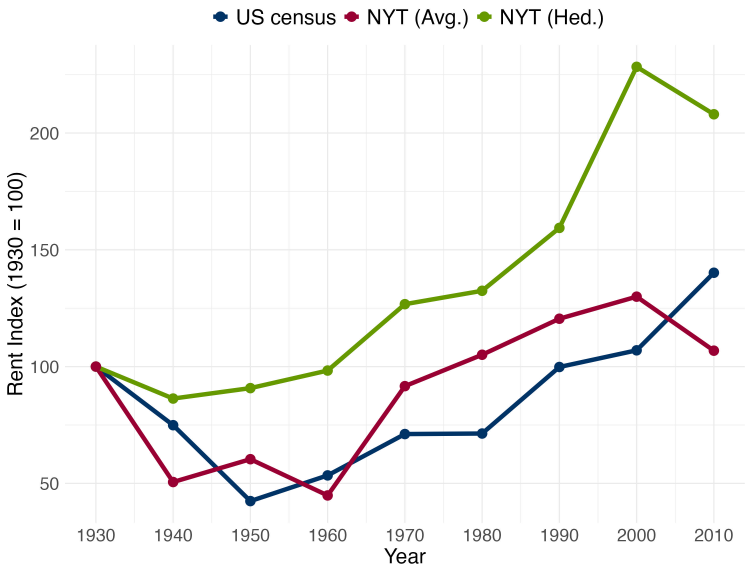
estimate the following hedonic pricing model that adjusts for observable unit characteristics and location-specific factors:

$$\log(rent_{ikmt}) = \alpha + \sum_{t=1940}^{2010} \beta_t \mathbb{1}(t = year) + \mathbf{X}_i'\boldsymbol{\delta} + \gamma_r + \gamma_k + \gamma_m + \epsilon_{ikmt} \tag{B.2}$$

The coefficients β_t capture changes in average advertised rents over time, relative to 1930. The vector of controls $\mathbf{X}_i$, with coefficient vector $\boldsymbol{\delta}$, includes indicators for unit characteristics that were consistently reported in the listings: whether gas, water, electricity, or air conditioning were included in rent; whether parking was available; and whether the unit was furnished. The specification also includes room fixed effects γ_r, census tract fixed effects γ_k, and calendar month fixed effects γ_m to account for differences in unit size, location, and seasonality.

Figure B5.3 displays the resulting rent indices. The census-based rent index shows a 45 percentage point decline until 1950, followed by a steady increase through 2000 and a slight uptick in 2010. The average rent per room index from NYT listings closely tracks this trend—declining in the 1940s and 1950s, then rising in subsequent decades. The NYT-based hedonic index exhibits a similar trajectory: a modest dip in the early period, strong growth through 2000, and a modest decline in 2010. Most importantly, the trends across both data sources run nearly parallel from 1950 to 2000. This suggests that, even after accounting for sample differences, both NYT listings and census rents provide a reliable signal of long-run rent dynamics.

Figure B5.3: Rent Indices



Note. Figure B5.3 plots the rent indices from two sources. The blue line shows the average of census tract-level median contract rents. The red line shows the average rent per room from the New York Times listings. The green line shows the hedonic index implied by the year coefficients β_t from Equation B.2, converted to an index with 1930 = 100, estimated using NYT rental listings and controlling for property characteristics, census tract fixed effects, and month-of-year dummies. All three series are indexed to 1930 (=100) for comparability. All rent prices were deflated to 2010 dollars using the CPI.

C Extended Results

C1 Balance in Pre-Treatment Characteristics

This section reports the comparisons of pre-treatment characteristics referenced in Section 4. The sample consists of the 166 treatment-site stacks first treated before 1970 that form the main estimation sample. All characteristics are measured in the last census before a tract's first public-housing completion (event time $\tau = -10$), the same baseline decade that serves as the omitted period in the event-study estimates. Table C1.1 reports two complementary diagnostics.

Panel A compares tracts across rings *within* treated-tract stacks, mirroring the identifying comparison in Equation 1: each characteristic is regressed on indicators for the public-housing tract and the first ring, with stack fixed effects, weighted by tract frequency within ring as in the event-study estimates, and with standard errors clustered by stack. The omitted category is the second-ring comparison group. Future public-housing tracts were somewhat denser, had lower rents and household incomes, and housed larger Black and Hispanic population shares than their outer-ring comparison tracts; the first ring sits between the two. Most differences are statistically significant but modest relative to the citywide gradients in Panel B: for example, baseline median contract rent is about 10 percent lower in the public-housing tract than in its comparison ring, against a 26 percent gap between treatment-site areas and the rest of the city. The joint tests reject exact balance, which is consistent with sites being selected rather than randomly assigned. As discussed in Section 4, the design does not require identical pre-treatment levels; stack-specific baseline-outcome controls absorb level differences, and the event-study leads in the main figures test for the differential pre-trends that would threaten identification.

Panel B compares treatment-site areas—the public-housing tract and both surrounding rings, observed in their baseline census—with all other New York City tracts observed in the same census years, using census-year fixed effects and standard errors clustered by tract. Treatment-site areas were markedly denser, poorer, more renter-occupied, and less White than the rest of the city, confirming that public housing was placed in selected neighborhoods. This selection motivates the ring design: the outer-ring comparison tracts resemble the treated tracts far more closely (Panel A) than the average city tract does (Panel B), and the monotone gradient from the public-housing tract to the second ring supports using nearby rings, rather than the rest of the city, as counterfactuals.

Table C1.1: Balance in pre-treatment characteristics

Panel A: Within project stacks (baseline census)				
	2nd ring mean	PH tract − 2nd ring	1st ring − 2nd ring	Obs.
Population (1,000s)	4.46	0.64**	-0.04	2,138
		(0.32)	(0.16)	
Pop. density (1,000s per km^2)	25.35	2.71*	0.16	2,138
		(1.62)	(0.80)	
Share White	0.87	-0.05***	-0.02***	2,138
		(0.02)	(0.01)	
Share Black	0.12	0.05***	0.02**	2,138
		(0.02)	(0.01)	
Share Hispanic	0.03	0.02***	0.01***	2,042
		(0.01)	(0.00)	
Housing units (1,000s)	1.47	0.17	-0.04	2,138
		(0.10)	(0.05)	
Share in owner-occupied housing	0.19	-0.03***	-0.01*	2,138
		(0.01)	(0.00)	
Median contract rent (100s, 2010$)	4.21	-0.44***	-0.25***	2,098
		(0.11)	(0.07)	
Household income (1,000s, 2010$)	22.00	-1.60*	-1.41**	2,002
		(0.96)	(0.56)	
Joint test (p-value)		0.039	0.005	
Projects		166		

Panel B: Project areas vs. rest of New York City				
	Rest-of-NYC mean	PH tract − NYC	1st ring − NYC	2nd ring − NYC
Population (1,000s)	2.54	2.69***	1.63***	1.54***
		(0.33)	(0.17)	(0.17)
Pop. density (1,000s per km^2)	13.14	13.41***	9.27***	8.16***
		(1.78)	(1.24)	(0.86)
Share White	0.98	-0.19***	-0.16***	-0.09***
		(0.02)	(0.02)	(0.01)
Share Black	0.01	0.19***	0.16***	0.09***
		(0.02)	(0.02)	(0.01)
Share Hispanic	0.01	0.07***	0.03***	0.02***
		(0.01)	(0.00)	(0.00)
Housing units (1,000s)	0.63	0.82***	0.47***	0.50***
		(0.11)	(0.05)	(0.05)
Share in owner-occupied housing	0.42	-0.27***	-0.21***	-0.19***
		(0.02)	(0.01)	(0.01)
Median contract rent (100s, 2010$)	6.10	-1.57***	-1.29***	-0.82***
		(0.16)	(0.15)	(0.15)
Household income (1,000s, 2010$)	26.40	-11.01***	-8.80***	-5.29***
		(0.82)	(0.55)	(0.50)
Joint test (p-value)		0.000		
Observations		5,168		

Note: All characteristics are measured in the last census before first treatment (event time $\tau = -10$). Panel A reports, for each characteristic, the weighted mean in the second-ring comparison group and the coefficients from a regression of the characteristic on indicators for the public-housing tract and the first ring, with stack fixed effects; estimates are weighted by tract frequency within ring, as in the event-study estimates, and standard errors (in parentheses) are clustered by stack. Panel B pools the baseline observations of all treatment-site-area tracts (public-housing tract, first ring, and second ring) with all other New York City tracts observed in the same census years and reports coefficients on the ring indicators from unweighted regressions with census-year fixed effects; standard errors are clustered at the tract level. Tracts assigned to several stacks enter Panel B once per baseline census year, with their closest ring. The joint test reports the p-value of a Wald test that all standardized characteristics jointly predict the corresponding group indicator (Panel A: ring versus second ring within stacks; Panel B: any treatment-site-area ring versus the rest of the city). Population, race, and owner-occupancy variables are person counts from the harmonized census extracts; rent and income are expressed in 2010 dollars. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C2 Heterogeneity by Construction Period

One of the paper's main claims is that the neighborhood response to public housing depends on the historical form in which it was built. The main sample contains 166 public-housing tracts first treated before 1970. The developments generating these first treatments were part of a mid-century program of slum clearance, large-site redevelopment, and high-density construction, implemented under comparatively strict tenant screening and often explicit racial allocation. By the end of the 1960s, that institutional setting was changing. NYCHA relaxed its admission rules, legal challenges weakened race-based and moral screening, federal support became less secure, and political opposition to large tower developments encouraged smaller and lower-density projects (Bloom, 2008; Jacobs, 1961; Newman, 1997). The year 1970 is therefore useful as a marker between two historically distinct construction regimes, even though it should not be interpreted as a sharp causal cutoff.

The projects themselves reflect this transition. Table B2.1 shows that median project size fell from 1,156 apartments for projects completed in 1941–50 and 1,246 in 1951–60 to 441 in 1961–70, 221 in 1971–80, 189 in 1981–90, and 51 after 1990. Building form, construction expenditure, and initial tenant composition changed at the same time, while the mean baseline White population of project tracts fell from approximately 3,843 for the pre-1970 cohort to 2,323 for later projects. Moreover, Section B3 shows that pre-construction neighborhood characteristics predict project scale, design, and opening tenant composition differently across the two cohorts. The empirical question is consequently not whether an otherwise identical project has a different effect after an arbitrary date. It is whether the bundled intervention implemented after 1970 produced a different neighborhood response, and how much of that difference is associated with observable changes in what was built and for whom.

I answer this question by pooling both cohorts and estimating a windowed version of the main event study. Let $P_p = \mathbb{1}\{Y_p \geq 1970\}$ identify the later cohort, and let D_{rhpkt} indicate that tract k lies in ring $r \in \{T, 1\}$ around treatment site p and is observed in event-time window h. The pretrend window covers 40 to 20 years before completion and the post-construction window covers 0 to 40 years after completion; the latter is the longest common post-treatment support available for sites first treated in 1970 or later. The implemented regression is

$$\begin{aligned} y_{kmpt} = &\sum_{r\in\{T,1\}} \sum_{h\in\{\text{pre,post}\}} D_{rhpkt} \left(\beta_{rh} + \theta_{rh} P_p + \widetilde{\mathbf{A}}_p' \boldsymbol{\gamma}_{rh}\right) \\ &+ \boldsymbol{\delta}_p' \mathbf{Z}_{kt} + \lambda_p y_{kp}^0 + \rho_{pmt} + \varepsilon_{kmpt}. \end{aligned} \tag{C.1}$$

The coefficient β_{rh} is the pre-1970 response in ring r and window h, while θ_{rh} is the difference for sites first treated in 1970 or later. The implied post-1970 response is therefore $\beta_{rh} + \theta_{rh}$, for which I report delta-method standard errors. In Panel A of Table C2.1, the attribute term is excluded, so $\boldsymbol{\gamma}_{rh} = 0$. Panel B includes the demeaned vector $\widetilde{\mathbf{A}}_p$—initial White, Black, and Hispanic tenant shares and maximum recorded building height—interacted with each ring-by-window indicator. Because the attributes are demeaned, β_{rh} in Panel B describes a treatment site with mean observed attributes. The second ring and all treated-ring observations for which neither reported window indicator is active form the omitted category; under this specification, this includes the decade immediately before com-

pletion and event times outside the two windows. Population and housing outcomes enter in levels, private outcomes subtract public-housing residents or units, and rents enter in logs. Stack-by-neighborhood-by-year fixed effects ρ_{pmt}, stack-specific infrastructure controls $\mathbf{Z}_{kt}$, stack-specific baseline slopes, ring-frequency weights, and stack-clustered standard errors follow Equation 1. Since project attributes were not randomly assigned across neighborhoods, Panel B is a conditional comparison rather than a causal decomposition of design and historical regime.

Panel A recovers the main paper's replacement pattern for the early regime. During the first four decades after completion, treatment raises total population in a pre-1970 public-housing tract by about 1,195 residents but reduces private population by 417; total housing units rise by 357 while private units fall by 423. Administrative placement and private-market response similarly move in opposite directions across racial groups: total Black and Hispanic populations increase by 685 and 498, whereas privately housed White and Black populations decline by 472 and 439. Census contract rent is essentially unchanged.

The post-1970 interactions reveal a different quantity response. Relative to the early cohort, the differences are −975 residents in total population, +821 in private population, and +539 in private units. The corresponding implied later-cohort effects—219, 404, and 116—are all statistically indistinguishable from zero. Total units nevertheless rise significantly by about 335, consistent with later projects adding subsidized units without detectable displacement of private housing. The later-cohort interaction also nearly eliminates the early private-Black decline, leaving an implied effect of −51. Prices move differently: Census rent falls by 0.150 log points, whereas estimates from the smaller New York Times sample are imprecise.

Panel B shows that observed project attributes absorb much of the demographic contrast but not the rent result. The cohort difference falls from +821 to −13 for private population and from +539 to an insignificant +190 for private units; the total-Black difference also becomes insignificant. Thus, later projects appear to have generated a more muted demographic response partly because the developments differed in built form and served different initial tenant populations; project size is not among the Panel B attributes, so the role of scale is not separately identified in this decomposition. Yet the Census-rent difference becomes −0.215 log points, leaving a precisely estimated implied later-cohort effect of −0.183; the implied private-Hispanic effect is also negative at roughly −375. First-ring estimates are small, but several pretrend coefficients are nonzero. The results should therefore be read as a comparison of historically distinct bundles of policy, design, tenant allocation, and location, not as the causal effect of crossing a 1970 cutoff. On that basis, focusing the remaining analysis on the pre-1970 regime avoids averaging its large replacement response with a later regime that added public housing without detectable private displacement, even though rents declined. This interpretation accords with evidence that federally assisted housing built in New York City between 1977 and 2000 did not reduce nearby property values (Ellen et al., 2007).

Table C2.1: Construction-Regime Comparison: Pre- and Post-1970 Effects

	Total pop.	Private pop.	Total units	Private units	White	Priv. White	Black	Priv. Black	Hispanic	Priv. Hispanic	Census rent (log)	NYT rent (log)
Panel A: Regime only												
PH tract, 0–40 yrs (T)	1,194.57***	-417.33*	356.68***	-422.75***	130.08	-471.55**	685.14***	-439.27***	497.80***	-49.70	0.007	-0.101
	(215.27)	(233.65)	(84.80)	(84.42)	(186.12)	(183.98)	(125.23)	(119.54)	(91.41)	(81.09)	(0.041)	(0.097)
× Post-1970	-975.49**	821.06*	-22.18	538.58***	-504.83	39.23	-482.25***	387.88**	-443.76**	-68.71	-0.157***	-0.530
	(383.16)	(466.47)	(155.17)	(155.94)	(308.38)	(307.26)	(168.96)	(163.36)	(173.63)	(167.07)	(0.053)	(0.409)
Implied post-1970 effect	219.09	403.72	334.50**	115.83	-374.75	-432.32*	202.89*	-51.40	54.04	-118.41	-0.150***	-0.631
	(316.97)	(403.74)	(129.95)	(131.11)	(245.88)	(246.09)	(113.41)	(111.34)	(147.62)	(146.06)	(0.034)	(0.398)
First ring, 0–40 yrs (T)	-6.89	33.82	0.36	6.54	-61.52	-62.09	-4.17	-7.47	32.49	33.52	0.022	-0.027
	(113.74)	(102.01)	(38.97)	(39.00)	(89.28)	(89.10)	(54.88)	(54.78)	(35.46)	(35.67)	(0.033)	(0.053)
× Post-1970	11.82	61.59	97.02	90.44	-68.22	-67.03	57.01	57.61	-49.91	-54.35	-0.036	0.074
	(196.29)	(152.69)	(59.48)	(59.16)	(176.32)	(176.20)	(82.02)	(81.91)	(72.46)	(72.45)	(0.045)	(0.117)
PH tract, −40 to −20 yrs (T)	128.00	401.06	53.64	63.06	173.35*	147.83*	15.97	15.48	-50.80	-23.62	0.013	-0.126
	(82.63)	(265.34)	(69.79)	(69.66)	(90.76)	(88.57)	(19.00)	(20.31)	(38.83)	(25.66)	(0.014)	(0.107)
× Post-1970	8.45	570.57	285.33*	281.16*	-121.93	-98.33	-3.32	-10.88	131.91	103.42	-0.013	0.019
	(118.73)	(460.18)	(151.47)	(151.36)	(104.00)	(102.13)	(42.23)	(43.14)	(86.61)	(81.16)	(0.015)	(0.127)
N	24,514	24,514	24,514	24,514	24,483	24,483	24,483	24,483	21,066	21,066	24,008	3,068
R^2	0.838	0.667	0.750	0.750	0.913	0.913	0.903	0.902	0.805	0.803	0.803	0.767
Panel B: With project attributes (demeaned)												
PH tract, 0–40 yrs (T)	1,159.32***	-218.44	405.78***	-312.63***	59.70	-469.64**	708.15***	-407.86***	488.66***	-13.94	0.032	-0.140
	(226.53)	(221.64)	(73.95)	(72.47)	(199.40)	(198.01)	(144.59)	(136.61)	(76.52)	(78.92)	(0.038)	(0.173)
× Post-1970	-834.59*	-13.34	-146.52	189.81	-230.52	-39.98	-363.70	289.37	-695.84***	-361.14*	-0.215***	-0.151
	(445.82)	(464.58)	(168.44)	(165.75)	(427.07)	(427.80)	(221.52)	(206.66)	(187.26)	(186.29)	(0.061)	(0.266)
Implied post-1970 effect	324.72	-231.78	259.26*	-122.82	-170.82	-509.62	344.45**	-118.49	-207.18	-375.08**	-0.183***	-0.291
	(350.62)	(382.49)	(150.93)	(151.28)	(317.19)	(321.40)	(147.82)	(143.52)	(156.84)	(153.25)	(0.050)	(0.225)
First ring, 0–40 yrs (T)	-1.32	111.84	32.53	35.81	-114.86	-113.80	15.15	10.78	50.06	49.11	0.025	-0.162
	(127.94)	(105.55)	(40.42)	(40.45)	(106.87)	(106.84)	(54.48)	(54.53)	(39.43)	(39.99)	(0.030)	(0.131)
× Post-1970	39.50	-30.22	80.80	73.75	33.32	30.73	40.84	41.44	-66.56	-72.05	-0.031	0.181
	(210.67)	(184.91)	(65.26)	(65.40)	(185.16)	(185.12)	(97.47)	(97.29)	(81.18)	(81.40)	(0.048)	(0.161)
PH tract, −40 to −20 yrs (T)	138.62*	524.24**	92.46	117.19*	154.06**	141.10*	17.76	22.97	-33.53	-23.14	0.006	-0.201
	(71.27)	(244.70)	(69.85)	(68.31)	(78.17)	(78.87)	(18.07)	(18.91)	(30.22)	(24.58)	(0.013)	(0.297)
× Post-1970	-124.25	-121.85	237.52	208.39	-224.63*	-218.26*	-29.47	-36.37	90.18	73.58	0.011	0.122
	(135.14)	(533.26)	(187.82)	(186.06)	(131.80)	(132.10)	(36.22)	(35.46)	(81.19)	(75.66)	(0.015)	(0.242)
N	24,514	24,514	24,514	24,514	24,483	24,483	24,483	24,483	21,066	21,066	24,008	3,068
R^2	0.838	0.668	0.750	0.750	0.913	0.913	0.904	0.903	0.806	0.804	0.804	0.768

Note: The table reports coefficients from Equation C.1, estimated on the pooled construction cohorts with a post-construction window covering 0 to 40 years and a pretrend window covering 40 to 20 years before completion. The second ring and treated-ring observations outside these windows form the omitted category. T reports the window effect for sites first treated before 1970; × Post-1970 reports the difference for sites first treated in 1970 or later; the implied post-1970 effect is their sum, with delta-method standard errors. Panel B adds the demeaned project-attribute vector (initial White, Black, and Hispanic tenant shares and maximum recorded building height), so its T coefficients describe a treatment site with mean attributes. Population and housing-unit outcomes are counts in levels; private outcomes subtract public-housing residents or units from the corresponding tract total; rents are in logs. The NYT rent specification uses the smaller listings sample and the Census-rent baseline control. Estimates use ring-frequency weights and stack-clustered standard errors. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C3 Heterogeneity by Income

The effects of affordable housing may depend on the economic conditions of the neighborhood in which it is built. To examine this dimension of heterogeneity, I extend the baseline stacked event study in Equation 1 by allowing treatment effects to vary with the pre-treatment income of the public-housing tract. Each treatment-site stack is assigned the mean real household income of that tract over all pre-treatment census waves. This stack-level measure is held fixed throughout the stack and divided into three groups across the 166 public-housing tracts first treated by 1969. Let $G_p \in \{1, 2, 3\}$ denote the resulting income group, ordered from lowest to highest. I estimate:

$$y_{kmpt} = \sum_{b=1}^{3} \sum_{r \in \mathcal{R}} \sum_{h \in \mathcal{H}} \beta_{r,h}^{(b)} \mathbb{1}\{G_p = b\} \mathbb{1}\{R(k,p) = r\} \mathbb{1}\{H(q_{pt}) = h\} + \boldsymbol{\delta}_p' \mathbf{Z}_{kt} + \lambda_p y_{kp}^0 + \rho_{pmt} + u_{kmpt}, \tag{C.2}$$

Here $\mathcal{R} = \{T, 1\}$ contains the public-housing tract and adjacent first ring, while $\mathcal{H} = \{\text{pre}, \text{med}, \text{long}\}$. Event time q_{pt} is measured in census periods: the pooled pre-period has $q_{pt} < -1$, the medium run has $0 \leq q_{pt} < 4$, and the long run has $q_{pt} \geq 4$. These correspond to at least 20 years before completion, 0–30 years after completion, and 40 to 60 years after completion; $q_{pt} = -1$ is omitted. The second ring is the comparison group.

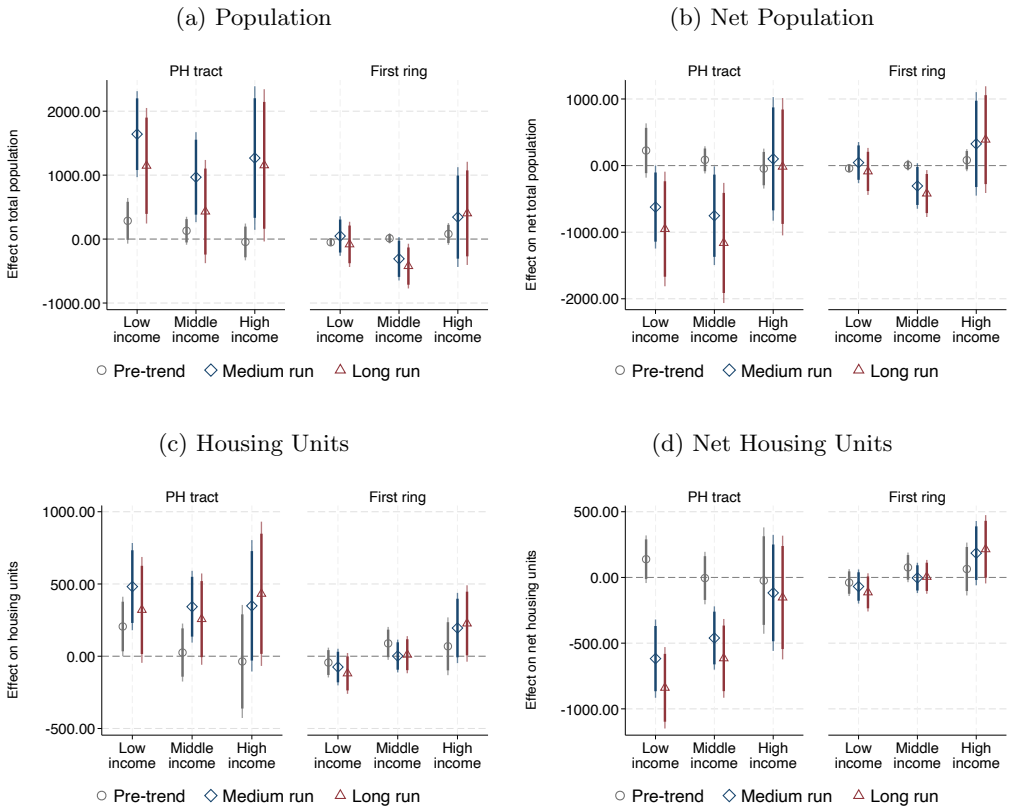
The remaining terms match the baseline specification. The vector $\mathbf{Z}_{kt}$ contains indicators for exposure by year t to urban renewal, an arterial road, and a subway opening, with stack-specific coefficients $\boldsymbol{\delta}_p$. The specification includes a stack-specific slope on the baseline outcome and stack-by-neighborhood-by-year fixed effects ρ_{pmt}; separate stack-year effects are absorbed because each ρ_{pmt} cell lies within a single stack-year cell, making the stack-year indicators linear combinations of the finer effects. For demographic outcomes, the baseline slope uses the corresponding gross outcome in the same scale even when the dependent variable is net of public-housing residents or units. For every rent outcome, it uses baseline log Census rent. Estimation uses years after 1920, stack-level clustering, and the stack-year ring-frequency weights, n_{rpt}/n_{pt}. The coefficients $\beta_{r,h}^{(b)}$ therefore trace the response by distance, horizon, and baseline-income group.

Income is expressed in constant 2010 dollars using the Consumer Price Index. The classification is not based on a single common baseline year: for each treatment site, income is averaged over all available pre-construction census waves. In practice, the contributing income observations come from 1930, 1950, and, for sites treated sufficiently late, 1960; earlier sites therefore contribute fewer pre-treatment waves. The income groups are formed directly among the 166 pre-1970 treatment sites. Rounded to the nearest dollar, the two boundaries are \$7,405 and \$16,387. Because several sites share the lower boundary value, the groups contain 60, 51, and 55 treatment sites rather than exactly equal thirds. The realized income ranges are \$3,657–\$7,405, \$7,477–\$16,387, and \$16,499–\$30,676; mean pre-treatment incomes are approximately \$6,000, \$12,700, and \$21,000 in the low-, middle-, and high-income groups, respectively. The relevant variation is a site's position in this pre-1970 cross-site distribution.

Figure C3.1 reports effects on population and the housing stock. Total population and housing units increase in the public-housing tract across all three income groups, reflecting

the addition of project units and residents. In the long run, gross population increases by approximately 1,100 to 1,200 residents in the lowest- and highest-income tertiles and by about 440 residents in the middle tertile; the corresponding housing-stock increases range from roughly 250 to 430 units. The net outcomes, which subtract public-housing residents and units, reveal a different pattern. In the lowest- and middle-income tertiles, the long-run private population declines by approximately 950 and 1,150 residents, while the private housing stock declines by about 840 and 620 units. The corresponding estimates in the highest-income tertile are close to zero. This pattern is consistent with clearance and redevelopment replacing more of the pre-existing private stock in lower-income neighborhoods. Estimates for the first ring are generally small, as are the pooled pre-treatment coefficients.

Figure C3.1: Heterogeneity by Baseline Income: Population and Housing Stock

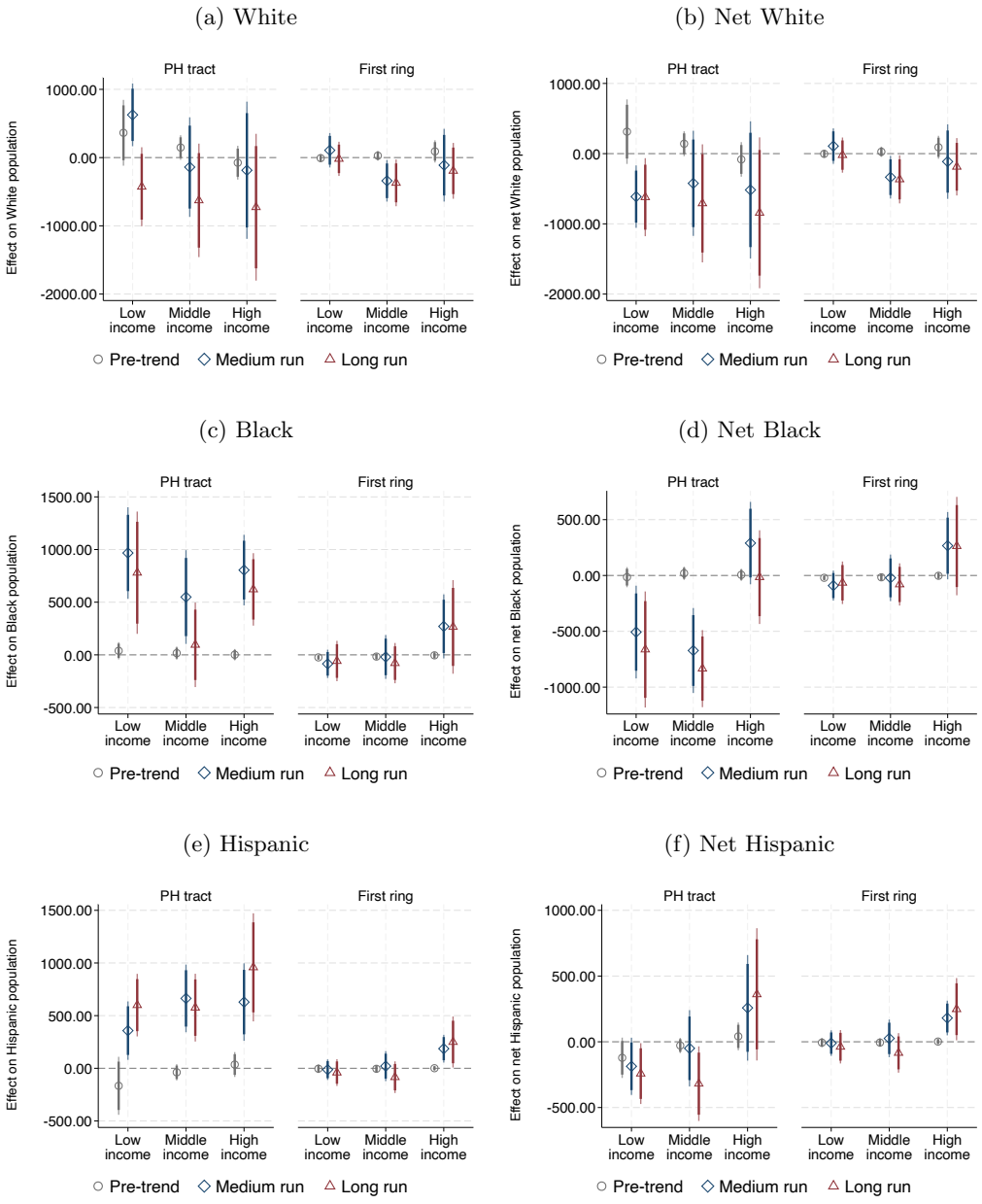


Note: The figure plots level coefficients $\hat{\beta}^{(b)}_{r,h}$ from (C.2), separately by stack-level baseline-income group. Panels (a) and (c) report effects on total population and housing units. Panels (b) and (d) subtract public-housing residents and units, respectively, and therefore report effects on the private population and housing stock. The two panels within each subfigure report the public-housing tract and adjacent first ring relative to the second ring. Markers distinguish the pooled pre-period (at least 20 years before completion), medium run (0–30 years), and long run (40 to 60 years); the census period immediately preceding completion is omitted. Regressions include stack-specific gross-baseline slopes, infrastructure-timing controls, and stack-by-neighborhood-by-year fixed effects. Standard errors are clustered by stack; vertical bars show 90% and 95% confidence intervals. Observations use stack-year ring-frequency weights, and the sample is restricted to sites first treated by 1969.

Figure C3.2 shows that public housing also changed the racial composition of the treated tract across the income distribution. The long-run White population declines in every tertile, both before and after subtracting public-housing residents. The net decline becomes larger with baseline income, increasing in magnitude from approximately 620 residents in the lowest-income group to 700 in the middle group and 850 in the highest-income group. Gross Black and Hispanic populations increase in all three groups. The long-run Black estimates are approximately 780, 95, and 620 residents from the lowest- to the highest-income tertile, while the Hispanic estimates are approximately 600, 570, and 960 residents. After subtracting residents living in public housing, however, the long-run Black and Hispanic effects are negative in the two lower-income groups and positive in the highest-income group. Thus, the addition of minority public-housing residents occurs throughout the income distribution, but the associated change in the privately housed population is most pronounced in initially higher-income tracts. Once again, effects in the first ring are limited.

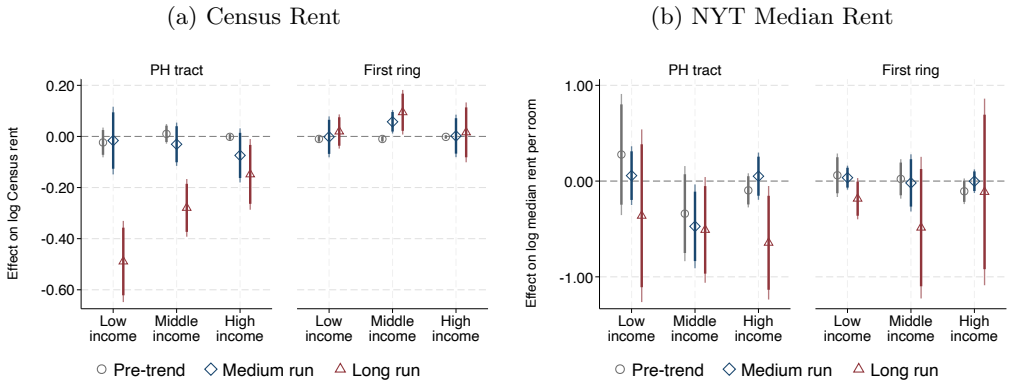
Figure C3.3 reports the corresponding results for rents. In the public-housing tract, census rents decline in the long run across all three income groups, but the magnitude decreases with baseline income. The estimates are approximately -0.49 log points in the lowest-income group, equivalent to a 39% decline, -0.28 log points in the middle group, and -0.15 log points in the highest-income group. The newspaper measure of median rent per room yields similarly negative long-run estimates in the low- and middle-income groups, at approximately -0.48 and -0.65 log points. The corresponding high-income public-housing-tract coefficient is not identified because that ring–horizon–income cell has insufficient listing support. Census-rent estimates in the first ring remain within approximately 0.1 log points of zero. Newspaper estimates are less stable: the low-income first-ring long-run coefficient is -0.35 log points, while other estimates are imprecise. The pooled-pre newspaper coefficients also differ across groups, including a precisely negative estimate in the high-income public-housing tract. The listing-rent gradient should therefore be interpreted descriptively rather than as clean causal heterogeneity.

Figure C3.2: Heterogeneity by Baseline Income: Racial Composition



Note: The figure plots level coefficients $\hat{\beta}^{(b)}_{r,h}$ from (C.2), separately by stack-level baseline-income group. Panels (a), (c), and (e) report effects on the total White, Black, and Hispanic populations. Panels (b), (d), and (f) subtract public-housing residents of the corresponding group and therefore report effects on the privately housed population. The two panels within each subfigure report the public-housing tract and adjacent first ring relative to the second ring. Markers distinguish the pooled pre-period (at least 20 years before completion), medium run (0–30 years), and long run (40 to 60 years); the census period immediately preceding completion is omitted. Regressions include stack-specific gross-baseline slopes, infrastructure-timing controls, and stack-by-neighborhood-by-year fixed effects. Standard errors are clustered by stack; vertical bars show 90% and 95% confidence intervals. Observations use stack-year ring-frequency weights, and the sample is restricted to years after 1920 and sites first treated by 1969.

Figure C3.3: Heterogeneity by Baseline Income: Rents



Note: The figure plots log-rent coefficients $\hat{\beta}^{(b)}_{r,h}$ from (C.2), separately by stack-level baseline-income group. Panel (a) reports the log of median Census rent, and Panel (b) reports the log of median rent per room from New York Times listings. The two panels within each subfigure report the public-housing tract and adjacent first ring relative to the second ring. Markers distinguish the pooled pre-period (at least 20 years before completion), medium run (0–30 years), and long run (40 to 60 years); the census period immediately preceding completion is omitted. Both regressions include a stack-specific slope on baseline log Census rent, infrastructure-timing controls, and stack-by-neighborhood-by-year fixed effects. Standard errors are clustered by stack; vertical bars show 90% and 95% confidence intervals. Observations use stack-year ring-frequency weights, and the sample is restricted to years after 1920 and sites first treated by 1969. The long-run high-income public-housing-tract coefficient in Panel (b) is not estimated because the corresponding listing cell lacks identifying support.

Taken together, the response is qualitatively similar across baseline-income groups: public housing raises gross population and minority counts, reduces the White population, and lowers Census rents within the treated tract. Census-rent declines are largest in the lowest-income neighborhoods, while changes in privately housed racial composition vary across the income distribution. Demographic and Census-rent responses are generally smaller in the first ring, consistent with effects concentrated on the project tract; the sparser newspaper estimates are less uniform.

This pattern differs from Diamond and McQuade (2019), who find that Low-Income Housing Tax Credit developments raise nearby property values in low-income neighborhoods but lower them in high-income neighborhoods. It also contrasts with amenity-based models of income sorting, in which changes in local amenities are capitalized most strongly in the neighborhoods where high-income households locate (Gaigné et al., 2022). Although rents and property values are not directly comparable, first-ring Census rents remain close to zero across all three income groups. The clearest Census-rent gradient instead occurs within the public-housing tract, where rents fall most in the lowest-income group. Large mid-century public-housing projects therefore appear to have changed the housing stock and demographic composition of the tracts in which they were built rather than generating a systematic income gradient across the surrounding neighborhood.

C4 Heterogeneity by Density

Density-Based Heterogeneity. A further dimension of heterogeneity is the density of the pre-existing housing stock, which captures how built-up a neighborhood was immediately before treatment. I calculate housing units per unit of tract area in the public-housing ring in the census period immediately before treatment ($q_{pt} = -1$) and average this measure to one value per treatment-site stack. I then form stack-level tertiles across the 166 public-housing tracts first treated by 1969. The cutoffs are approximately 3,207 and 10,173 housing units per square kilometer (about 13 and 41 units per acre), producing groups of 56, 55, and 55 treatment sites. Mean baseline densities are approximately 1,210, 6,489, and 17,141 housing units per square kilometer in the low-, middle-, and high-density groups.

Let $Q_p \in \{1, 2, 3\}$ denote the resulting density group, ordered from lowest to highest. I estimate the same pooled-horizon specification used for income heterogeneity:

$$\begin{aligned} y_{kmpt} = &\sum_{b=1}^{3} \sum_{r \in \mathcal{R}} \sum_{h \in \mathcal{H}} \beta_{r,h}^{(b)} \mathbb{1}\{Q_p = b\} \mathbb{1}\{R(k,p) = r\} \mathbb{1}\{H(q_{pt}) = h\} \\ &+ \boldsymbol{\delta}_p' \mathbf{Z}_{kt} + \lambda_p y_{kp}^0 + \rho_{pmt} + u_{kmpt}. \end{aligned} \tag{C.3}$$

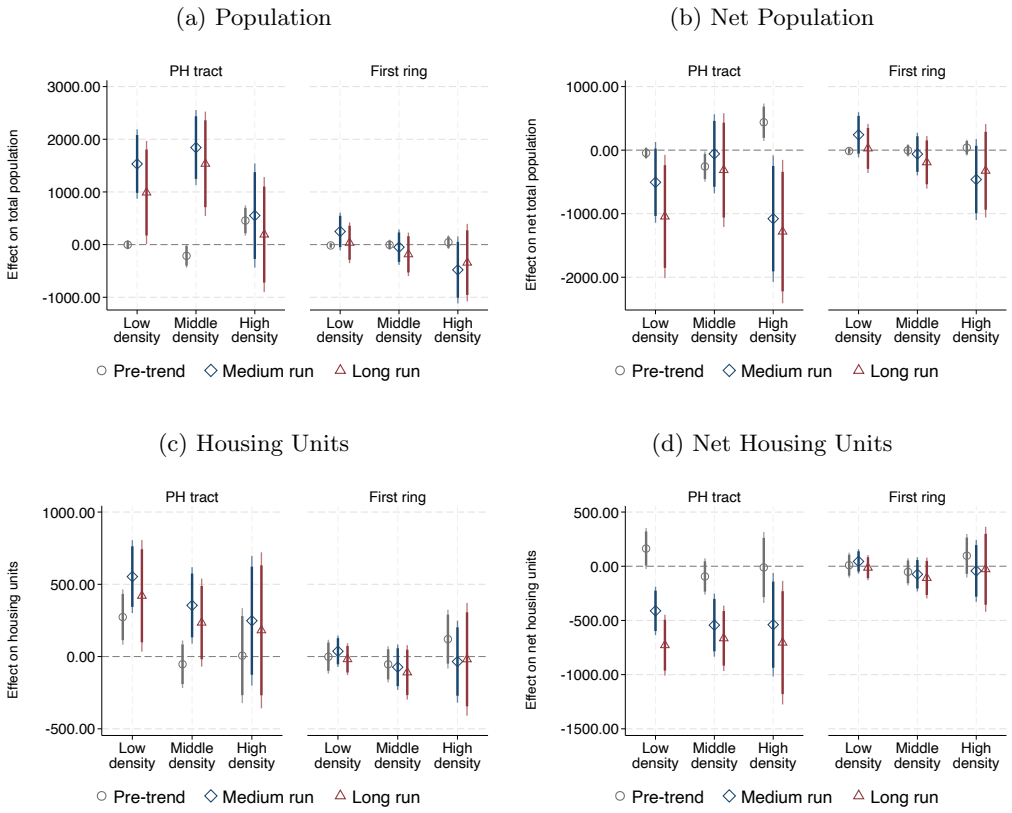
The notation and horizons follow Equation C.2. In particular, $\mathcal{R} = \{T, 1\}$ contains the public-housing tract and adjacent first ring, while $\mathcal{H} = \{\text{pre}, \text{med}, \text{long}\}$ pools census-period event time into $q_{pt} < -1$, $0 \le q_{pt} < 4$, and $q_{pt} \ge 4$; the second ring and $q_{pt} = -1$ are omitted. The vector $\mathbf{Z}_{kt}$ contains the urban-renewal, arterial-road, and subway-timing indicators with stack-specific coefficients. The specification also includes stack-specific baseline-outcome slopes and stack-by-neighborhood-by-year fixed effects. Demographic net-outcome regressions use the corresponding gross baseline in the same scale, while all rent regressions use baseline log Census rent. Estimation uses years after 1920, stack-level clustering, and stack-year ring-frequency weights n_{rpt}/n_{pt}.

The level estimates in Figure C4.1 do not show a uniformly monotonic density gradient. In the long run, total population in the public-housing tract rises by approximately 990, 1,540, and 190 residents across the low-, middle-, and high-density groups. After subtracting public-housing residents, the corresponding estimates are about −1,050, −310, and −1,280. The private housing stock declines in all three groups by similar amounts—approximately 730, 670, and 710 units. Thus, lower baseline density does not consistently predict a larger population or housing-stock response once the tertiles are defined within the pre-1970 cohort.

The racial estimates in Figure C4.2 are likewise non-monotonic. Long-run total Black population rises by roughly 790 and 910 residents in the low- and middle-density groups but is close to zero in the high-density group; private Black population declines in all three groups. Total Hispanic population increases with baseline density, while the private Hispanic estimate is negative in the two lower-density groups and close to zero in the highest group. The long-run private White decline is largest in the high-density group. Moreover, the high-density group exhibits statistically meaningful pooled pre-period coefficients for total and private population and for White population. These patterns warrant interpreting the demographic differences as descriptive heterogeneity rather than clean causal effects of baseline density.

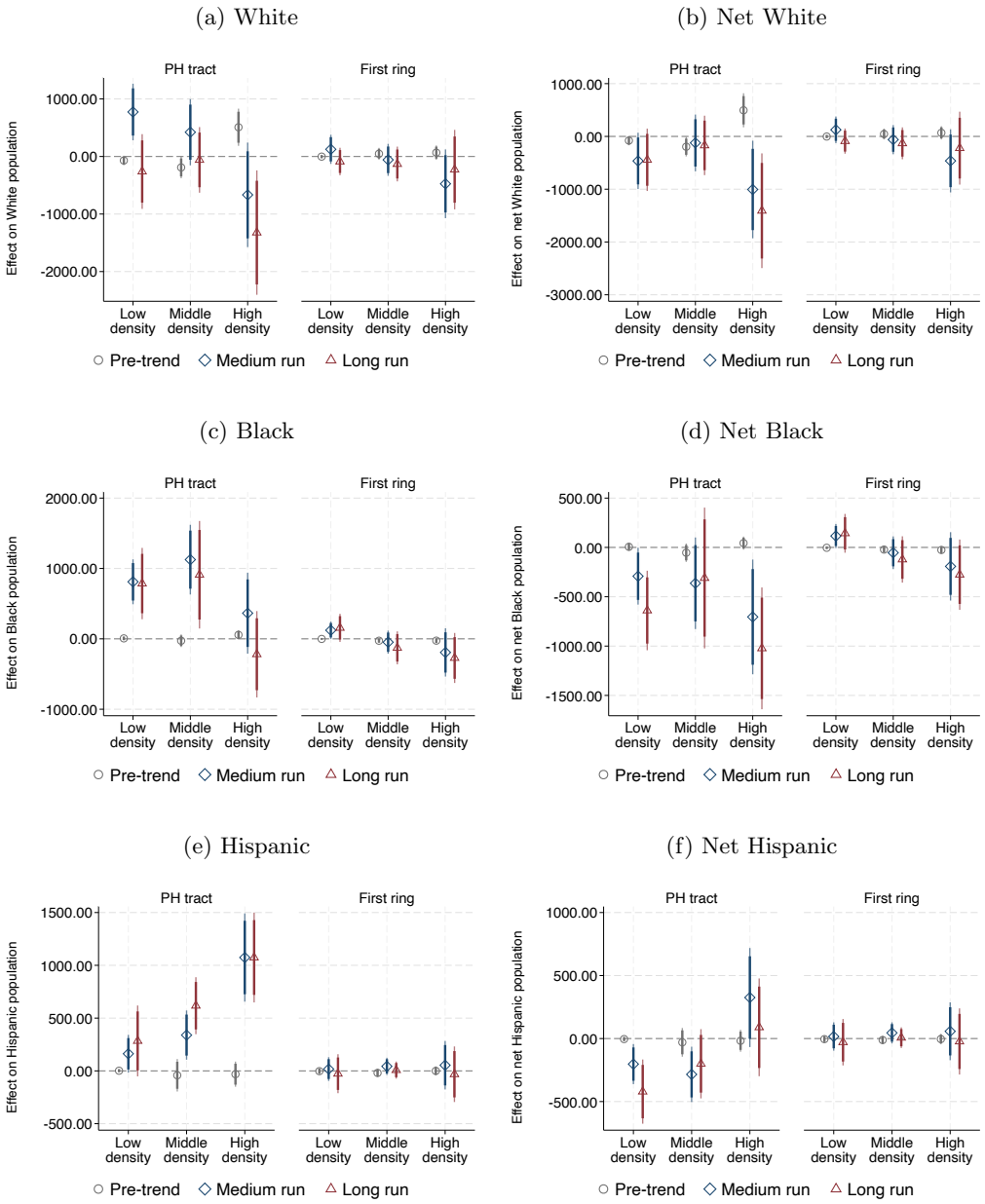
The clearest gradient appears in Census rents (Figure C4.3). Long-run public-housing-tract coefficients become steadily less negative as baseline density rises: −0.52, −0.33, and −0.20 log points. In the first ring they are approximately −0.04, +0.07, and +0.19. This pattern is consistent with a project representing a larger proportional shock where the pre-existing stock is sparse. The New York Times estimates do not provide the same monotonic evidence: the low-density public-housing-tract estimates are large, positive, and imprecise, whereas the middle- and high-density long-run estimates are negative. The pooled-pre low-density project-tract listing coefficient is not estimated. Taken together, the density exercise supports a Census-rent gradient but not a general claim that every demographic and housing-stock response attenuates with baseline density. Conceptually, this margin is best viewed as complementary to treatment intensity rather than as an independently assigned neighborhood characteristic.

Figure C4.1: Baseline Density: Population and Housing Stock



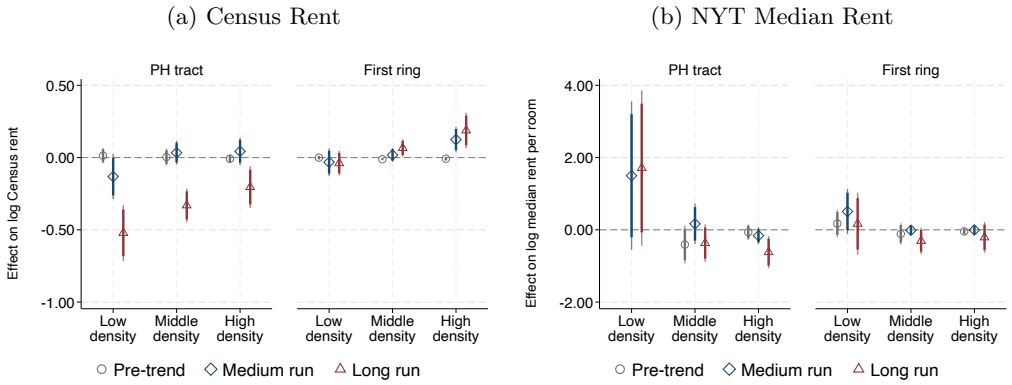
Note: The figure plots level coefficients $\hat{\beta}^{(b)}_{r,h}$ from Equation C.3, separately by stack-level baseline housing-density group. Panels (a) and (c) report effects on total population and housing units. Panels (b) and (d) subtract public-housing residents and units, respectively, and therefore report effects on the private population and housing stock. The two panels within each subfigure report the public-housing tract and adjacent first ring relative to the second ring. Markers distinguish the pooled pre-period (at least 20 years before completion), medium run (0–30 years), and long run (40 to 60 years); the census period immediately preceding completion is omitted. Regressions include stack-specific gross-baseline slopes, infrastructure-timing controls, and stack-by-neighborhood-by-year fixed effects. Standard errors are clustered by stack; vertical bars show 90% and 95% confidence intervals. Observations use stack-year ring-frequency weights, and the sample is restricted to years after 1920 and sites first treated by 1969.

Figure C4.2: Baseline Density: Racial Composition



Note: The figure plots level coefficients $\hat{\beta}^{(b)}_{r,h}$ from Equation C.3, separately by stack-level baseline housing-density group. Panels (a), (c), and (e) report effects on total White, Black, and Hispanic population. Panels (b), (d), and (f) subtract public-housing residents of the corresponding group and therefore report effects on the privately housed population. The two panels within each subfigure report the public-housing tract and adjacent first ring relative to the second ring. Markers distinguish the pooled pre-period (at least 20 years before completion), medium run (0–30 years), and long run (40 to 60 years); the census period immediately preceding completion is omitted. Regressions include stack-specific gross-baseline slopes, infrastructure-timing controls, and stack-by-neighborhood-by-year fixed effects. Standard errors are clustered by stack; vertical bars show 90% and 95% confidence intervals. Observations use stack-year ring-frequency weights, and the sample is restricted to years after 1920 and sites first treated by 1969.

Figure C4.3: Baseline Density: Rents



Note: The figure plots log-rent coefficients $\hat{\beta}^{(b)}_{r,h}$ from Equation C.3, separately by stack-level baseline housing-density group. Panel (a) reports the log of median Census rent, and Panel (b) reports the log of median rent per room from New York Times listings. The two panels within each subfigure report the public-housing tract and adjacent first ring relative to the second ring. Markers distinguish the pooled pre-period (at least 20 years before completion), medium run (0–30 years), and long run (40 to 60 years); the census period immediately preceding completion is omitted. Both regressions include a stack-specific slope on baseline log Census rent, infrastructure-timing controls, and stack-by-neighborhood-by-year fixed effects. Standard errors are clustered by stack; vertical bars show 90% and 95% confidence intervals. Observations use stack-year ring-frequency weights, and the sample is restricted to years after 1920 and sites first treated by 1969. The pooled-pre low-density public-housing-tract coefficient in Panel (b) is not estimated.

C5 The Value of Design Quality by Neighborhood Income

Section 6.1 estimates a single citywide valuation of each design attribute. If higher-income households are more sensitive to local amenities (Gaigné et al., 2022), the market response to design quality should be stronger where baseline incomes are higher. To test this, I augment the demeaned attribute specification in Equation 4 with interactions between the two physical-design attributes – maximum recorded building height and construction cost per room – and a binary indicator for the neighborhood's pre-treatment income:

$$\begin{aligned} y_{kmpt} = &\sum_{r\in\mathcal{R}}\sum_{h\in\mathcal{H}} \theta_{r,h} D_{r,h,kpt} + \sum_{a\in\mathcal{A}}\sum_{r\in\mathcal{R}}\sum_{h\in\mathcal{H}} \gamma_{a,r,h}\widetilde{A}_{ap} D_{r,h,kpt} \\ &+ \sum_{r\in\mathcal{R}}\sum_{h\in\mathcal{H}} \kappa_{r,h}\, H_p\, D_{r,h,kpt} + \sum_{a\in\{\text{height, cost}\}}\sum_{r\in\mathcal{R}}\sum_{h\in\mathcal{H}} \varphi_{a,r,h}\, H_p\, \widetilde{A}_{ap}\, D_{r,h,kpt} \\ &+ \boldsymbol{\eta}_t'\mathbf{B}_{kp} + \boldsymbol{\delta}_p'\mathbf{Z}_{kt} + \lambda_p y^0_{kp} + \rho_{pmt} + u_{kmpt}. \end{aligned} \tag{C.4}$$

The moderator H_p equals one if the public-housing tract's pre-treatment real household income – averaged over all pre-construction census waves, as in Appendix C3 – lies above the median across the 166 public-housing tracts first treated by 1969. The split assigns 83 treatment sites to each group; the cutoff is approximately \$12,600 in 2010 dollars, with group means of roughly \$7,400 and \$18,800. Because the moderator is measured before construction and held fixed within the stack, it cannot be contaminated by the treatment's own effect on incomes, and its main effect is absorbed by the stack fixed effects. The $\kappa_{r,h}$

terms allow the *average* treatment effect to differ by income group, so the triple coefficients $\varphi_{a,r,h}$ isolate differences in the *valuation of design*: $\gamma_{a,r,h}$ is the attribute slope in below-median-income neighborhoods, and $\gamma_{a,r,h} + \varphi_{a,r,h}$ the slope in above-median-income neighborhoods. All remaining terms, the sample, the weights, and the clustering reproduce Equation 4; the baseline-outcome slope uses the gross demographic count for the population outcomes and baseline log Census rent for the rent outcomes.

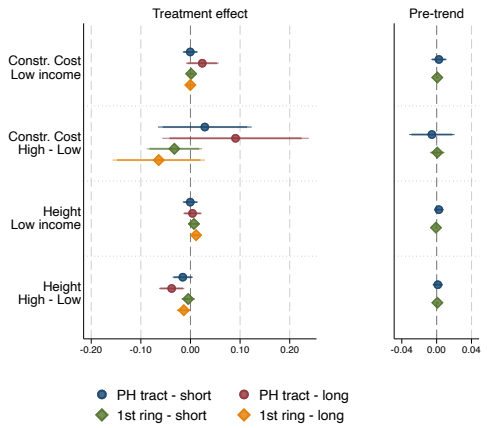
Figure C5.1 reports the design-attribute coefficients. Two results stand out, and they are two sides of the same mechanism. First, building height is capitalized into rents only in higher-income neighborhoods. In below-median-income neighborhoods the long-run effect of a one-story increase in maximum recorded height on Census rents in the public-housing tract is zero (0.004, s.e. 0.009), while the high-income differential is −0.038 (s.e. 0.013): a one-story increase lowers long-run rents by close to four percent where baseline incomes are high, and not at all where they are low. A smaller differential of the same sign appears in the first ring (−0.013, s.e. 0.007). Second, the demographic response mirrors this pattern in reverse. The long-run White population loss associated with height is concentrated in low-income neighborhoods (−0.123 per story, s.e. 0.038); the high-income differential of +0.139 (s.e. 0.058) offsets it almost exactly, leaving no height response of the private White population in high-income areas. The first-ring Black population shows the same offset (differentials of +0.040 and +0.041, both significant at five percent, against negative low-income slopes), and the Hispanic estimates point in the same direction. Where incomes are high, taller projects are absorbed through prices; where incomes are low, the adjustment operates through population sorting rather than rents.

Construction cost per room shows no comparable income gradient. The low-income cost slopes replicate the retention pattern of Section 6.1 – costlier projects retain more White, Black, and Hispanic residents – but the income differentials are imprecisely estimated throughout, so substantial heterogeneity in either direction cannot be ruled out. The pooled pre-period interactions in the right-hand panels are uniformly small and insignificant: neither design nor design-by-income predicts outcomes before construction, which is the placebo test implied by the triple-difference design. The New York Times listing measure cannot support this exercise in the treated tract: only eight above-median-income projects retain listings in the long run, and the corresponding cells are dropped. Its one well-supported design estimate, a first-ring long-run height slope of −0.040 (s.e. 0.015), corroborates the Census-rent height penalty.

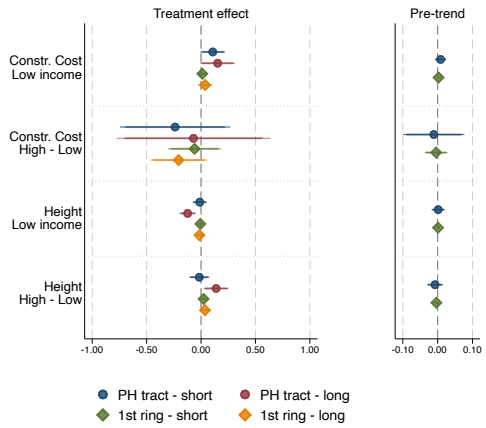
These estimates inherit the interpretation of Section 6.1: NYCHA did not assign design randomly across neighborhoods, so the coefficients describe conditional cross-project heterogeneity rather than the causal effect of exogenously altering a project's design. With that caveat, the pattern supports an amenity-sorting mechanism in the spirit of Gaigné et al. (2022): the market price of design quality is place-dependent, and the valuation of avoiding tower-style construction accrues where amenity demand – proxied by baseline income – is highest. High-income neighborhoods in this sample are also disproportionately located in Manhattan and denser at baseline, so the income split should be read as a bundle of correlated place characteristics rather than income in isolation.

Figure C5.1: The Value of Design Quality by Neighborhood Income

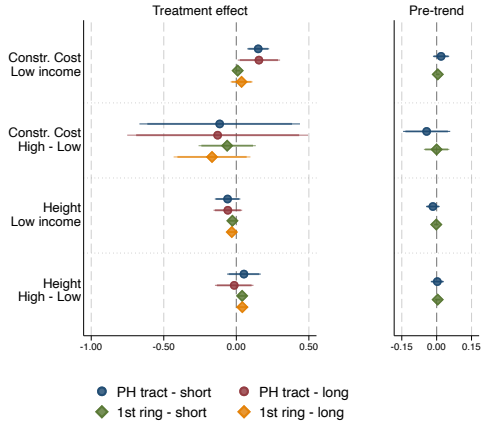
(a) Log Census rent



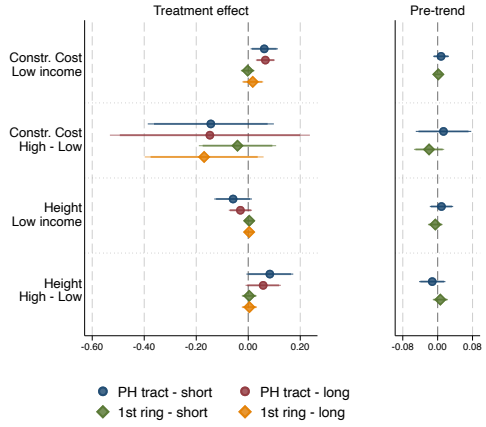
(b) Log private White population



(c) Log private Black population



(d) Log private Hispanic population



Note: The figure plots the design-attribute coefficients from Equation C.4. In each panel, the "Low income" rows report the attribute slope $\hat{\gamma}_{a,r,h}$ in below-median-income neighborhoods and the "High − Low" rows the income differential $\hat{\varphi}_{a,r,h}$; their sum is the slope in above-median-income neighborhoods. Panel (a) reports log median Census rent; Panels (b)–(d) report the log privately housed White, Black, and Hispanic populations. The left side of each panel reports post-completion effects for public-housing tracts and first-ring tracts in the medium run (0–30 years) and long run (40 to 60 years); the right side reports the corresponding pooled pre-period coefficients. The omitted period is the census decade immediately preceding completion, and the comparison group is the second ring. Construction cost is CPI-deflated cost per room in 2010 dollars, scaled by \$5,000; height is the maximum recorded number of stories; both are demeaned across the estimation sample. Regressions include the full demeaned attribute block, income-group-specific ring-by-horizon main effects, baseline-demographics-by-year controls, stack-specific baseline-outcome slopes, infrastructure-timing controls, and stack-by-neighborhood-by-year fixed effects. Standard errors are clustered at the stack level; horizontal bars show 90% and 95% confidence intervals. Observations use stack-year ring-frequency weights, and the sample is restricted to years after 1920 and sites first treated by 1969.

C6 Alternative Estimators

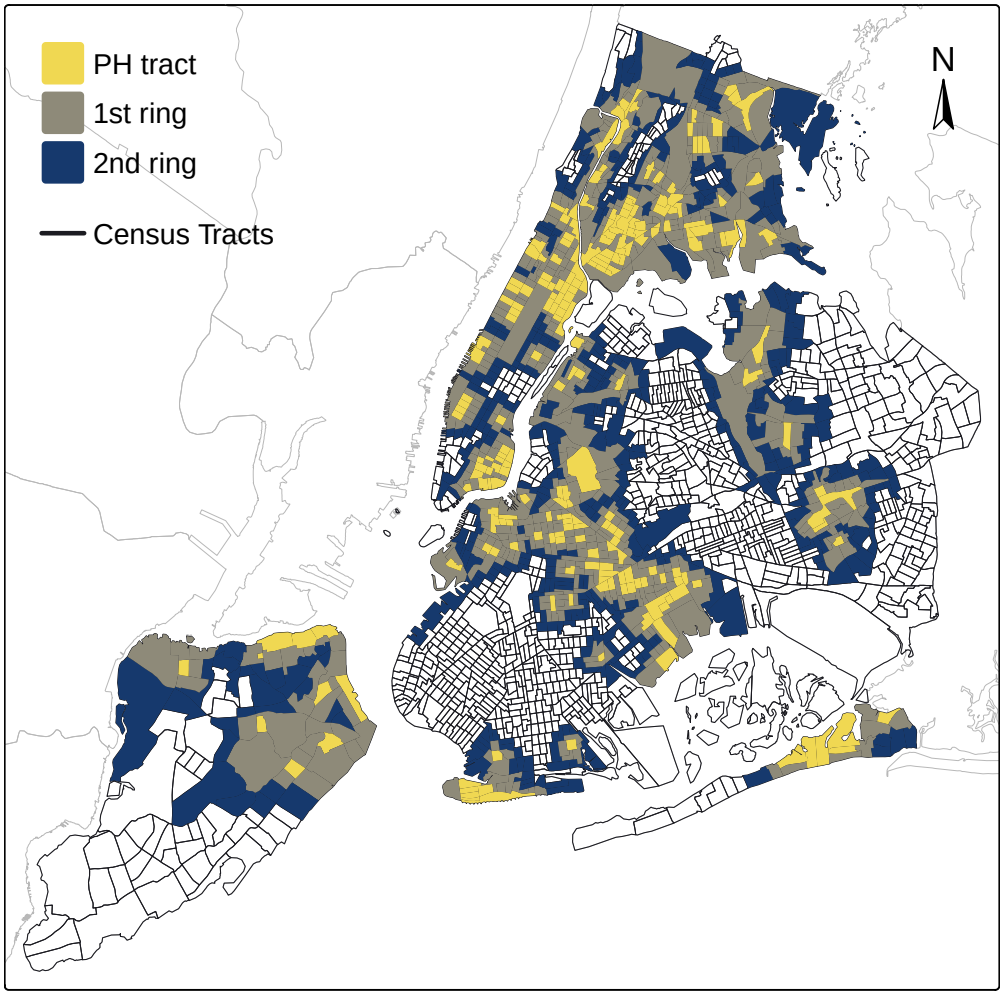
The baseline estimates using Equation 1 compare each public-housing tract with nearby, never-treated tracts in the second ring around the project. This local comparison is attractive because public housing was built in neighborhoods that differed substantially from the rest of the city, but the results could nevertheless depend on the particular comparison tracts used as controls. I therefore assess the robustness of the long-run estimates using two complementary approaches applied to the unstacked tract panel. First, I estimate dynamic treatment effects using the estimator of Chaisemartin and D'Haultfœuille (2024) (DCDH), which preserves the same comparison group while allowing treatment effects to differ across cohorts and over time. Second, I estimate synthetic difference-in-differences (Arkhangelsky et al., 2021) (SDID), which instead constructs a weighted counterfactual from the broader pool of never-treated tracts by matching pre-treatment outcome trajectories. Together, these approaches assess whether the baseline findings are robust to both the estimation method and the construction of the counterfactual.

Both estimators use the decennial tract panel and the sample of public-housing tracts first treated before 1970. In both specifications, immediately adjacent tracts are excluded to limit contamination from spillovers, and all regressions control for contemporaneous urban renewal, arterial-road construction, and subway expansion. Dynamic treatment effects are estimated from four decades before to six decades after project completion, with the decade before completion as the reference period. DCDH relies on clean comparisons between treated and never-treated tracts and allows for non-parametric neighborhood (NTA) trends, whereas SDID reweights both never-treated tracts and pre-treatment periods—after removing borough-by-year means among never-treated tracts—to reproduce the pre-treatment trajectories of the treated tracts. Because SDID is estimated using demeaned outcomes whereas DCDH is not, the two estimators need not coincide in levels, but their dynamic treatment effects are directly comparable.

Figure C6.1 maps the tract classification underlying this sample construction. Gold tracts contain new-construction public housing, grey tracts form the first ring of adjacent tracts, and navy tracts constitute the second ring. Of the census tracts in the panel, 232 ever contain new-construction public housing—166 were first treated before 1970, which form the treated group here, and 66 were first treated in 1970 or later—while 514 tracts belong to the first ring and 489 to the second. For the SDID estimation, the donor pool consists of all never-treated tracts outside the first ring: 158 treated tracts (the pre-1970 public housing tracts that additionally satisfy the balanced-panel and single-adoption requirements) are compared with 1,416 never-treated donor tracts, while the remaining 590 tracts—first-ring tracts, tracts treated after 1970, and tracts not observed in every census decade—are excluded.

Figure C6.2 illustrates how SDID selects the unit weights ω from this donor pool, using the population outcome as an example. Never-treated donor tracts are shaded by their unit weight relative to the uniform weight 1/1,416: red tracts are upweighted and blue tracts are downweighted relative to a standard difference-in-differences, which weights all donors equally. Treated tracts are shown in gold and excluded tracts in light grey. The weights are estimated separately for each adoption decade over the common donor pool and then averaged across cohorts. Two features stand out. First, the weights are broadly dispersed rather

Figure C6.1: Tracts by Distance Relationship to Public Housing



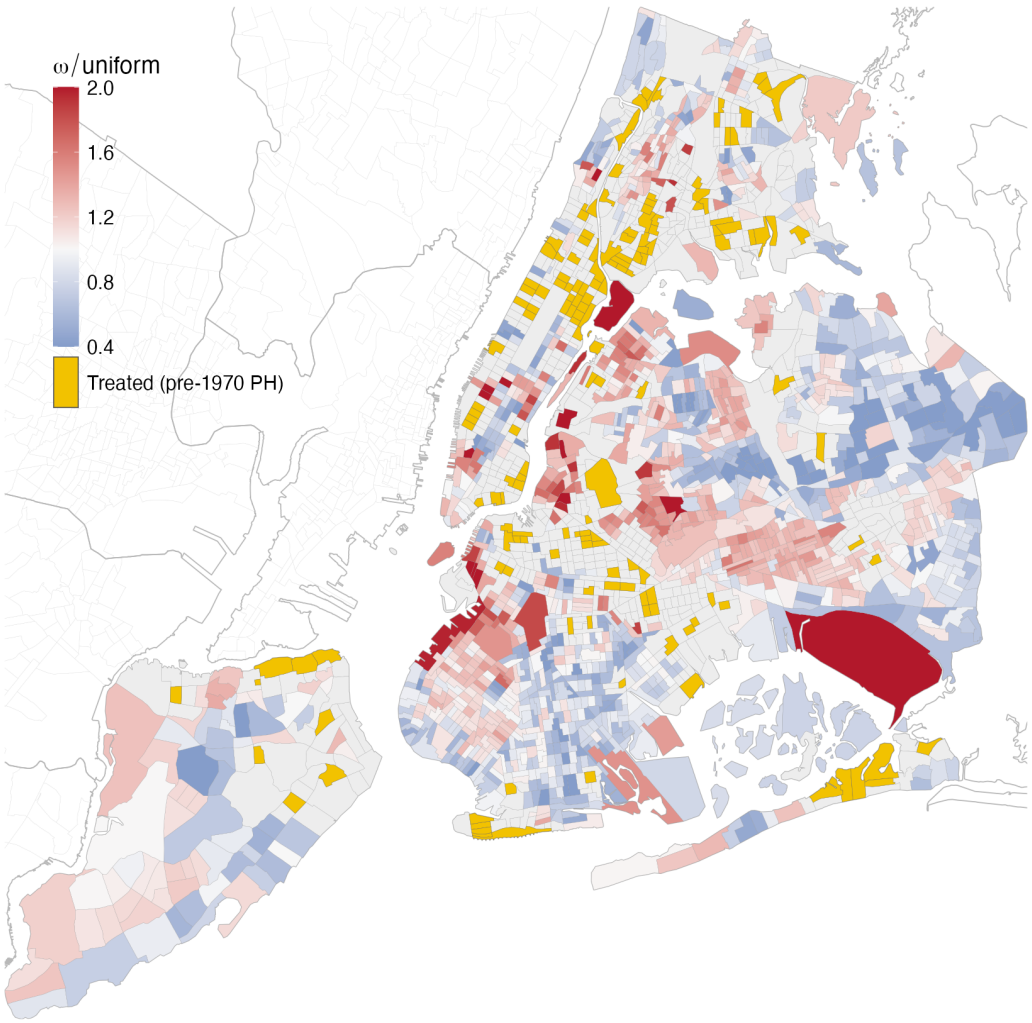
Note. Figure C6.1 classifies 2010 census tracts by their distance relationship to public housing, as used in the stacked panel setup. Gold tracts contain new-construction public housing (232 tracts; 166 first treated before 1970 and 66 first treated in 1970 or later), grey tracts are adjacent to a public-housing tract (first ring, 514 tracts), and navy tracts are adjacent to first-ring tracts (second ring, 489 tracts). All remaining tracts are unshaded.
Source. NYCHA Development Data Book and US Decennial Census. Details on the construction of the data set can be found in Section 3.

than concentrated on a handful of donors—most tracts receive weights close to uniform—so the SDID estimates do not hinge on a small number of comparison tracts. Second, donor tracts near the treated areas—in upper Manhattan, the South Bronx, and central Brooklyn—tend to receive above-uniform weights, while many tracts in eastern Queens receive below-uniform weights, consistent with SDID favoring donors whose (borough-demeaned) population levels and pre-treatment trends resemble those of the dense neighborhoods in which public housing was built.

Figure C6.3 and Figure C6.4 display the results; each panel plots the outcome both in total and net of public-housing residents (or units). Across outcomes the placebo coefficients are small and statistically insignificant, and the two estimators track one another closely after treatment, indicating that the baseline patterns are not an artifact of the TWFE design. Public housing raises the total population and total housing stock on impact, but the population and housing stock net of the projects fall steadily over the following decades—evidence that the developments displaced private residents and replaced more private housing than the projects themselves added.

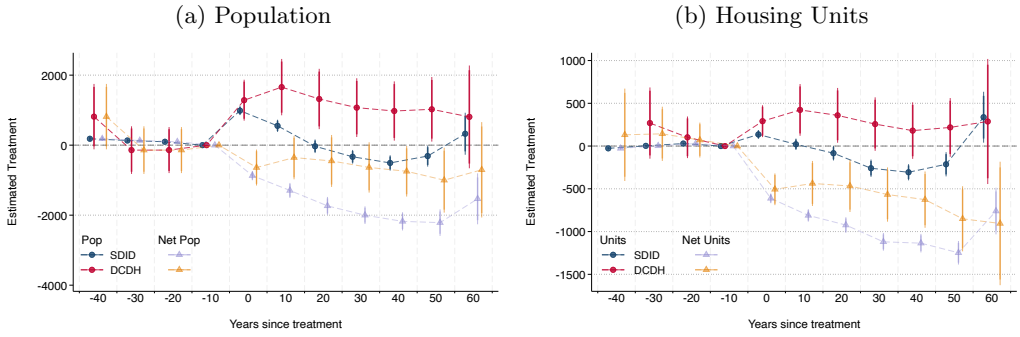
The total Black population rises sharply while the total White population declines over time; once public-housing tenants are excluded, both the private Black and the private White populations fall, whereas the Hispanic population net of public housing is essentially unchanged. Rents also decline in the long run: under both estimators the census average-rent series falls by roughly 0.3–0.5 log points (about 25–40 percent) six decades after construction, while the noisier New York Times series is centered near zero. The listing-rent series is included for completeness only: few tracts have continuously observed listings over the pre-treatment window on which the SDID matching and the DCDH comparisons rely, so these designs have little support for the asking-rent outcome, and I do not draw inferences about private asking rents from them. The Census-rent series provides the informative robustness comparison for prices. These magnitudes and dynamics mirror the baseline estimates in Figure 2, Figure 3, and Figure 4.

Figure C6.2: SDID Donor Weights: Population, Tracts First Treated Before 1970



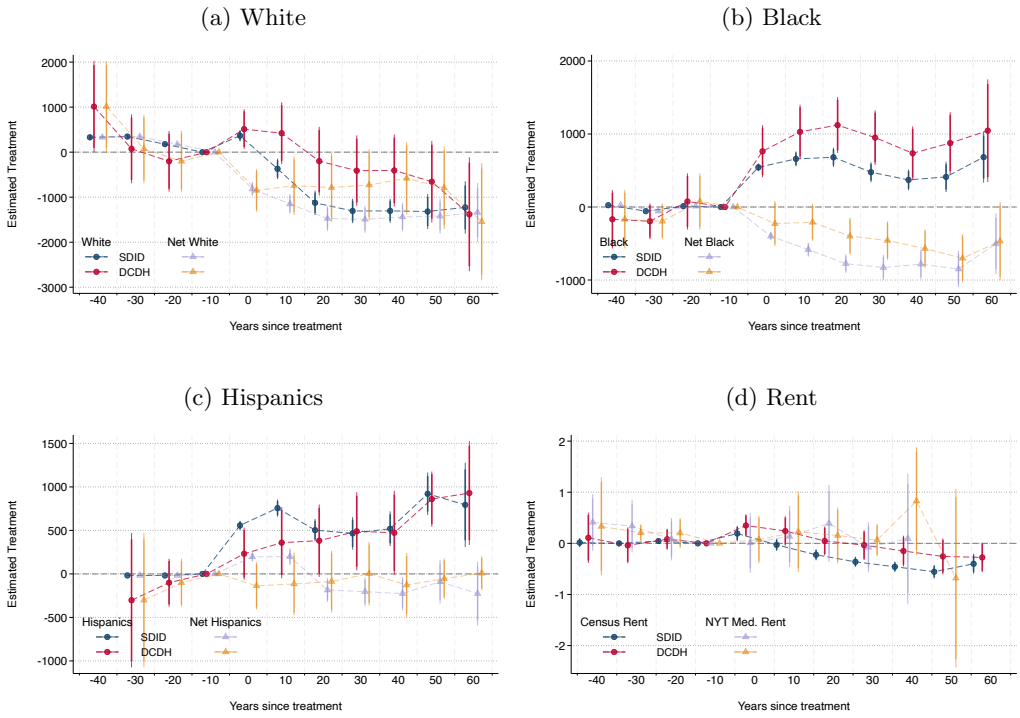
Note. Figure C6.2 maps the SDID unit weights ω for the population outcome and the sample of public-housing tracts first treated before 1970. Never-treated donor tracts (1,416) are shaded by the ratio of their weight to the uniform weight 1/1,416; the color scale is truncated at 0.4 and 2. Treated tracts (158) are shown in gold; tracts excluded from the estimation sample (590)—first-ring tracts, tracts treated after 1970, and tracts not observed in every census decade—are shown in light grey. Weights are estimated with the `synthdid` package separately for each adoption decade over the common never-treated donor pool, after removing borough-by-year means among never-treated tracts, and averaged across cohorts.
Source. US Decennial Census and NYCHA Development Data Book. Details on the construction of the data set can be found in Section 3.

Figure C6.3: Alternative Estimators: Population & Housing Stock



Note: Figure C6.3 plots dynamic treatment effects estimated with synthetic difference-in-differences (SDID; Arkhangelsky et al., 2021) and the estimator of Chaisemartin and D'Haultfœuille (2024) (DCDH). Panel (a) reports effects on total population and on population after subtracting public-housing residents; Panel (b) reports effects on total housing units and on housing units after subtracting public-housing units. The latter series therefore measure changes in the private population and housing stock. Circles denote total outcomes and triangles denote private outcomes; navy and lavender are SDID estimates, while red and orange are DCDH estimates. Both estimators use the decennial tract panel and compare public-housing tracts first treated before 1970 with never-treated tracts. Tracts in the immediately adjacent ring are excluded to limit spillover contamination. Event time runs from 40 years before to 60 years after completion, with the decade before completion as the reference period. All specifications control for urban-renewal, arterial-road, and subway exposure. SDID reweights control tracts and pre-treatment periods after removing borough-by-year means among never-treated tracts; DCDH uses clean treatment comparisons and allows for non-parametric neighborhood (NTA) trends. Vertical bars show 90% and 95% confidence intervals, based on a placebo bootstrap with 200 replications for SDID and tract-clustered standard errors for DCDH.

Figure C6.4: Alternative Estimators: Race and Rents



Note: Figure C6.4 plots dynamic treatment effects estimated with synthetic difference-in-differences (SDID; Arkhangelsky et al., 2021) and the estimator of Chaisemartin and D'Haultfœuille (2024) (DCDH). Panels (a)–(c) report effects on the total White, Black, and Hispanic populations and on each population after subtracting residents of the corresponding group who live in public housing; the latter series therefore measure effects on privately housed residents. Panel (d) reports effects on the log of average Census rent and the log of median rent per room from New York Times rental listings. Circles denote total population or Census rent and triangles denote private population or NYT rent; navy and lavender are SDID estimates, while red and orange are DCDH estimates. Both estimators use the decennial tract panel and compare public-housing tracts first treated before 1970 with never-treated tracts. Tracts in the immediately adjacent ring are excluded to limit spillover contamination. Event time runs from 40 years before to 60 years after completion, with the decade before completion as the reference period. All specifications control for urban-renewal, arterial-road, and subway exposure. SDID reweights control tracts and pre-treatment periods after removing borough-by-year means among never-treated tracts; DCDH uses clean treatment comparisons and allows for non-parametric neighborhood (NTA) trends. Vertical bars show 90% and 95% confidence intervals, based on a placebo bootstrap with 200 replications for SDID and tract-clustered standard errors for DCDH.

C7 Spatial-Inference Robustness

One concern with the stacked difference-in-differences design proposed in Section 4 is that its treatment-site sub-experiments may not be statistically independent. Although each stack is organized around a different public-housing tract, the same census tract can enter the comparison rings of several nearby stacks. Outcomes may also respond to unobserved local shocks that extend across tract and stack boundaries. Clustering standard errors by stack allows arbitrary dependence within each treatment site, but it does not capture dependence connecting different stacks through repeated tract use or geographic proximity. Ignoring these links could therefore understate statistical uncertainty. To address this concern, I recompute inference in two complementary ways: first using two-way clustering by stack and tract, which accounts directly for repeated tract use, and then using Conley spatial-HAC standard errors, which allow correlation across nearby locations.

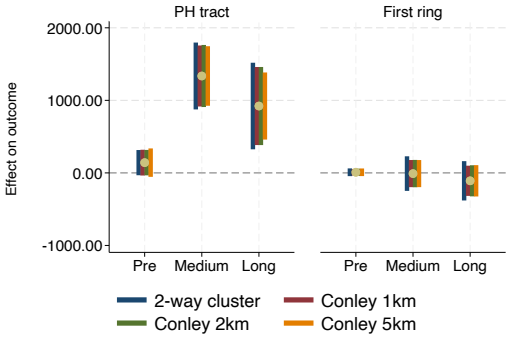
Both exercises retain the sample, controls, fixed effects, frequency weights, and pre-1970 treatment-site cohort of the baseline event-study specification in Equation 1; for the Conley calculation, the fixed effects and controls are partialled out before applying the spatial correction, so point estimates are identical across methods and only the estimated uncertainty changes. I report Conley standard errors for 1, 2, and 5 kilometre distance cutoffs with a uniform spatial kernel, allowing serial correlation through all eight available decennial lags.

Figure C7.1 compares the resulting 95% confidence intervals for the main population and housing-stock outcomes, Figure C7.2 reports the racial-composition outcomes, and Figure C7.3 reports the rent results. No inference method produces uniformly wider intervals: their relative width varies somewhat across coefficients, but the intervals are generally similar. More importantly, the substantive conclusions are unchanged. In the public-housing tract, total population and housing units increase after construction, whereas private population and private housing units decline; estimates for the first ring remain substantially smaller. The demographic results likewise preserve the contrast between total and privately housed populations.

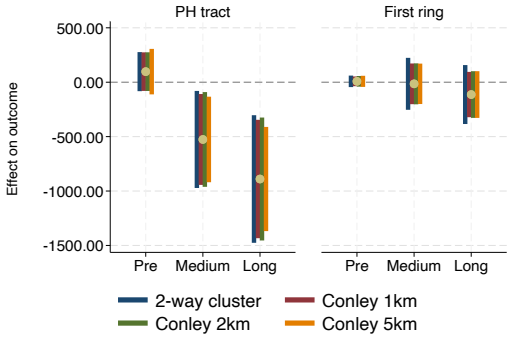
The rent panels show the same long-run decline under both data sources and every inference method. In the public-housing tract, the common point estimates are approximately -0.41 log points for Census rent and -0.52 log points for median *New York Times* rent per room, and all four confidence intervals exclude zero. The first-ring Census-rent estimate is close to zero, while the listing-rent estimate is about -0.25 log points and remains negative across the alternative confidence intervals. The pre-period and medium-run rent estimates are generally close to zero or imprecisely estimated. Thus, the main population, housing-stock, and rent findings are not driven by repeated use of the same tracts across project stacks or by unaccounted spatial correlation among nearby observations.

Figure C7.1: Spatial-Inference Robustness: Population and Housing Stock

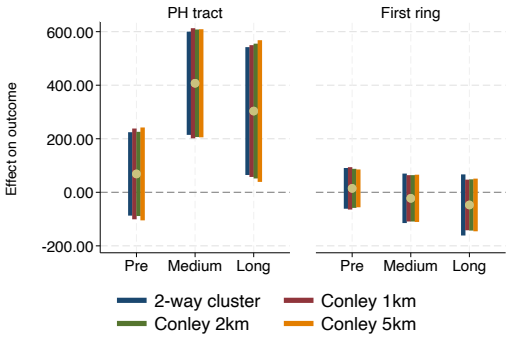
(a) Total population



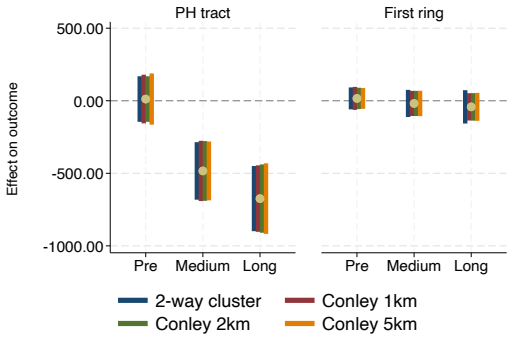
(b) Private population



(c) Total housing units



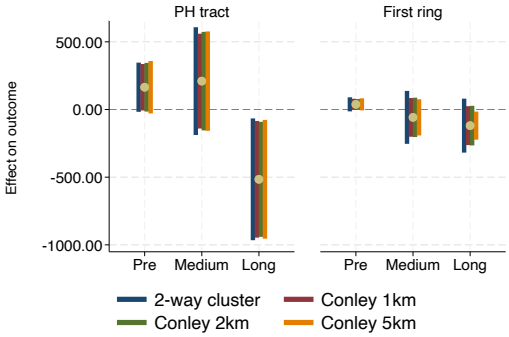
(d) Private housing units



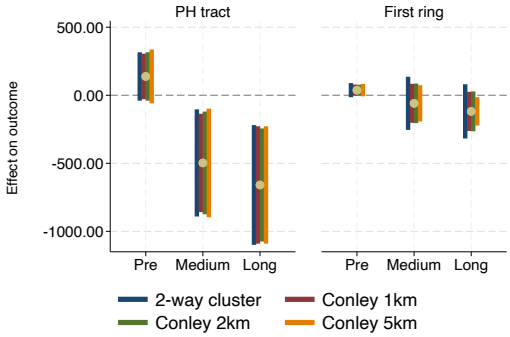
Notes: Figure C7.1 reports level coefficients from the main event-study specification in Equation 1 for treatment sites first treated before 1970. Panels (a) and (c) report total population and housing units; Panels (b) and (d) subtract public-housing residents or units from the corresponding tract totals. The left subgraph within each panel shows the public-housing tract and the right subgraph the first ring; the second ring is the comparison group. "Pre" pools event times earlier than the census decade immediately preceding completion, which is omitted; "Medium" covers 0–30 years after completion; and "Long" covers 40 to 60 years. Markers show the common point estimates, and vertical bars show 95% confidence intervals: *two-way clustering by stack and tract*; *Conley 1 km*; *Conley 2 km*; and *Conley 5 km*. The Conley estimates use a uniform spatial kernel and permit serial correlation through all eight available decennial lags. All estimates use the same controls, fixed effects, frequency weights, and pre-1970 estimation sample as Equation 1.

Figure C7.2: Spatial-Inference Robustness: Racial Composition

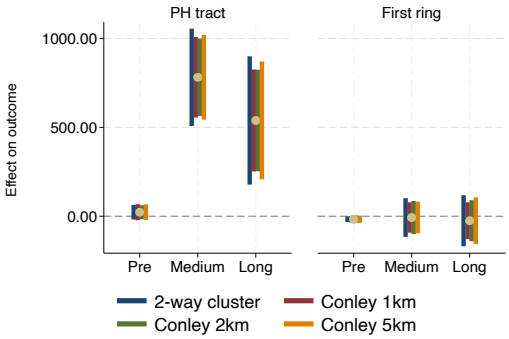
(a) Total White population



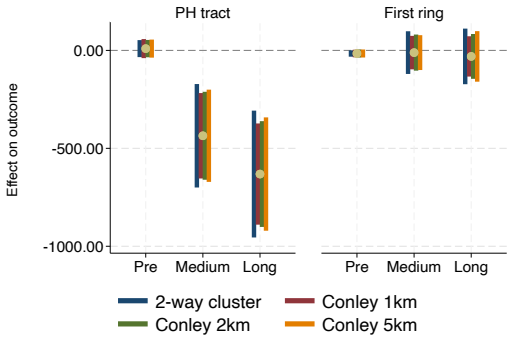
(b) Private White population



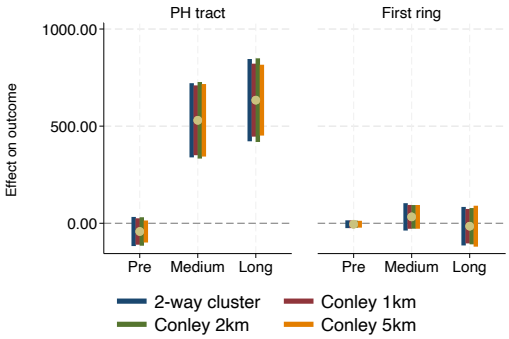
(c) Total Black population



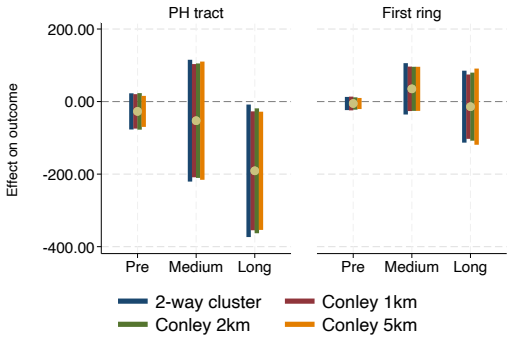
(d) Private Black population



(e) Total Hispanic population



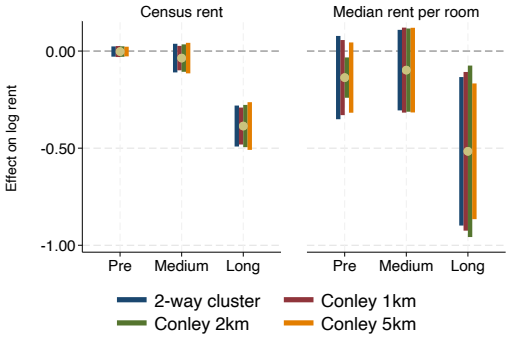
(f) Private Hispanic population



Notes: Figure C7.2 reports level coefficients from the main event-study specification in Equation 1 for treatment sites first treated before 1970. Panels (a), (c), and (e) report total White, Black, and Hispanic population; Panels (b), (d), and (f) subtract public-housing residents of the corresponding group and therefore report the privately housed population. The left subgraph within each panel shows the public-housing tract and the right subgraph the first ring; the second ring is the comparison group. "Pre" pools event times earlier than the census decade immediately preceding completion, which is omitted; "Medium" covers 0–30 years after completion; and "Long" covers 40 to 60 years. Markers show the common point estimates, and vertical bars show 95% confidence intervals: *two-way clustering by stack and tract*; *Conley 1 km*; *Conley 2 km*; and *Conley 5 km*. The Conley estimates use a uniform spatial kernel and permit serial correlation through all eight available decennial lags. All estimates use the same controls, fixed effects, frequency weights, and estimation sample as Equation 1.

Figure C7.3: Spatial-Inference Robustness: Rent Outcomes

(a) Public-housing tract



(b) First ring



Notes: Figure C7.3 reports log-rent coefficients from the main event-study specification in Equation 1. Panel (a) reports the public-housing tract and Panel (b) the first ring. Within each panel, the left subgraph reports log median Census rent and the right subgraph log median rent per room from *New York Times* listings. The second ring and the census decade immediately preceding completion are the reference categories. "Pre" pools earlier event times, "Medium" covers 0–30 years after completion, and "Long" covers 40 to 60 years. Markers show the common point estimates, and vertical bars show 95% confidence intervals: *two-way clustering by stack and tract*; *Conley 1 km*; *Conley 2 km*; and *Conley 5 km*. The Conley estimates use a uniform spatial kernel and permit serial correlation through all eight available decennial lags. All estimates use the same controls, fixed effects, frequency weights, and pre-1970 estimation sample as Equation 1.

C8 MWTP Sensitivity to Housing-Expenditure Share

The baseline MWTP estimates in Section 6.2 calibrate the housing-expenditure share to $\alpha = 0.30$. The common calibration facilitates comparisons across groups and time; it is not intended to imply that housing-expenditure shares were actually constant from 1930 to 2010. Because no historical data provide consistent housing-expenditure shares by group and tenure over the full sample period, I do not estimate a separate value for each horizon. Instead, I evaluate the MWTP estimates over a deliberately wide range of plausible values. I first show that the baseline calibration lies within the range of published New York rent-share benchmarks and then examine how the medium-run public-housing-tract estimates vary with α.

Although no single source measures housing-expenditure shares consistently by group and tenure over the full 1930–2010 period, several published benchmarks suggest that the baseline calibration is reasonable. Figure C8.1 assembles four New York series spanning 1900-2023. The median gross rent-to-income ratio among New York City renter households ranges from about 20 to 34 percent between 1965 and 2023 in the New York City Housing and Vacancy Survey (Gaumer, 2024). At the beginning of the century, tenement-era budget studies report rent shares of roughly one quarter of family income, with estimates between 22 and 36 percent (Chapin, 1909; DeForest and Veiller, 1903). Rent records from the 97 Orchard Street tenement similarly imply housing-expenditure shares of 20 to 31 percent between 1900 and 1932 (Steinberg, 2016; Ford, 1936). The only series that lies persistently above these values divides rent by a single worker's full-time wage rather than household income and therefore overstates the household expenditure share by construction (Ferrara, 2026). Overall, the household-based evidence clusters around 20-35 percent throughout the century. The baseline calibration of $\alpha = 0.30$ therefore lies comfortably within the historical range, while the sensitivity analysis below considers values well outside it.

Since

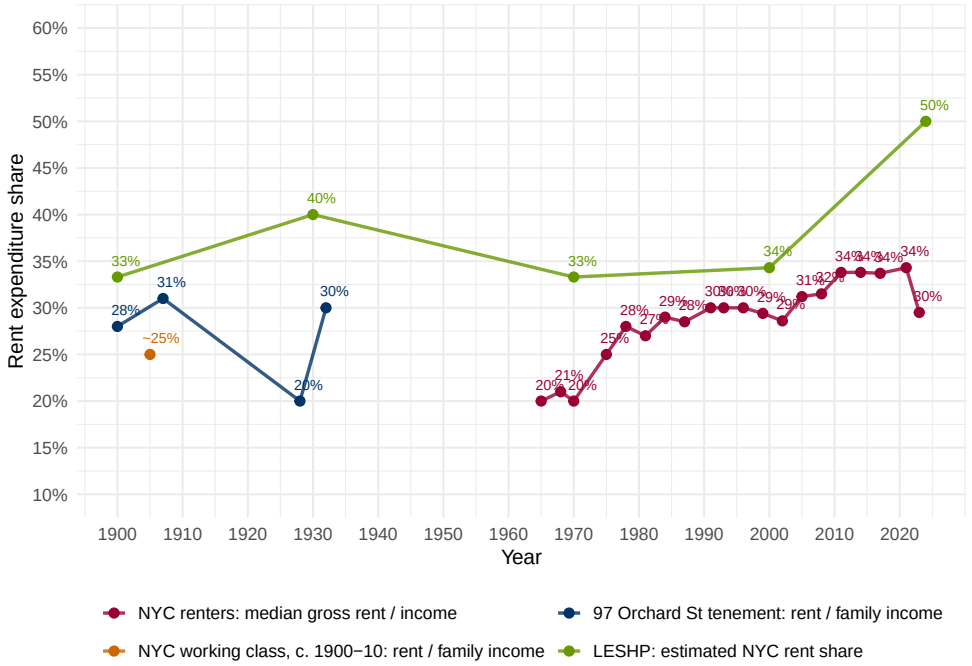
$$\widehat{MWTP}^{g}_{a,R,h}(\alpha) = \frac{\widehat{\gamma}^{g,pop}_{a,R,h} + \alpha\widehat{\gamma}^{rent}_{a,R,h}}{\alpha} = \frac{\widehat{\gamma}^{g,pop}_{a,R,h}}{\alpha} + \widehat{\gamma}^{rent}_{a,R,h},$$

the exercise is partly mechanical: a lower α places more weight on the population response relative to rent and therefore expands the absolute value of MWTP whenever the population coefficient is far from zero. A higher α compresses the population component and moves the estimate toward the rent effect.

Figure C8.2 traces this sensitivity for the public housing tract at the 0–30 year horizon, using the headline specification of Section 6.2. Each panel corresponds to one project attribute and plots the MWTP of White, Black, and Hispanic private-market residents against α over a grid from 0.05 to 0.60; the dashed vertical line marks the baseline $\alpha = 0.30$. All values are expressed as exact percent changes, $100\,(e^{m} - 1)$, following Equation A.20, where m is the MWTP in log-rent units. Because the exact transform is bounded below by -100 percent but unbounded above, the mechanical amplification at small α is asymmetric: large negative estimates compress toward the -100 percent bound, while positive estimates—most visibly construction cost and ground coverage—blow up as $\alpha \to 0$. As α increases, every curve flattens toward the pure rent effect $\widehat{\gamma}^{rent}_{a,R,h}$.

Figure C8.2 illustrates how the MWTP estimates change as the expenditure share parameter α varies. The figure focuses on the public housing tract at the 0–30 year horizon

Figure C8.1: Rent Expenditure Shares in New York

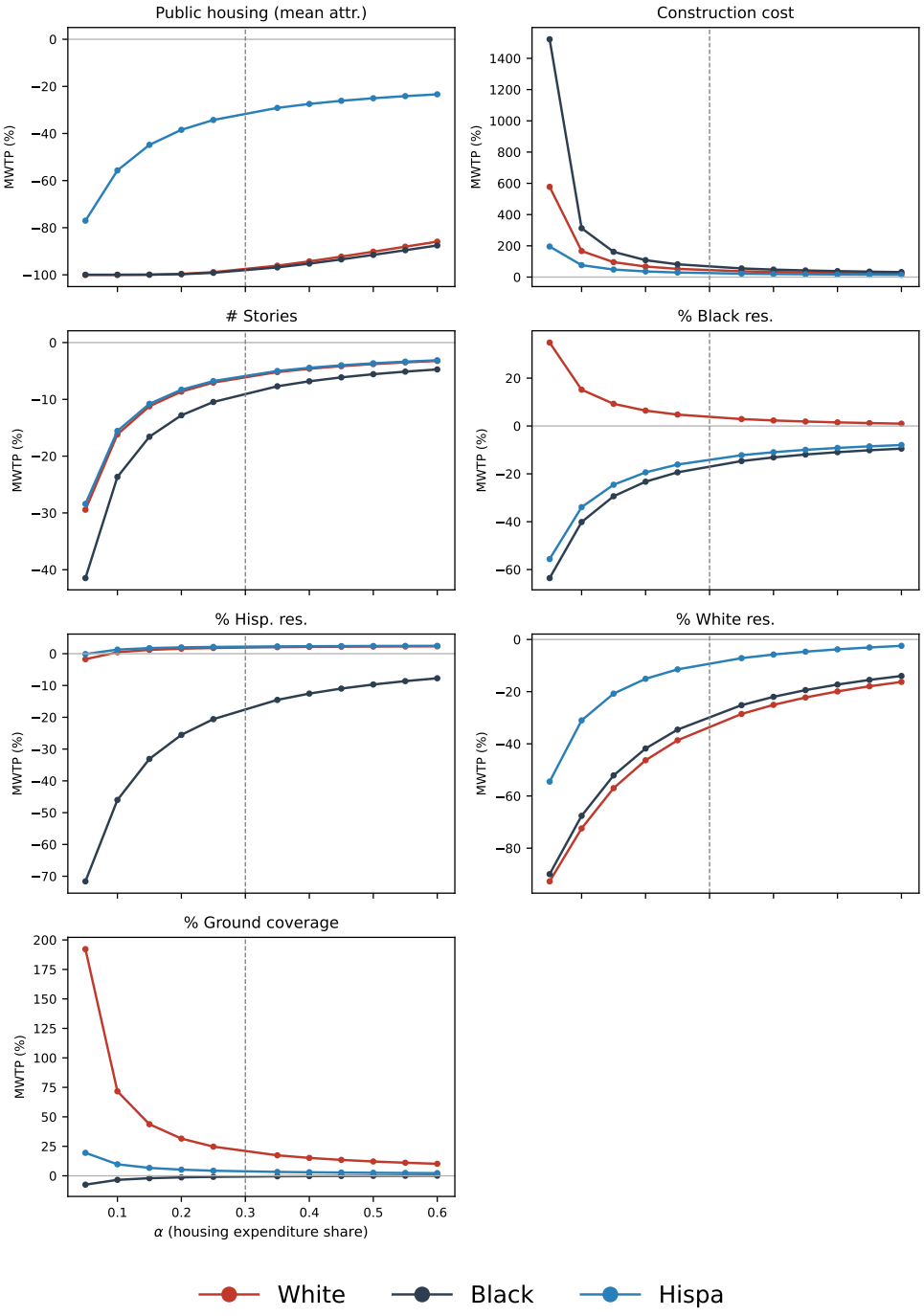


Note. Figure C8.1 plots published estimates of the rent expenditure share of New York households, 1900–2023. ● *NYC renters:* median gross rent-to-income ratio among all New York City renter households, from the 2023 New York City Housing and Vacancy Survey report (Gaumer, 2024). ● *NYC working class, c. 1900–10:* typical share from tenement-era budget studies, approximately 25 percent with a reported range of 22–36 percent (Chapin, 1909; DeForest and Veiller, 1903). ● *97 Orchard Street tenement:* rent relative to family income on the Lower East Side; the 1900 and 1907 shares divide surveyed tenement rents by typical local earnings of $35–45 per month, while the 1928 and 1932 shares are reported directly (Steinberg, 2016; Ford, 1936). ● *Lower East Side History Project:* estimated share of income spent on rent in New York City, reproduced from the historical series presented by Ferrara (2026). The source does not document the income definition or the detailed construction of the historical observations, so the series is treated as an illustrative benchmark rather than as directly comparable to the household-survey estimates.

using the headline specification from Section 6.2. Each panel shows one project attribute and plots the MWTP of White, Black, and Hispanic private-market residents for values of α between 0.05 and 0.60. The dashed vertical line marks the baseline value, $\alpha = 0.30$. All estimates are reported as exact percentage changes, $100(e^m - 1)$, where m is the MWTP in log-rent units (Equation A.20). Smaller values of α place more weight on the population response, so the estimates become more extreme. Because the exact percentage transformation cannot fall below -100 percent but has no upper bound, large negative estimates approach -100 percent, whereas large positive estimates, notably for construction cost and ground coverage, increase rapidly as α becomes small. As α increases, the influence of the population response diminishes and the estimates move closer to the rent effect alone.

The figure shows that the main conclusions of Section 6.2 are robust to the choice of the expenditure share parameter α. Across the entire range of values considered, the mean pre-1970 project is valued negatively by both White and Black private-market residents, while the Hispanic valuation remains smaller in magnitude and gradually moves toward zero as α increases. Likewise, construction cost is valued positively, whereas maximum recorded building height and the initial White resident share are valued negatively throughout. Although the signs and relative ranking of the estimates remain stable, their magnitudes vary substantially with α, as implied by the MWTP formula.

Figure C8.2: Sensitivity of MWTP to the Housing-Expenditure Share



Note. Figure C8.2 plots the MWTP of White, Black, and Hispanic private-market residents for each project attribute in the public housing tract at the 0–30 year horizon, as a function of the housing-expenditure share α (grid from 0.05 to 0.60). Estimates use the reduced-form coefficients of the headline specification for the pre-1970 cohort with demeaned attributes; height is the maximum recorded number of stories, and the number of apartments is excluded and analyzed separately as treatment intensity. MWTP is converted to percent of rent using the exact transform $100\,(e^{m}-1)$ of Equation A.20, where m is the MWTP in log-rent units. The dashed vertical line marks the baseline value $\alpha = 0.30$ used in Section 6.2.

C9 Efficiency of Construction Investment

In this appendix, I ask whether additional construction investment was justified by the neighborhood losses it prevented. Holding fixed the decision to build a project of a given size, I compare the additional construction cost per room with the present value of the resulting private-market rent externality, following the accounting approach of Diamond and McQuade (2019). The comparison holds project size fixed and focuses exclusively on neighborhood externalities. It excludes any benefits to public-housing tenants from higher-quality buildings, as well as fiscal operating costs and general-equilibrium relocation; including tenant benefits would increase the estimated return to construction investment.

Let $\hat{\beta}_{ih}$ denote the estimated effect of construction cost per room (per \$5,000, 2010 dollars) on log private median listing rent per room in ring $i \in \{0, 1\}$ (public-housing tract and first ring) and horizon $h \in \{s, l\}$ (0–30 and 40–60 years), taken from the headline attribute specification of Section 6.1. An increase of Δ dollars per room changes log rents by $\Delta \ln p_{ih} = \hat{\beta}_{ih}\, \Delta/5000$. The annual dollar effect is obtained by applying the estimated rent change to the pre-completion private rent bill of ring i for the median pre-1970 project,

$$RB_i \;=\; 12 \cdot \bar{p}_i^{\,0} \cdot N_i^0,$$

where $\bar{p}_i^{\,0}$ is the pre-completion real median contract rent and N_i^0 the number of pre-completion private occupied units, summed over the ring's tracts ($RB_0 \approx$ \$4.8 million and $RB_1 \approx$ \$15.8 million per year). Because the base is measured before completion, income-based public-housing rents never enter it. The annual externality flow is $F_{ih} = RB_i\,[\exp(\Delta \ln p_{ih}) - 1]$, and its present value applies the medium-run flow to years 0–34 and the long-run flow to years 35–64 at discount rate r:

$$PV(r) \;=\; \sum_{i=0}^{1}\Big(F_{is}\sum_{t=0}^{34}(1+r)^{-t} \;+\; F_{il}\sum_{t=35}^{64}(1+r)^{-t}\Big).$$

The incremental outlay for the median project (about 2,966 rooms) is $C = \Delta \times 2{,}966$, and the object of interest is the ratio $\Theta(r) = PV(r)/C$: the capitalized neighborhood externality avoided per marginal construction dollar. Standard errors follow from the delta method, treating the four reduced-form coefficients as independent.

Table C9.1 reports the implied present value of the neighborhood rent externality together with the corresponding cost-benefit ratio. At a 5 percent discount rate, increasing construction investment by \$1,000 per room raises construction expenditure by about \$3.0 million for the median project. Applying the estimated rent effect to the treated tract yields a present value of roughly \$1.8 million. Applying the same calculation to the first ring yields a present value of -\$2.2 million. Because the first ring has roughly three times the baseline private rent bill of the treated tract, its contribution dominates the aggregate calculation, implying a combined neighborhood externality of approximately -\$0.4 million and a cost-benefit ratio of $\Theta = -0.14$ (s.e. 1.02). Across discount rates between 3 and 7 percent, the combined point estimate ranges from -0.28 to -0.08. The confidence intervals are wide, however, encompassing both sizeable negative values and values well above one.

The estimates come with three important caveats. First, project attributes were not randomly assigned: pre-construction neighborhood conditions predict construction investment (Appendix B3), so the coefficients capture conditional cross-project associations rather

Table C9.1: Ballpark cost–benefit of the construction-investment margin

	Δ = \$1,000 per room			Δ = IQR = \$3,379 per room		
Discount rate	3%	5%	7%	3%	5%	7%
PV externality, treated tract (\$m)	3.41	1.78	1.07	12.23	6.31	3.75
PV externality, first ring (\$m)	-4.23	-2.19	-1.30	-13.97	-7.26	-4.34
PV externality, total (\$m)	-0.82	-0.41	-0.24	-1.74	-0.94	-0.59
	(4.61)	(3.02)	(2.29)	(15.95)	(10.26)	(7.73)
Incremental construction cost (\$m)	2.97	2.97	2.97	10.02	10.02	10.02
Externality per marginal construction \$	-0.28	-0.14	-0.08	-0.17	-0.09	-0.06
	(1.55)	(1.02)	(0.77)	(1.59)	(1.02)	(0.77)

Notes: Table C9.1 reports the present value of the private-market rent externality avoided by raising construction investment per room by Δ (left panel: \$1,000; right panel: the cross-project interquartile range of \$3,379), for the median pre-1970 project, against the incremental construction outlay. All amounts in 2010 dollars. Rent effects use the coefficients of construction cost per room on log *New York Times* median listing rent per room from the headline attribute specification (demeaned attributes, apartment count excluded, baseline-outcome control), for the public-housing tract and the first ring at the 0–30 and 40–60 year horizons. The rent base is the pre-completion private rent bill: 12 times pre-completion real median contract rent times private occupied units (net of public housing, scaled by tract occupancy), summed within ring and taken at the cross-project median. Flows apply the medium-run coefficient to years 0–34 and the long-run coefficient to years 35–64. The incremental cost multiplies Δ by the median project's rental rooms (apartments times rooms per apartment, NYCHA records). Delta-method standard errors in parentheses treat the four coefficients as independent.

than causal effects. Second, rent declines do not represent pure social losses. They partly reflect transfers between landlords and tenants, and the calculation holds the baseline rent bill fixed over time. Third, I abstract from benefits within the project itself. If higher construction investment also improved housing services for tenants or reduced future maintenance and renovation costs for the housing authority, the overall return to construction investment would exceed the neighborhood component I measure here. The exercise nevertheless shows that the neighborhood externalities are economically meaningful relative to construction costs. At the same time, the estimates are too imprecise—and the calculation too partial from a welfare perspective—to determine whether the historical level of construction investment was too low or too high.

Appendix References

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